Build or Subsidize? The Welfare Effects of Deployment and Affordability Subsidies in U.S. Broadband

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Job Market Paper

Abstract

Should broadband subsidies make existing service cheaper, or build networks where none exist? I evaluate the $14.2 billion Affordable Connectivity Program (ACP) and the $42.45 billion Broadband Equity, Access, and Deployment (BEAD) program in a common welfare framework. I combine cross-state event studies with a tract-level model of demand, pricing, and entry estimated on 170,419 products across 66,454 tracts. To scale entry estimation, I bound provider-tier fixed costs with moment inequalities that avoid solving the full equilibrium game. ACP lowered prices 7–10 percent at peak and shifted demand toward faster plans; BEAD's effects are smaller and lag construction. In 25-year counterfactuals, BEAD yields at least $12 in welfare per subsidy dollar under conservative assumptions and about $26 in the baseline, while ACP's measured gains roughly offset outlays. The programs solve different margins: build where deployment is uneconomic, then target affordability where low-income households remain price constrained.

Keywords: Broadband subsidies; digital divide; entry and market structure; moment inequalities; welfare and distributional incidence.

JEL classification: C57; D22; H42; H54; L13; L96

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1 Introduction

Should broadband subsidies make service affordable, or build it where it does not exist? The 2021 Infrastructure Investment and Jobs Act (IIJA) bet roughly $65 billion on both at once, and its two flagship programs pull on opposite ends of the market. The Affordable Connectivity Program (ACP), a $14.2 billion demand-side subsidy, gave eligible low-income households monthly discounts of up to $30, lowering the consumer-facing price where service already existed. The Broadband Equity, Access, and Deployment (BEAD) program, a $42.45 billion supply-side grant and the largest federal deployment effort in U.S. history, lowers the fixed cost of building service where private deployment never materialized. The stakes are concrete: in 2020, roughly 21 million American households lacked fixed broadband,[1] a shortfall that hits hardest on low-income and rural communities and reaches into job search, schooling, and healthcare.

The two instruments target distinct market failures, and how far each travels depends on a third force: market structure. On the demand side, affordability binds: by early 2021, 15 percent of home-broadband users reported trouble paying their internet bill, and subscription rates among households below 200 percent of the federal poverty line ran about 20 percentage points below the national average (Pew Research Center, 2021). On the supply side, deployment binds: fixed costs make broadband privately unprofitable in low-density areas even when the social return is large, the classic wedge that justifies public subsidy (Tirole, 1988; Bresnahan and Reiss, 1991; Kearns, 2026). Market structure is not a third failure but the condition that governs how far the other two reach: the average census tract in my data has 2.36 fixed-broadband providers, and the median within-tract Herfindahl–Hirschman index (HHI) is 5,472, more than twice the conventional threshold for concentrated markets. Where providers are few, a subsidy that enlarges the addressable market or lowers the cost of entry can move posted prices and product menus, equilibrium responses that subscriber counts alone cannot reveal.

This paper asks three questions. First, did ACP and BEAD change broadband prices, product offerings, and local market structure after IIJA? Second, how large are the welfare gains from each program once firms can enter or reposition? Third, are demand- and supply-side subsidies substitutes, or do they reach different parts of the digital divide? The questions are live: ACP exhausted its funding in June 2024, leaving Congress to decide whether a successor is worth financing, while BEAD, still building, is the largest federal deployment program in U.S. history. Evaluations that count new subscribers or dollars per connection

[1] Federal Communications Commission (2020) reported 21.3 million Americans lacking access to fixed broadband at 25/3 Mbps based on FCC Form 477 data.

cannot answer them in local markets with few providers and many infra-marginal households.

I answer with reduced-form and structural evidence. The reduced form uses continuous-treatment event studies that compare states with different exposure to ACP eligibility and BEAD funding before and after IIJA. ACP exposure is the share of households below 200 percent of the federal poverty line, measured from the 2020 American Community Survey (ACS) before launch; BEAD intensity is the log of allocation per unserved location, fixed by federal formula over pre-grant counts. Both intensities are predetermined with respect to post-IIJA outcomes. Because all states are treated at once, identification comes from cross-state differences in intensity rather than in timing, so the negative-weighting and contamination problems of staggered-adoption designs do not arise. The identifying assumption is that, conditional on state and year fixed effects and state-specific linear trends, states with different exposure would have trended similarly absent the programs; pre-period coefficients near zero support it. One data feature biases the BEAD estimates toward zero: the FCC's Urban Rate Survey samples providers in urban tracts, under-representing exactly the rural markets BEAD targets. I therefore read the reduced-form BEAD estimates as a conservative lower edge and rely on the structural counterfactuals to recover the completed-deployment margin.

Event studies establish that the programs moved prices, menus, and concentration, but they cannot quantify welfare. A falling price is consistent with pass-through, strategic repricing, or a shifting plan mix, and the reduced form cannot separate them; state averages wash out the tract-level variation in structure and cost that drives the response; and no event study can value a design that was never implemented or split welfare into consumer and producer components. The structural model takes up exactly these questions. It is a two-stage game played in each census tract: providers first choose which speed tiers to offer, paying a fixed cost per activated position, then set prices in Bertrand–Nash competition. I estimate a nested-logit demand system using quadratic differentiation instruments in the spirit of Gandhi and Houde (2026), recover markups and marginal costs from the pricing first-order conditions, and bound entry fixed costs with the scalable discrete-game method of Fan and Yang (2024), which requires only that observed configurations be undominated.[2] Bounding fixed costs rather than point-estimating them is what makes the problem tractable: I never solve the entry game inside the estimation loop, infeasible with tens of thousands of markets and many potential positions, and I never impose an equilibrium selection rule to do

[2]The key estimates line up with external benchmarks. The mean own-price elasticity of -4.99 falls within the product-level broadband and pay-TV range (-5.9 in Goetz (2019); -4.1 to -5.4 in Crawford et al. (2019)). Kearns (2026) reports smaller elasticities of -1.48 to -3.30 in Seattle; the difference likely reflects Seattle's served-market setting and a demand specification allowing income heterogeneity. Recovered fixed-cost bounds are of the same order as the deployment costs Kearns (2026) recovers from revealed decisions. Section 5 reports the full comparisons.

so. The reduced-form and the structural analysis are linked by construction: the tract-level eligibility share that defines ACP in the counterfactuals is the same variable that measures treatment intensity in the event study, and the model's 2020 baseline predates both programs, so the counterfactuals start from the undisturbed market whose response the event studies trace.

Both halves of the analysis draw on official federal data. For the reduced form, I build a balanced panel of 52 U.S. states (the 50 states, the District of Columbia, and Puerto Rico) from 2014 to 2025, combining FCC Urban Rate Survey prices, ACP eligibility exposure from the ACS, and BEAD allocations from federal records. For the structural analysis, the demand sample contains 170,419 observed broadband products across 66,454 census tracts at the tract-provider-speed-tier level. The entry sample is larger, adding feasible but unoffered positions implied by each provider's county footprint: 727,601 potential positions across 70,854 tracts.

The two programs operate through different margins. In the event study, moving from the 25th to the 75th percentile of ACP exposure lowers average prices by roughly 7 to 10 percent at peak, with effects on high-speed plans of 15 to 18 percent that persist through the second and third post-launch years; the low-speed plan share contracts and the high-speed share expands, as subsidized households raise demand for faster service and providers upgrade their menus. BEAD effects are smaller and arrive with a two-to-three-year lag, consistent with its institutional clock: allocations announced June 2023, state proposals due that December, construction only afterward. The timing is itself informative: effects appear where the construction timeline predicts and not before, pointing to entry as the operative channel rather than a near-null result. These estimates mark the leading edge of the program, not its full effect, because BEAD construction extends through 2026 and beyond.

The counterfactuals translate these patterns into welfare. A 75 percent BEAD fixed-cost subsidy induces roughly seven to nine thousand new firm-market entries and brings service to most of the nine thousand tracts that begin with no provider, raising monthly consumer surplus from about \$14 billion to about \$15.5 billion. A \$30 ACP discount raises monthly consumer surplus to about \$14.7 billion, operating almost entirely through demand expansion: the consumer-facing price falls by about nine dollars while the gross posted price barely moves, and take-up rises by roughly 1.5 percentage points. Holding gross prices fixed isolates the demand-expansion channel an affordability transfer is designed to deliver; it also excludes the equilibrium price reductions the event study documents, so the ACP welfare estimate is a lower bound rather than a full accounting of the program's effect.

Distributional results answer the third question: the programs' gains barely overlap, so they are complements rather than substitutes. Baseline market structure determines BEAD's gains: monopoly tracts gain about \$31 per household per month, five times the gain in tracts with three or more providers, but gains are nearly flat across income quartiles. ACP's gains run the opposite way: the lowest-income quartile gains more than twice the highest, but gains are largely invariant to provider count. Each instrument reaches a part of the market the other cannot, BEAD the deployment margin and ACP the affordability margin, with limited overlap in served monopoly and duopoly markets.

The cost-benefit analysis sets these gains against fiscal outlays, giving two benefit-cost ratios: one for BEAD's one-time capital investment in unserved markets, one for ACP's recurring transfer in served ones. The object of interest is not which ratio is larger but the sequencing it implies. Benefits here are the discounted stream of consumer and producer surplus the model attributes to each program, and nothing else; I make no appeal to spillovers into employment, schooling, or health, so the ratios understate the full social return. At a 3 percent discount rate over a 25-year horizon, standard for long-lived broadband assets (Grunvalds et al., 2017), the 75 percent BEAD subsidy stays above 12 even under the most conservative assumptions I can impose and reaches about \$26 per dollar spent in the baseline; because lower subsidy rates reach easier markets, the ratio only rises as the subsidy falls. I therefore lead with the floor: BEAD clears any plausible cost-benefit bar by a wide margin, and the exact multiple is second-order. ACP's recurring outlays roughly break even, a measured benefit-cost ratio near one: the welfare gains I can price offset outlays, with a small positive residual from marginal subscribers and a pure transfer to infra-marginal households. Because this figure holds gross prices fixed, it omits the equilibrium price cuts the event study documents and so is itself a lower bound; ACP's true ratio exceeds one, though by how much the structural model is not designed to say. Together the two ratios support a sequencing principle: build where private deployment is uneconomic, then target affordability where price remains the binding constraint.

This paper makes three contributions. First, to the evaluation of broadband subsidies, it moves beyond adoption and take-up to causal evidence on equilibrium outcomes such as prices, product menus, entry, and concentration (Rosston et al., 2010; Nevo et al., 2016; Flamm and Varas, 2022; Espín and Rojas, 2024), and it evaluates ACP and BEAD together: the two programs draw on the same federal appropriation yet are typically studied apart, and joint estimation on one demand system, one cost structure, and one baseline is what makes the central trade-off answerable at all. Second, to the economics of infrastructure in markets with large fixed deployment costs: because the household's price constraint and the firm's fixed-cost constraint rarely bind in the same market, demand- and supply-side subsidies emerge as complements from the deployment technology rather than from program design, and the return to each is ordered by local market structure. Third, methodologically, it brings scalable entry bounds to infrastructure: closest to Eizenberg (2014), Ciliberto et al. (2021), and Fan and Yang (2024), I use moment inequalities to recover fixed costs and entry responses across tens of thousands of local markets, a scale at which solving the entry game inside estimation would be infeasible.

The rest of the paper proceeds as follows. Section 2 describes the institutional setting, data, and local market structure. Section 3 presents the reduced-form evidence. Section 4 develops the structural model. Section 5 reports estimation results and diagnostics. Section 6 presents counterfactual simulations and distributional effects. Section 7 reports the cost-benefit analysis and sensitivity. Section 8 concludes.

2 Industry Background and Data

2.1 Industry Background

2.1.1 Current Landscape

The technologies available to a household depend primarily on location, which determines both local competitive pressure and the geography of the access gap. Four main technologies deliver broadband service. Digital subscriber line (DSL) transmits data over legacy copper telephone lines. Cable modem service uses coaxial television networks. Fiber-optic broadband carries data through optical fibers using light signals. Fixed wireless access connects households through radio links from nearby transmission towers. These technologies differ substantially in speed, reliability, and infrastructure requirements. Fiber generally provides the highest speeds and lowest latency but requires heavy capital investment in network deployment. Fixed wireless offers lower throughput yet deploys more cost-effectively in rural areas, requiring no extensive trenching or new wired infrastructure. Providers compete on speed, data allowances, and price. Telephone and cable incumbents still dominate, but national fiber entrants, fixed-wireless operators, and a handful of municipal systems now contest local markets.[3]

Two facts define the market and explain why Congress intervened. Service is expensive relative to household income, and most households face at most two or three providers (Flamm and Varas, 2022). Both facts hold sharply in my data. The average census tract has just 2.36 fixed-broadband providers. The median within-tract provider HHI is 5,472, more than double the 2,500 threshold conventionally associated with concentrated markets. The gap falls unevenly. Fiber, the platform with the best performance, concentrates in cities and suburbs. Rural households still depend on aging copper DSL whose speeds and reliability fall

[3] Wilson (2025) studies the effects of municipal broadband entry on private internet service provider investment. See Appendix A for an overview of broadband technology types.

well short of the federal broadband standard.

These conditions produce the digital divide: a gap in reliable internet access that tracks income and geography closely. Its consequences run through job search, healthcare, and schooling (Forman et al., 2005; Goldfarb and Prince, 2008; Goolsbee and Klenow, 2002). Federal policy has responded along the same two margins that define the underlying failures: lowering the price of service that already exists and paying to build service where private providers have not.

2.1.2 Federal Broadband Subsidies

The 2021 Infrastructure Investment and Jobs Act (IIJA) committed roughly \$65 billion to broadband through two programs that address opposite sides of the market. The Affordable Connectivity Program (ACP), funded at \$14.2 billion, subsidized household bills directly. The Broadband Equity, Access, and Deployment (BEAD) program, funded at \$42.45 billion, subsidizes deployment in underserved areas.

ACP succeeded the Emergency Broadband Benefit (EBB), which reached nearly nine million low-income households at up to \$50 per month between February and December 2021. ACP scaled far larger. At its peak, it enrolled 23 million households with a monthly discount of up to \$30 (\$75 on qualifying Tribal lands) and a one-time device discount of up to \$100.[4] Its funding lapsed in June 2024 when Congress declined to reappropriate. This full cycle, launch in 2022 through funding exhaustion in 2024, falls within the event-study window. The program's termination also motivates the central policy question of whether a successor affordability program is worth financing. The welfare and fiscal-efficiency analysis in Sections 6–7 addresses this question directly.

BEAD is the largest federal program ever to subsidize broadband providers directly. A federal formula allocates funds to states by the count of underserved locations: broadband-serviceable locations with no service, or with service slower than 25/3 Mbps or latency above 100 milliseconds.[5] Because the formula is based on pre-grant counts of unserved locations, treatment intensity is fixed in advance of post-IIJA broadband outcomes. That is why the effects documented below arrive late. IIJA passed in November 2021; NTIA announced allocations only in June 2023; state proposals were due that December; subgrantee selection, permitting, and construction followed only after that. Deployment will extend through at least 2026. The reduced-form event-study sample runs through 2025, so it captures only these early, lagged effects, not BEAD's full deployment cycle. I anchor the counterfactuals to a pre-IIJA baseline instead: they represent the program's steady-state effects once deployment

[4] https://www.fcc.gov/acp
[5] https://broadbandusa.ntia.doc.gov

is complete, not a forecast for any single year.

2.1.3 Economic Rationale

Each program targets a distinct market failure; that distinction structures the empirical strategy and the counterfactual analysis. On the demand side, affordability binds: even where service exists, low income and limited digital literacy keep many households below the participation margin. By early 2021, 15 percent of U.S. home-broadband users reported trouble paying for high-speed internet service, and 2020 ACS data show that subscription rates among households below 200 percent of the federal poverty line ran about 20 percentage points below the national average (Pew Research Center, 2021). A demand-side subsidy addresses this gap directly, lowering the effective price to help price-constrained households subscribe.

On the supply side, deployment binds: fixed costs are high enough to make broadband privately unprofitable in low-density or low-income areas even when the social return to connectivity is large. This wedge between private and social returns to infrastructure is the classic case for public subsidy (Bresnahan and Reiss, 1991; Kearns, 2026). BEAD lowers the provider's private cost of entry, but its net effect depends on how much private deployment it displaces.

The two failures barely overlap, making the programs complements rather than substitutes. A demand subsidy does nothing where no infrastructure exists. In wholly unserved areas, no discount can induce adoption of a product that is not offered. A deployment subsidy has limited reach on affordability where competitive infrastructure already exists and price, not access, is the binding constraint. The boundary softens in served monopoly or duopoly markets. In those markets, a second provider may face high entry costs, and BEAD-induced entry can discipline prices through Bertrand competition. Each program therefore addresses the failure the other leaves unresolved in most markets, with some overlap at the frontier.

Market structure opens a third channel. With most households facing one to three providers (Flamm and Varas, 2022), broadband is an oligopoly, and a demand subsidy that enlarges the addressable market can move posted prices and product menus, equilibrium responses that subscriber counts miss. The reduced-form event study tests for these responses; the counterfactuals then hold gross provider prices fixed, isolating the demand-expansion channel from provider price adjustment.

2.2 Data Sources and Estimation Samples

This section draws on two distinct samples: a state-year panel for the reduced-form event studies, and a tract-provider-speed-tier structural sample for the entry and demand model. Both rely on the data sources described below. I construct a new tract-level dataset of the U.S. broadband market on the eve of IIJA, the paper's central data input. No single federal source contains the information needed to estimate broadband supply, demand, and entry jointly: different agencies collect plan-level prices, product-level deployment, household demographics, and adoption at different units of observation. I assemble five publicly available federal sources into a nationally representative pre-IIJA dataset linking prices, product characteristics, market structure, and adoption at census-tract resolution. Three construction steps make the merged data usable at this scale: a brand-first price-matching hierarchy that assigns a price to each of the 727,601 potential product positions (provider-tier-tract triples, defined formally in Section 4), a manual normalization of provider names into 1,732 canonical corporate identifiers, and a market-share construction validated against an independent national benchmark. I describe each step and its source below.

2.2.1 Data Sources

FCC Urban Rate Survey (URS). The FCC conducts the Urban Rate Survey annually. The survey draws a stratified random sample of Internet Service Providers (ISP) operating in urban census tracts and collects each plan's advertised download speed, upload speed, technology type, and monthly price.[6] It is one of the few nationally representative sources of plan-level broadband prices and is now a standard input to structural work on broadband competition and infrastructure policy (Espín and Rojas, 2024; Kearns, 2026). The first step assigns a survey price to every tract-level product. I match prices through a six-level brand-first hierarchy that proceeds from a state-provider-tier match down to a national tier average, selecting the URS plan closest in download speed within each level, and deflate prices to constant 2025 dollars using the CPI-U. The hierarchy achieves complete coverage: all 727,601 potential product positions receive a price; no position therefore drops out of the entry analysis for lack of price data. Coverage preserves match quality: 63.9 percent of positions match on the provider's own plans, and only 1.4 percent fall to the coarsest national-tier level. Appendix D documents the full hierarchy and the match-quality distribution.

The URS samples only urban tracts, under-representing the rural markets BEAD is designed to serve. As a result, the state-year event-study estimates for BEAD tend toward

[6] https://www.fcc.gov/economics-analytics/industry-analysis-division/urban-rate-survey-data-resources

zero: they capture urban and suburban response, but rural deployment effects fall outside the sample frame. I treat the URS-based BEAD estimates as conservative lower bounds on the program's full impact.

FCC Form 477 Deployment Data. Form 477, collected semi-annually, requires all ISPs to report the maximum advertised download and upload speeds of every fixed residential broadband service they offer at the census-block level.[7] I restrict to residential fixed broadband and aggregate to the census-tract × provider × speed-tier level. The December 2020 release defines the structural baseline. It identifies which provider-tier products are observed in each tract and which are feasible but not offered, the two inputs to the Fan–Yang moment inequalities. Form 477 deployment records also construct the county-footprint potential entry set by identifying all provider-tier pairs active anywhere in the county.

Form 477 marks a block as served if a provider *could* serve any household at that speed, overstating actual availability in multi-dwelling-unit buildings and rural blocks. Since I use December 2020 data before the post-IIJA transition to the Broadband Data Collection system, this overstatement is uniform across the cross-section. It therefore does not differentially affect the treatment variation used for identification. Provider names in Form 477 vary substantially across subsidiaries and legacy brands, and treating each filing name as a distinct firm would badly overstate competition. The second construction step consolidates these names into 1,732 canonical corporate identifiers using a manually constructed crosswalk, attributing markups, market structure, and entry decisions to the economic decision-maker, not its filing labels (Appendix C).

American Community Survey (ACS). The Census Bureau's ACS provides annual small-area estimates of demographic and socioeconomic conditions at the census-tract level.[8] I use the ACS 2019 five-year release, the pre-IIJA vintage closest to the 2020 structural baseline. At the tract level, I extract the number of housing units, median household income, poverty rate (Table C17002, households below 200 % FPL), broadband subscription rate (Table B28002), renter share, college-educated share, age-65+ share, and no-internet share. These variables enter the model as market-size measures, eligibility proxies for ACP counterfactuals, and demand-side controls in both the market-share construction and the structural demand estimation. The poverty rate also serves as the ACP treatment intensity FPL200_i in the reduced-form event study, directly bridging the two parts of the empirical strategy.

[7] Data available at https://broadband477map.fcc.gov/#/data-download; 2014–2023.

[8] Data available at https://www.census.gov/programs-surveys/acs.

FCC Broadband Adoption Data. The FCC publishes tract-level fixed broadband adoption statistics derived from Form 477 subscription filings. The data report the share of households with fixed broadband connections by speed threshold in five binned ranges (10, 30, 50, 70, and 90 percent of households).[9] I use the ≥10/1 Mbps adoption bin to anchor the outside-option share $s_{0t} = 1 - S_t^{\text{fixed}}$ in each tract. This outside-option share is the key object the CSS market-share construction, the third construction step, requires. The 10/1 Mbps threshold aligns with the speed-tier definition used throughout the structural model (Tier H: ≥10/1 Mbps; Tier L: below). The construction passes an external check it was not built to satisfy: the construction implies 93.7 million subscribing households at ≥10/1 Mbps, within about 1 % of the FCC national benchmark of 94.5 million (Table 3).

BEAD Program Allocation Records. A federal formula determines state-level BEAD allocations, announced by the NTIA in June 2023, from the certified count of broadband-serviceable locations lacking reliable service at 25/3 Mbps or entirely unserved.[10] The event study exploits this formula property: certified unserved-location counts, not post-IIJA broadband outcomes, determine funding intensity. I measure treatment intensity as log BEAD, the log of total BEAD allocation per certified unserved location, a quantity the formula fixes before the 2022–2025 outcome window.

2.2.2 Sample construction and the two estimation samples

These five sources feed two estimation samples, each matched to a distinct empirical question.

The *state-year policy panel* covers 52 states (including the District of Columbia and Puerto Rico) from 2014 to 2025 ($N = 624$ state-year cells). It pairs Urban Rate Survey plan-level outcomes, log prices by speed tier, plan-composition shares, and provider concentration, with ACP eligibility exposure (FPL200 from the 2020 ACS) and BEAD funding intensity (log BEAD from NTIA). The resulting data supports the continuous-treatment event studies in Section 3.

The *tract-level structural sample* is a single 2020 cross-section built around the Form 477 deployment file. I merge URS prices, ACS demographics, and FCC adoption shares, then restrict to tracts containing at least one positively priced product. The resulting *demand estimation sample* contains 170,419 products (provider-tier-tract triples with observed tiers) across 66,454 census tracts and 1,732 canonical providers. The *entry sample* augments the observed products with county-footprint feasible positions, yielding 727,601 potential

[9] December 2020 release available at https://docs.fcc.gov/public/attachments/DOC-395959A1.pdf.

[10] State allocation totals available at https://www.ntia.gov/funding-programs/internet-all/broadband-equity-access-and-deployment-bead-program/program-documentation/state-allocation-totals.

provider-tier-tract positions across 70,854 tracts.[11] The county-footprint rule captures nearby but not-yet-offered configurations without positing entry from providers with no local infrastructure. The two samples differ in unit of observation and question: the demand sample identifies preference and price-sensitivity parameters from observed market shares; the entry sample identifies which positions providers offer and which they do not, 557,182 of the 727,601 potential positions being unoffered.

Data quality, not only identification, motivates this split between a state-year panel and a single-year structural cross-section. No consistent tract-year panel spans 2014–2025: the FCC's post-2021 transition from Form 477 to the Broadband Data Collection system redefined geographic units and coverage measurement in ways that would inject spurious dynamics into a long tract panel. The state-year aggregation absorbs that structural break; the 2020 cross-section predates both ACP enrollment and any BEAD-induced deployment; the structural baseline, and every counterfactual built on it, therefore reflects a market untouched by either program. Appendix B reports file-level details, merge keys, and sample restrictions; the full replication package is available upon request.

2.3 Descriptive Statistics

Table 1 summarizes the demand estimation sample. The typical plan costs \$74.06 per month. High-speed plans dominate the 2020 product menu: 90.8 % of products fall into Tier H, reflecting that most carriers have retired legacy sub-10 Mbps DSL. Despite the preponderance of high-speed products, consumer choice remains thin. Just 2.36 providers serve the average tract, offering 2.56 products, and over half of tracts have at most two providers (Table 4). Median household income is \$65,422. The poverty share below 200 % FPL, the ACP-eligible population, is 32.6 %.

[11]The demand estimation sample (66,454 tracts) is smaller than the full entry sample (70,854 tracts) because 4,400 tracts with observed products did not match when merging demand and tract-level covariates. The resulting sample remains large enough to reliably recover the demand and supply estimates.

Table 1: Summary Statistics — Estimation Sample

Variable	Mean	Std. Dev.	P25	P75
Panel A. Product and Market Characteristics				
Price ($/month)	74.06	19.64	63.66	85.99
Download speed (Gbps)	0.555	0.767	0.026	1.000
Upload speed (Gbps)	0.313	0.748	0.006	0.500
Tier H indicator	0.908	—	—	—
Products per tract	2.56	1.39	2	3
Providers per tract	2.32	1.16	2	3
Panel B. Tract Demographics				
Population (tract)	4,533	2,416	2,957	5,615
Median household income ($)	65,422	32,562	44,083	78,750
Share renters (%)	35.24	22.36	17.79	49.02
Share college educated (%)	29.91	19.12	15.54	40.47
Share age 65+ (%)	16.64	7.63	11.76	20.31
Share below 200% FPL (%)	32.55	18.18	18.56	43.71

Notes: Price is reported in dollars per month. Tier H denotes service with download speed ≥ 10 Mbps and upload speed ≥ 1 Mbps. I measure demographic variables at the census-tract level from the 2019 ACS 5-year estimates and merge them to each product observation. The poverty measure draws from ACS Table C17002 (income-to-poverty ratio below 2.0). Statistics cover 170,419 product-level observations across 66,454 tract markets.

2.4 Market Share Construction

Estimating broadband demand confronts one binding data limitation: product-level market shares go unobserved. The FCC discloses provider locations, technologies, and advertised speeds, but never the subscriber count for a given provider-speed product in a given tract, which is required to estimate the demand system.

I recover these shares by combining observed adoption totals with product availability, following the logic of Crawford et al. (2019). Two steps suffice. FCC tract-level adoption data, reported in binned ranges for any fixed broadband and for service at or above 10/1 Mbps, anchor total adoption and the outside option. Logit weights built from product characteristics and demographics then allocate that total across the available provider-speed-tier products. These shares are inferences, not observed counts; treat them as such. *Implementation details, the allocation equation, and validation diagnostics are in Appendix E.*

Table 2 reports summary statistics for the constructed market shares.

Table 2: Market Share Summary Statistics

Variable	Mean	Std. Dev.	P25	P75
Product market share $\hat{s}_{jt}$	0.322	0.208	0.182	0.417
Within-tier share $\hat{s}_{j\|g,t}$	0.456	0.245	0.278	0.527
Outside-option share s_{0t}	0.191	0.149	0.100	0.300
Provider HHI	5,472	2,576	3,463	5,415

Notes: All shares satisfy $\sum_j s_{jt} + s_{0t} = 1$ in every tract. I infer product shares from FCC adoption information and product-level data (see Appendix E.2). Within-tier share equals $\hat{s}_{j|g,t} = \hat{s}_{jt} / \sum_{k \in g} \hat{s}_{kt}$, where $g \in \{H, L\}$ denotes the speed tier. I compute the Herfindahl–Hirschman Index (HHI) from provider shares among fixed-broadband subscribers within tracts, multiplying the sum of squared shares by 10,000 so that the index is on the standard 0–10,000 scale. The demand estimation sample contains 170,419 observed provider-tier products in 66,454 census tracts.

The constructed sample reproduces national aggregates closely. Table 3 benchmarks the sample against FCC totals: the 70,854 tracts cover 118.6 million households (95.9 % of the national benchmark), and the inferred adoption shares imply 93.7 million residential fixed connections at ≥10/1 Mbps, within about 1 % of the FCC national figure. Appendix E reports the model-implied market shares by provider as a further internal consistency check.

Table 3: Aggregate Validation of the Sample

Moment	Sample	FCC national benchmark	Coverage
Households	118.59 million	123.67 million	95.90%
Residential fixed connections, ≥10/1 Mbps	93.70 million	94.45 million	99.21%

Notes: The sample contains 70,854 census tracts; the demand estimation sample contains 66,454 tracts. Residential fixed connections equal tract households multiplied by the FCC-based 10/1 adoption-share bin. The structural model defines product speed tiers by advertised 10/1 Mbps availability. FCC national benchmarks are from Federal Communications Commission (2022). Coverage is the structural-sample value divided by the FCC national benchmark. These moments validate the scale of the sample against national residential broadband aggregates; they are not provider-level subscriber counts.

Broadband market structure. How many providers serve a tract governs both the competitive pressure on price and the room for subsidy-induced entry, so the distribution of provider counts is a key element of the analysis. Table 4 reports it.

Most tracts are monopolies or duopolies. Nearly a fifth have exactly one fixed-broadband provider, and nearly two-thirds have at most two. The 20.77 percent of households in

Table 4: Provider Competition in the Sample

Number of providers in tract	Share of tracts (%)	Share of households (%)
One provider	21.63	20.77
Two providers	42.07	40.94
Three or more providers	36.30	38.30

Notes: Provider counts cover canonical providers with positive constructed product shares in the tract. Sample: 66,454 census tracts, December 2020.

monopoly tracts face no intra-market price competition, and only 36.30 percent of tracts reach three providers or more. This thin competition is itself part of what the federal programs target, and precisely what gives subsidy-induced entry room to move consumer welfare.

3 Reduced-Form Evidence

3.1 State-Year Panel and Treatment Variables

The event study traces how broadband prices, product menus, and concentration moved in states more exposed to each program. The panel is balanced over 52 states (including Puerto Rico and the District of Columbia) from 2014 to 2025, yielding $N = 624$ state-year cells. I build outcomes from FCC Urban Rate Survey plan records. I aggregate them to state-year plan-count-weighted average prices and plan-composition shares, then deflate prices to 2025 USD by the CPI-U. Six outcomes span three economic margins: prices (log average price and log prices by speed tier), product mix (plan shares below and above the 25/3 Mbps threshold), and concentration (provider HHI). Table 5 reports descriptive statistics.

Panel structure and outcome construction. The study window spans both program rollouts, with enough pre-treatment years to check parallel trends. For both ACP and BEAD, 2021 is the enactment year, so the sample runs seven years before (2014–2020), the enactment year (2021), and four years after (2022–2025). For ACP, the post-enactment period splits into program implementation (2022–2024) and one follow-up year (2025). For BEAD, 2021 is when IIJA passed, but the program's own rollout starts later: NTIA allocations arrive in 2023, and deployment continues past the end of the sample. This gives five pre-treatment leads to detect violations of parallel trends, and four to five post-treatment years to trace

Table 5: Descriptive Statistics: Main Outcomes

Outcome	N	Mean	SD	Min	Median	Max
Panel A. Prices						
Log avg. real price	597	4.583	0.190	3.827	4.556	5.254
Log price (<25/3 Mbps)	563	4.200	0.210	3.314	4.223	5.073
Log price (≥25/3 Mbps)	587	4.703	0.223	4.169	4.673	5.483
Panel B. Plan Composition						
Plan share (<25/3 Mbps)	597	0.261	0.260	0.000	0.171	1.000
Plan share (≥25/3 Mbps)	597	0.739	0.260	0.000	0.829	1.000
Panel C. Market Structure						
Provider HHI	597	0.524	0.238	0.146	0.477	1.000

Notes: The unit of observation is state × year, 2014–2025 ($N = 624$ state-year cells before dropping observations with missing outcomes). These six are the main outcomes in the event studies and difference-in-differences regressions. Prices are plan-count-weighted averages, deflated to 2025 USD by the CPI-U. Plan shares and the HHI lie in $[0, 1]$.

dynamic effects.

One feature of the data biases the BEAD estimates toward zero. The Urban Rate Survey samples urban tracts; the panel therefore captures urban and suburban conditions and under-represents the rural markets BEAD targets. If BEAD's price and concentration effects concentrate in rural areas, as the structural results below imply, the URS-based estimates understate its true impact, making them a conservative lower bound. ACP, which works through demand expansion in already-served and largely urban markets, aligns better with the URS frame; the survey captures its effects more directly.

Both treatment intensities are predetermined and time-invariant, which limits the endogeneity concerns that plague contemporaneous treatment measures. The ACP eligibility share (FPL200) comes from the 2020 ACS, before launch. A federal formula fixes the BEAD intensity ($\log \text{BEAD}$) from pre-grant unserved-location counts. Neither responds to ISP behavior or state outcomes over the window. This supports the assumption that exposure is uncorrelated with differential trends.

ACP exposure. I measure state-level ACP eligibility exposure as the share of households with incomes below 200 % of the federal poverty level (FPL200), drawn from the 2020 American Community Survey. Time-invariant and pre-determined, it varies across states and proxies for the potential demand stimulus ACP can generate. The program became active in

2022.

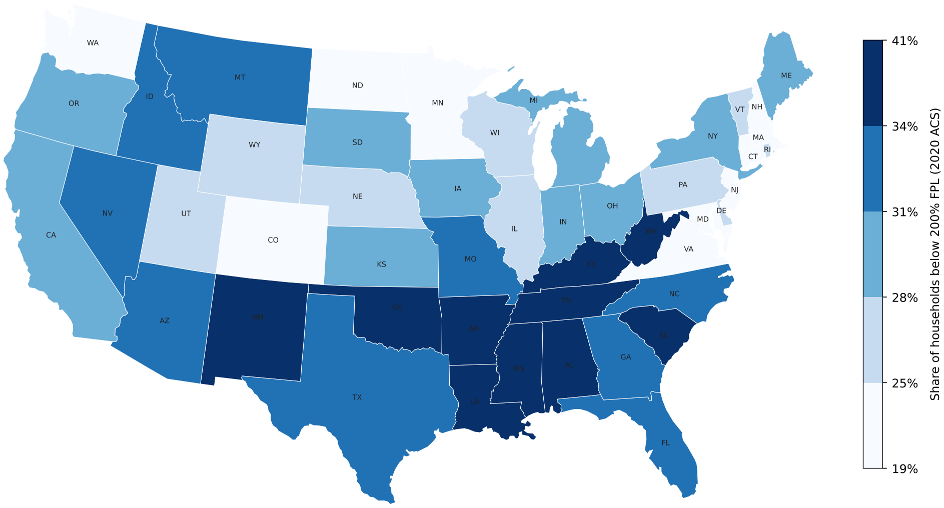

Figure 1: State-level ACP exposure: share of households with incomes below 200 % of the federal poverty level (FPL200), measured from the 2020 American Community Survey. Darker shading indicates higher eligibility exposure. Color breaks at sample quintiles.

BEAD exposure. I measure state-level BEAD intensity as the log of BEAD allocation (USD) per unserved broadband location (log BEAD). A higher value indicates that a state receives more funding per targeted location, implying a more intensive supply-side intervention. The federal allocation formula also fixes this measure, making it time-invariant. The NTIA released BEAD allocations in 2023.

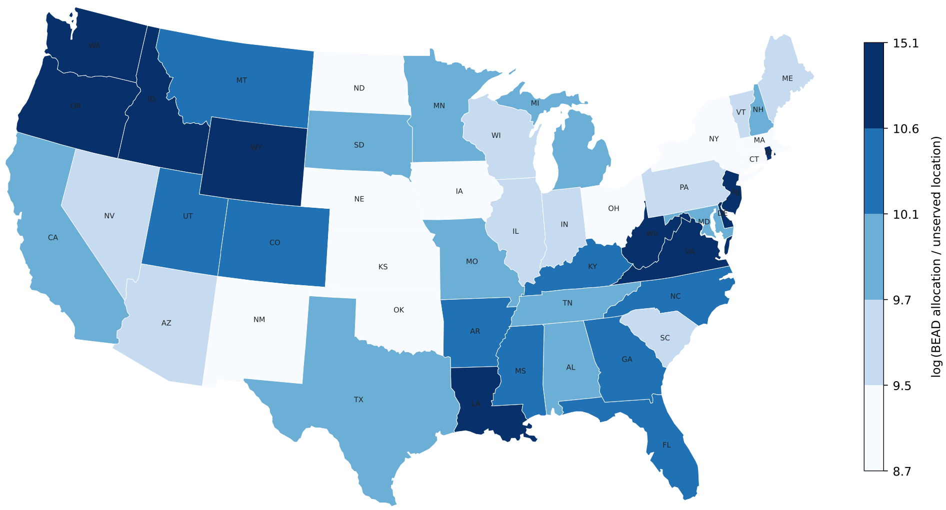

Figure 2: State-level BEAD exposure: log of BEAD allocation (USD) per unserved broadband location (log BEAD), based on NTIA allocation announcements (June 2023). Darker shading indicates higher funding intensity per targeted location. Color breaks at sample quintiles.

3.2 Identification Strategy

For program $P \in \{\text{ACP}, \text{BEAD}\}$, let D_i^P denote the time-invariant treatment intensity for state i and let τ index event time relative to the relevant policy year. The dynamic specification is:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{\substack{\tau=-5 \\ \tau\neq-1}}^{\bar{\tau}^P} \beta_\tau^P \Big(D_i^P \times \mathbf{1}[t = \bar{t}^P + \tau]\Big) + \gamma_i\, t + \varepsilon_{it}, \tag{3.1}$$

where α_i are state fixed effects, λ_t are year fixed effects, and $\bar{t}^P$ is the event-year normalization for program P. For ACP, $\bar{t}^{\text{ACP}} = 2022$, the first full implementation year. For BEAD, $\bar{t}^{\text{BEAD}} = 2021$, the IIJA enactment year. The static post indicator instead begins in 2023, when the NTIA announced allocations. The omitted period is $\tau = -1$, corresponding to the year immediately before the relevant event year; all estimates are therefore relative to that baseline. I weight regressions by the number of plans n_{it} and cluster standard errors at the state level.

The role of fixed effects. State fixed effects α_i absorb all time-invariant determinants of broadband outcomes at the state level, including permanent differences in geographic coverage, regulatory environment, and demographic composition. Identification therefore exploits *within-state* variation in outcomes over time, purged of level differences across states. Year fixed effects λ_t absorb common national shocks to broadband markets, including economy-wide shifts in input costs, federal regulatory changes, and aggregate demand trends. The interaction of treatment intensity with year indicators then captures whether states more intensively exposed to each program moved differentially relative to the national trend.

State-specific trends. For both ACP and BEAD, I include state-specific linear time trends $\gamma_i t$ because treatment intensity may correlate with pre-existing state-level broadband trajectories. For ACP, states with higher poverty rates mechanically have higher eligibility exposure. They may also have followed different broadband price and adoption paths before the program launched. For BEAD, states with more unserved locations tend to be lower-density and lower-income. They may therefore have experienced systematically different infrastructure investment trends before IIJA enactment. The digital divide literature has documented that broadband adoption and pricing trends differ systematically across income and demographic groups (Goldfarb and Prince, 2008; Espín and Rojas, 2024). Without state-specific trends, such pre-existing divergence would contaminate both sets of estimates. State-specific trends absorb linear state-level deviations from the national year effects. Non-linear departures from each state's pre-existing trend that coincide with program rollout then identify the treatment coefficients $\hat{\beta}_\tau^{\text{ACP}}$ and $\hat{\beta}_\tau^{\text{BEAD}}$.

Unlike standard staggered-adoption event studies, this design exposes all states to ACP and BEAD simultaneously. Identification comes from cross-state differences in treatment intensity, ACP eligibility shares and BEAD funding intensity, rather than from differences in treatment timing. Consequently, the negative-weighting and contamination problems that arise when already-treated units serve as controls for later-treated units (Sun and Abraham, 2021; Goldsmith-Pinkham et al., 2020) do not apply in this setting. The event-study coefficients therefore measure how outcomes evolve over time in states with higher exposure relative to states with lower exposure.

Assumption 1 (Conditional parallel trends)**.** *Conditional on state fixed effects, year fixed effects, and state-specific linear time trends, states with different treatment intensities D_i^P would have experienced the same outcome trends absent the policy. Formally,* $\mathbb{E}[\varepsilon_{it} \mid D_i^P, \alpha_i, \lambda_t, \gamma_i] = 0$ *for all* $\tau < 0$.

The identifying assumption is that states with higher treatment intensity D_i^P were not on systematically different trajectories for reasons unrelated to the policy. Pre-program demographic composition and federal allocation formulas predetermine both FPL200 and $\log \text{BEAD}$. Differential broadband trends before launch among states with higher exposure therefore pose the main threat. The pre-period coefficients $\hat{\beta}_\tau^P$ for $\tau < 0$ serve as direct falsification tests of Assumption 1.

A secondary concern is anticipation of program benefits: ISPs or households might alter behavior before formal program launch in expectation of the policy. Under the IIJA timeline, the NTIA announced BEAD in November 2021 but did not finalize allocations until 2023. Anticipatory responses would therefore appear as effects at $\tau = 1$–2 (2022–2023), which is largely consistent with what I find: early effects are near zero. For ACP, the predecessor Emergency Broadband Benefit program ran from February to December 2021. The $\tau = -1$ base period may therefore absorb some anticipatory ISP behavior, leading me to understate ACP's total effect. Neither concern appears to drive the main qualitative results.

3.3 ACP: Event Study Results

Figure 3 presents event study estimates from (3.1) for six key outcomes. I normalize event time to 2022 ($\tau = 0$); the dashed vertical line marks November 2021 (IIJA enactment). The omitted period is $\tau = -1$ (year 2021).

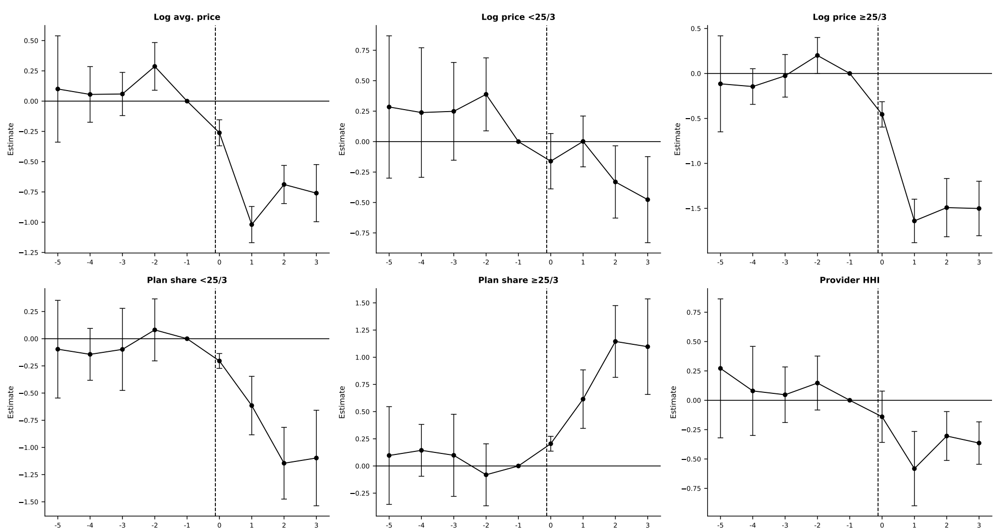

Figure 3: ACP event study estimates. Treatment intensity: share of households below 200% FPL (FPL200_i). TWFE with state-specific linear trends, weighted by number of plans, standard errors clustered by state. Event time $\tau = 0$ corresponds to 2022 (ACP launch); the dashed vertical line marks November 2022. The omitted period is $\tau = -1$ (2021). Bars indicate 95% confidence intervals with standard errors clustered at the state level.

Pre-trends. Pre-period coefficients are statistically indistinguishable from zero across most outcomes, as the parallel trends assumption requires. Conditional on state and year fixed effects and state-specific trends, the pre-launch estimates for high-speed prices, plan shares, and provider HHI are all close to zero. The one exception is the log price of plans below 25/3 Mbps, which runs somewhat high at $\tau = -5$ to $\tau = -3$, though with confidence intervals wide enough to make little of it.

Post-ACP effects. I find that ACP lowered prices and shifted providers toward faster-speed plans. After the 2022 launch, average broadband prices fall in higher-eligibility states. The log-average-price coefficient $\hat{\beta}_1^{\text{ACP}}$ is roughly -1.0 per unit of FPL200 share. With an interquartile range in state eligibility of about 0.10 (0.27 at P25, 0.37 at P75), a high-eligibility state saw prices fall about 10 percent more than a low-eligibility state at the peak ($\tau = 1$). The gap settles near 7 percent through $\tau = 2$–3. The effect is largest for high-speed plans: the response of -0.15 to -0.18 implies a 15–18 percent interquartile gap that persists through $\tau = 2$–3.

The product mix shifts in tandem: the low-speed plan share contracts and the high-speed share expands through $\tau = 2$–3, as expected when subsidized households raise demand for faster service and ISPs upgrade their menus to meet it. Provider HHI also declines in more exposed states, though imprecisely.

These estimates are consistent with equilibrium responses to demand expansion. Prices fall and menus tilt toward faster plans in states where ACP enlarges the eligible population, which is inconsistent with a simple mechanical increase in subscriptions at unchanged prices and menus. The reduced form cannot disentangle whether the price movement reflects pass-through, strategic repricing, or composition; the structural model below isolates the demand-expansion channel that drives it.

3.4 BEAD: Event Study Results

Figure 4 presents event study estimates for BEAD. I normalize event time to 2021 (IIJA enactment, $\tau = 0$); the omitted period is $\tau = -1$ (year 2020). The specification follows (3.1) with state and year fixed effects, state-specific linear time trends, plan-count weights, and state-clustered standard errors, as Section 3 describes.

Pre-trends. Pre-period coefficients are small and statistically indistinguishable from zero across all outcomes. States that ultimately received more BEAD funding per unserved location were on similar trajectories in prices, technology, and market structure before the program was announced.

Post-BEAD effects. BEAD's estimated effects are smaller and emerge only at the edge of the event window, consistent with the program's construction timeline. The window runs through $\tau = 4$ (2025), and meaningful effects require networks to reach customers before entrants can affect incumbent prices. Through the first two post-enactment years

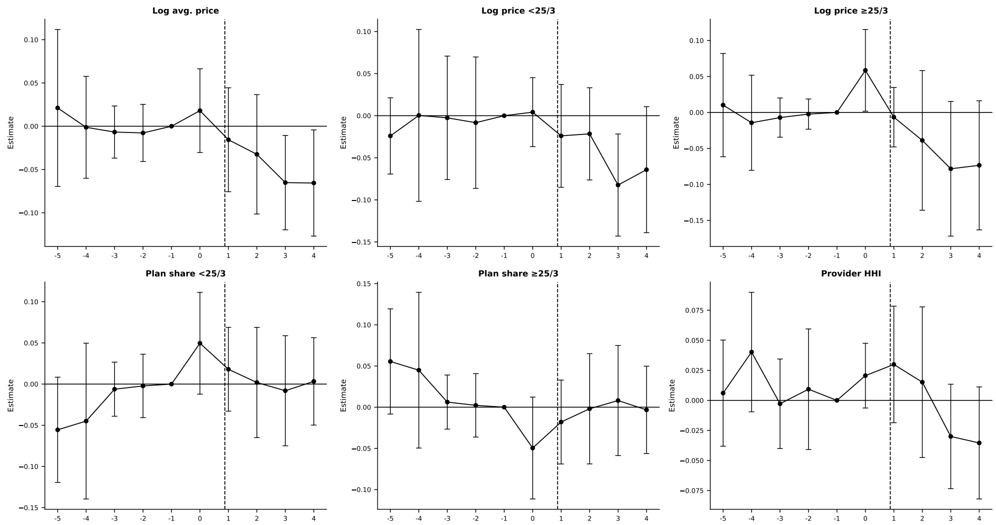

Figure 4: BEAD event study estimates. Treatment intensity: log BEAD allocation per unserved location ($\log \text{BEAD}_i$). TWFE with state-specific linear trends, weighted by number of plans, standard errors clustered by state. Event time $\tau = 0$ corresponds to 2021 (IIJA enactment); the dashed vertical line marks November 2022. The omitted period is $\tau = -1$ (2021). Bars indicate 95% confidence intervals with standard errors clustered at the state level.

($\tau = 1$–2, 2022–2023), every outcome is statistically indistinguishable from zero. Prices begin to move only at $\tau = 3$–4 (2024–2025), falling roughly 0.05–0.07 log points per log unit of BEAD intensity. Provider HHI turns negative over the same window. Plan composition does not respond throughout the window. The timing is informative. Effects appear where the construction timeline predicts, and not before, which points to entry as the operative channel rather than a near-null result. Equally important, the event window closes before the program does. With construction extending through 2026 and beyond, these estimates capture only the earliest funded deployments; no event window available captures the program's full effect. Quantifying it requires the structural counterfactuals of Section 6, which simulate the completed program against the pre-IIJA baseline; treat them as steady-state effects.

3.5 Summary Difference-in-Differences Estimates

Pooling the post-period into static two-way fixed-effects difference-in-differences estimates (Table F.1, Appendix F) summarizes the average effect and checks the event-study dynamics: genuine dynamic patterns should reappear as significant pooled coefficients of the same sign.

ACP shows significantly lower prices in higher-eligibility states (log average price -0.561, log high-speed price -0.922), a higher high-speed plan share ($+0.390$), and lower HHI (-0.306), the same price and product-mix adjustments the event study traces. BEAD's six estimates are all small and insignificant, consistent with effects that emerge only in the later years but that the urban URS frame attenuates. The flat plan-composition estimates are consistent with entry as the operative channel for BEAD rather than repositioning of incumbent product menus.

3.6 What the Reduced-Form Evidence Establishes

The event studies show that ACP is associated with lower prices and faster plan mixes, but BEAD effects emerge only after a two- to three-year lag. Several questions remain for the structural model. A falling price is consistent with direct pass-through, strategic repricing, or a shifting plan mix. The reduced form cannot distinguish among these mechanisms. State averages also wash out the tract-level variation in market structure, provider presence, and cost that underlies the response. No event study can measure the welfare gains from a policy design that never took effect. Nor can it decompose welfare into consumer and producer components or recover the entry and product-choice margins behind the observed price and concentration patterns. Standard evaluations that focus on new subscribers or cost per connection miss these equilibrium responses (Goolsbee and Guryan, 2006). The structural model quantifies them. The two analyses are complementary: the reduced form documents that the programs moved observable outcomes, and the structural model traces the channels and measures counterfactual welfare.

4 Structural Model

4.1 Model Setup and Timing

Markets and products. Markets are indexed by t and firms (or providers) by f. A *market* is a U.S. census tract in 2020, before the IIJA broadband programs launched.[12] A *product position* is a provider-speed-tier *triple* (f, g, t), where $g \in \{H, L\}$ denotes the IIJA high-speed tier and low-speed tier.

I distinguish three product sets. The observed set $\mathcal{J}_t^{obs}$ contains provider-tier products actually offered in tract t; these products enter the demand estimation and markup recovery. The potential set $\mathcal{J}_t^0$ contains provider-tier positions that are feasible in tract t. Feasible but unobserved positions are elements of $\mathcal{J}_t^0 \setminus \mathcal{J}_t^{obs}$. They do not enter the demand estimation as active products, but the model needs them for entry and fixed-cost identification. The

[12] I use census-tract data and pre-2021 FCC Form 477 deployment data to ensure the baseline predates IIJA programs. This timing is important for the counterfactuals: simulated price discounts and fixed-cost reductions are applied to markets that do not already reflect ACP or BEAD effects.

indicator Y_{fgt} equals one if provider f offers tier g in tract t and zero otherwise. The vector $Y_t = \{Y_{fgt}\}_{(f,g)\in\mathcal{J}_t^0}$ summarizes the product configuration. I also define Y_{ft}, the product portfolio that provider f offers any tier in tract t. Provider entry in this model is *product positioning*. A provider that already serves some tracts in a county expands its product line by activating a tier in an additional tract or offering a second speed tier in an existing tract. The model thus captures both geographic expansion and product-line decisions within the local market.

The *outside option* is no fixed-broadband subscription. Market size M_t is the number of households in the tract, and the market share of the outside option is $s_{0t} = 1 \sum_{j\in\mathcal{J}_t^{obs}} s_{jt}$.

Timing and information. The following two assumptions formalize the timing and the information structure.[13]

Assumption 2 (Timing)**.** *The game proceeds in two stages:*

1. *Product-positioning stage. Firms observe demand shifters X_t, cost shifters W_t, rivals' observed footprints, and the fixed-cost shocks ζ_{fgt} for all firms. Based on this information, each firm simultaneously chooses its portfolio $Y_{ft} \in \mathcal{Y}_{ft}$.*
2. *Pricing stage. Conditional on the realized product configuration Y_t, firms observe demand shocks ξ_{jt} and marginal-cost shocks ω_{jt} and simultaneously set monthly plan prices in a complete-information Bertrand-Nash game.*

[13] Several studies use this timing assumption. Examples include Fan and Yang (2024) in the brewery industry, Eizenberg (2014) in the personal computer industry, and Hidalgo (2024) in the broadband industry.

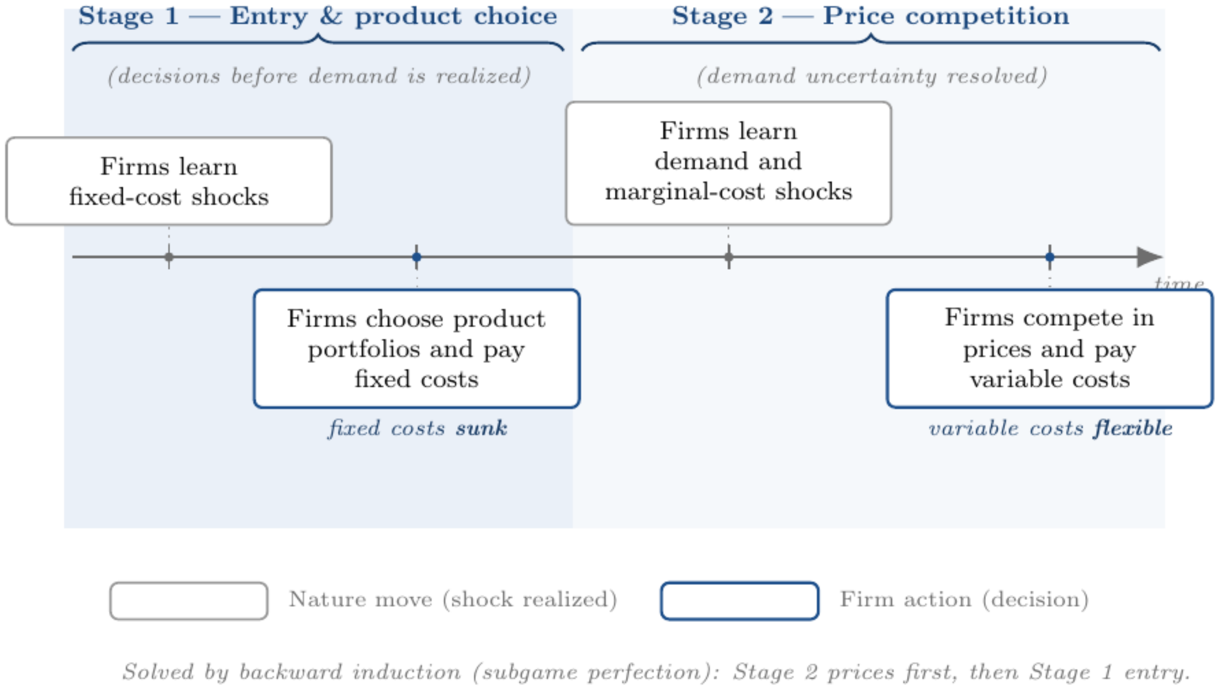

Figure 5: Timing of the structural entry and pricing game.

Assumption 3 (Information). *1. Firms. At the product-positioning stage, fixed-cost shocks ζ_{fgt}, market characteristics (X_t, W_t), and rivals' potential market footprints are all common knowledge. Firms do not yet know the demand and cost shocks (ξ, ω) that will be realized in the second-stage pricing game; only their joint distribution is common knowledge. Firms therefore compute expected variable profits by integrating over (ξ, ω).*

2. Econometrician. The shocks (ξ, ω, ζ) are unobserved. The econometrician observes market characteristics (X_t, W_t) and the product choices Y_t.

Splitting portfolio choice from pricing is standard in this literature (Mazzeo, 2002; Seim, 2006; Fan, 2013; Eizenberg, 2014) and reflects a difference in horizons. An ISP commits to building infrastructure months or years before it posts a price list. Complete information on fixed costs is a simplifying assumption, but a plausible one in this setting. Deployment costs depend on local permitting, right-of-way negotiations, and contractor availability, all largely observable within local infrastructure markets. ISPs publicly advertise prices and rivals monitor them; a complete-information pricing stage is therefore also a natural approximation.

Firm f's first-stage objective is

$$\max_{Y_{ft} \in \mathcal{Y}_{ft}} \mathbb{E}_{\xi,\omega}\left[r_t^f\left(Y_{ft}, Y_{-ft}\right) \mid I_{ft}\right] - \sum_{g \in \{H,L\}} Y_{fgt} FC_{fgt}, \tag{4.1}$$

where $Y_{ft} = (Y_{fHt}, Y_{fLt})$, Y_{-ft} denotes rivals' portfolios, and $r_t^f(\cdot)$ is variable profit from the pricing game. Because a provider can offer at most two tiers in a tract, $\mathcal{Y}_{ft}$ contains four portfolios: no product, high-speed only, low-speed only, and both tiers.

The obstacle to estimation is dimensionality. With about ten feasible provider-tier positions per tract, the configuration space Y_t contains hundreds to thousands of portfolios. Solving the entry game inside an estimation loop across 70,854 markets is infeasible. The scalable discrete-game method of Fan and Yang (2024) bypasses this problem. The estimation method avoids solving the game and delivers bounds for the incremental profit from each position. These bounds imply moment inequalities for the fixed-cost parameters. Solving the game is only required when evaluating counterfactual policies.

4.2 Consumer Demand

Household h's indirect utility from product j in market t is

$$U_{hjt} = \delta_{jt} + \nu_{hg} + (1-\rho)\varepsilon_{hjt}, \tag{4.2}$$

where δ_{jt} is mean utility, ν_{hg} is a group-level taste shock, and ε_{hjt} is Type-I extreme value.[14] Mean utility is

$$\delta_{jt} = X'_{jt}\beta + \alpha p_{jt} + \xi_{jt}. \tag{4.3}$$

The vector X_{jt} includes download speed, upload speed, their squares, and a high-speed tier indicator. The parameter α is the marginal utility of price and ξ_{jt} is unobserved quality.

All fixed-broadband products sit in a single inside nest and compete against the outside option of no subscription. The single nest is a deliberate tractability choice, defensible on two grounds. First, it applies uniformly across all 70,854 tracts. A speed-tier nest would be degenerate wherever providers offer only one tier: the within-nest share equals one and the structure collapses to logit. Many rural tracts carry only a low-speed tier; a tier nest would therefore require dropping or special-casing those markets. Second, the single nest yields a unique closed-form Bertrand-Nash price equilibrium. The entry counterfactuals require this tractability because the positioning game requires recomputing equilibrium prices at every iteration across tens of thousands of markets and fixed-cost draws. Richer structures, such as

[14] This paper departs from Fan and Yang (2024), who employ a random-coefficients demand model. Although such models allow for richer substitution patterns, they may generate non-unique pricing equilibria in multiproduct settings (Nocke and Schutz, 2018), thus complicating welfare and entry counterfactuals. Therefore, I use the nested logit model, which delivers a unique closed-form pricing solution and ensures a stable mapping from demand primitives to prices (e.g., Berry (1994)). Tractability is essential for evaluating firms' endogenous product offerings and entry decisions. I nonetheless rely on Fan and Yang (2024)'s framework to recover fixed-cost parameters in the entry stage.

random coefficients or a second nest, can admit multiple pricing equilibria in multiproduct markets (Nocke and Schutz, 2018).

The Berry inversion (Berry, 1994) gives

$$\log(s_{jt}) - \log(s_{0t}) = X'_{jt}\beta + \alpha p_{jt} + \rho \log(s_{j|g,t}) + \xi_{jt}, \tag{4.4}$$

where $s_{j|g,t} = s_{jt}/(1 - s_{0t})$. In the tract-level data, the left-hand side is the mean utility recovered from product-level implied shares.

Price and the within-inside share are both endogenous. I instrument for both using the quadratic differentiation instruments of Gandhi and Houde (2026): for characteristic $k \in \{\text{download speed, upload speed}\}$,

$$z^{\text{within}}_{jt,k} = \sum_{l \neq j,\ \text{same tier}} (x_{lt,k} - x_{jt,k})^2, \qquad z^{\text{cross}}_{jt,k} = \sum_{l \neq j,\, \text{other tier}} (x_{lt,k} - x_{jt,k})^2, \tag{4.5}$$

yielding four excluded instruments that measure how differentiated product j is from its rivals in speed-characteristic space. The within-tier and cross-tier sums provide distinct identifying variation for the two endogenous variables: same-tier differentiation shifts the within-nest share, while cross-tier differentiation shifts the overall inside share, so the two instrument groups are not collinear in their effects on $(p_{jt}, \ln s_{j|in,t})$. Both endogenous variables are projected onto the full instrument set in the first stage. Although Gandhi and Houde (2026) develop these instruments primarily for random-coefficients demand, the within/cross construction maps directly onto the nested structure here. The preferred specification absorbs county and provider fixed effects; identification of $\hat{\alpha}$ and $\hat{\rho}$ comes from within-county, within-provider variation in relative prices and shares across tracts.

4.3 Oligopoly Price Competition

In the pricing stage, all product configurations Y_t are realized and common knowledge. Firm f's variable profit is

$$r^f_t = \sum_{j \in \mathcal{J}^f_t} (p_{jt} - mc_{jt})\, M_t\, s_{jt}(p_t). \tag{4.6}$$

Differentiating with respect to p_{jt} and setting to zero gives the Bertrand-Nash first-order condition:

$$s_{jt} + \sum_{k \in \mathcal{J}^f_t} (p_{kt} - mc_{kt}) \frac{\partial s_{kt}}{\partial p_{jt}} = 0. \tag{4.7}$$

Stacking over all products yields the linear system:

$$s_t + (H_t \odot \Omega_t)(p_t - mc_t) = 0, \tag{4.8}$$

where H_t is the $J_t \times J_t$ matrix of demand derivatives with $(H_t)_{jk} = \partial s_{kt}/\partial p_{jt}$, and Ω_t is the ownership matrix with $(\Omega_t)_{jk} = 1$ if the same firm offers both j and k. I solve the system for the markup vector:

$$\hat{\mu}_t^{BN} = -(H_t \odot \Omega_t)^{-1} s_t, \tag{4.9}$$

then recover marginal costs as $\hat{mc}_{jt} = p_{jt} - \hat{\mu}_{jt}^{BN}$. I then estimate the cost regression

$$\log(\hat{mc}_{jt}) = X'_{jt}\gamma + \omega_{jt}. \tag{4.10}$$

4.4 Product Positioning and Fixed Costs

A product position is a provider × speed-tier pair in a census tract, for provider f, speed tier $g \in \{H, L\}$, and census tract t:

$$Y_{fgt} = \mathbf{1}\{\text{provider } f \text{ offers speed tier } g \text{ in tract } t\}. \tag{4.11}$$

This definition treats entry as product positioning rather than entry into the national broadband industry. It captures both geographic expansion into a tract and product-line expansion within a tract. For example, a provider that already offers the low-speed tier in tract t enters the high-speed position if it also begins offering the high-speed tier; it need not offer the low tier first, since the two are independent entry decisions.

Potential product positions. I organize the observed data at the tract × canonical-provider × speed-tier level; each observed row therefore corresponds to $Y_{fgt} = 1$. To estimate fixed costs, I also need feasible positions that are not chosen. I define the potential set using local provider footprints rather than assuming every national provider can enter every tract. Let $c(t)$ denote the county containing tract t. The baseline potential set is

$$\mathcal{J}_t^0 = \left\{(f, g) : \exists\, t' \in c(t) \text{ such that } Y_{fgt'} = 1\right\}. \tag{4.12}$$

Thus, provider-tier position (f, g) is feasible in tract t if that same provider-tier pair appears somewhere else in the county. This restriction is economically natural: broadband expansion depends on nearby network presence, local permitting, and rights of way. It also keeps the potential choice set local enough to avoid treating providers with no plausible footprint as potential entrants.

Feasible but unobserved positions in $\mathcal{J}_t^0$ have no observed price or characteristics, since no provider offers them. To evaluate the profit a provider would earn from entering such a position, I impute its characteristics using a fallback hierarchy: I use the provider's own average characteristics for that tier in that state if available, the provider's national average for that tier if not, and the overall tier average as a last resort. Observed positions keep their actual observed characteristics; only unobserved positions are imputed.

Table 6 reports the implied dimensions. The average tract contains 3.20 observed products but 10.24 potential provider-tier positions. A full market-level portfolio model would require evaluating $2^{|\mathcal{J}_t^0|}$ configurations per market, motivating the Fan and Yang (2024) approach below.

Table 6: Dimension of Observed and Potential Product Sets

Product-set definition	Mean	P25	Median	P75	P90
Observed products in tract, $\mathcal{J}_t^{obs}$	3.20	2	3	4	5
Observed providers in tract	2.32	1	2	3	4
County-footprint positions, $\lvert\mathcal{J}_t^0\rvert$	10.24	6	9	13	19
County providers	8.05	4	7	10	16

Notes: The unit of observation is a census tract. Observed products are provider-tier products offered in the tract. The county-footprint potential set includes provider-tier positions observed somewhere in the county where the tract belongs.

I use the potential product set exclusively for the entry model and fixed-cost identification. Demand estimation and markup recovery use only the observed product set.

4.5 Multiple Equilibria and Equilibrium Selection

Product positioning generically has multiple Nash equilibria: a single fixed-cost draw can support several mutually best-response configurations. Researchers have recognized this multiplicity since Bresnahan and Reiss (1991) and Tamer (2003); it is central to Ciliberto et al. (2021) and Fan and Yang (2024). It normally obstructs estimation because it severs the map from parameters to a unique prediction.

It does not obstruct estimation here. The moment-inequality approach asks only that the observed configuration be *not dominated*, a condition every Nash equilibrium satisfies; the fixed-cost bounds therefore remain valid whatever selection rule operates in the data. This is the decisive advantage of the Fan and Yang (2024) method over likelihood approaches that must assume a unique equilibrium.

The counterfactual stage does require solving for post-policy equilibria. I handle this as follows. For each market and each draw of fixed costs, I run iterated best-response from three starting points: the observed pre-policy configuration, the empty configuration, and the full feasible configuration.[15] I declare the algorithm converged when no firm can profitably deviate from the current portfolio, verified by a firm-by-firm Nash check. If the algorithm finds multiple fixed points from different starting configurations, I retain all of them. The reported national-level bounds combine equilibrium multiplicity and fixed-cost parameter uncertainty. They do so by stacking market-level lower and upper outcomes across equilibria and fixed-cost draws and reporting the 2.5th and 97.5th percentile bounds.

Markets where the algorithm finds no fixed point within the iteration limit (fewer than one percent of markets) receive the pre-policy configuration. This conservative fallback ensures the bounds remain valid. It is likely to understate the response of hard-to-solve markets. Appendix I (Figure I.2) reports extensive convergence diagnostics. They show that the algorithm converges in the vast majority of markets and that convergence failures are not systematically correlated with market size or policy intensity.

Appendix G discusses the main measurement assumptions underlying the structural analysis (inferred product-level market shares, imputed URS prices, and a static 2020 baseline), along with supporting robustness checks.

4.6 Identification

Identification proceeds in three steps, but only two require a separate argument. Demand parameters come from the Berry inversion. Fixed costs come from the discrete product-positioning game. Marginal costs sit in between: once demand is identified, the Bertrand–Nash first-order conditions in (4.7) pin them down mechanically, with no additional identifying argument of their own. The substantive identification challenges are demand and fixed costs.

Demand. The linearized Berry inversion in (4.4) contains two endogenous regressors. Price p_{jt} correlates with unobserved quality ξ_{jt}. The within-nest log share $\ln s_{j|in,t}$ correlates with ξ_{jt} mechanically and biases $\hat{\rho}$ toward one (Berry, 1994). I instrument both using the Gandhi and Houde (2026) quadratic differentiation instruments defined in (4.5). I absorb county and provider fixed effects prior to estimation; identification therefore comes from within-county, within-provider variation in relative prices and shares across tracts, using the local competitive configuration of rival speed characteristics as instruments. The exclusion restriction is that pre-existing infrastructure footprints, uncorrelated with the product-level demand residual

[15] Fan and Yang (2024) consider two starting points: the empty and the full feasible configuration.

after these absorptions, determine rivals' speed configurations.

Fixed costs: the Fan–Yang moment inequalities. Fixed costs are latent; I identify them from revealed product-positioning decisions. For each potential position $(f, g, t) \in \mathcal{J}_t^0$, define the lower and upper expected variable-profit bounds:

$$\Delta_{fgt}^{\text{lower}} = \mathbb{E}\Big[r_t^f(Y_{fgt} = 1,\, Y_{-fgt} = \mathbf{1}) - r_t^f(Y_{fgt} = 0,\, Y_{-fgt} = \mathbf{1})\Big], \tag{4.13}$$

$$\Delta_{fgt}^{\text{upper}} = \mathbb{E}\Big[r_t^f(Y_{fgt} = 1,\, Y_{-fgt} = \mathbf{0})\Big], \tag{4.14}$$

where $r_t^f(\cdot)$ denotes variable profit in the pricing game, and Y_{-fgt} is the vector of rivals' offer decisions at position g in tract t. The lower bound sets all rivals active ($Y_{-fgt} = \mathbf{1}$, maximum competition); the upper bound sets no rivals active ($Y_{-fgt} = \mathbf{0}$, monopoly). I approximate both expectations by $B = 100$ paired draws of demand and cost shocks $(\hat{\xi}, \hat{\omega})$ from the estimated residual distributions. The bound Δ^{lower} is the expected incremental variable profit under maximum competition (all feasible rivals active); Δ^{upper} is the expected variable profit under monopoly.

Assumption 4. *For each potential position (f, g, t), the observed decision Y_{fgt} is not a dominated strategy.*

Nash optimality of the observed configuration along with Assumption 4 imply one-sided restrictions on fixed costs. Under the parameterization $FC_{fgt} := W_{fgt}\theta + \sigma_{b(t)}\zeta_{fgt} = \mu_{fgt}^C(\theta) + \sigma_{b(t)}\zeta_{fgt}$ with $\zeta_{fgt} \sim N(0,1)$ and $\mu_{fgt}^C(\theta) = \theta_{b(t)} + \theta_H \mathbf{1}\{g = H\}$

$$\Phi\left(\frac{\Delta_{fgt}^{\text{lower}} - \mu_{fgt}^C(\theta)}{\sigma_{b(t)}}\right) \leq \Pr(Y_{fgt} = 1) \leq \Phi\left(\frac{\Delta_{fgt}^{\text{upper}} - \mu_{fgt}^C(\theta)}{\sigma_{b(t)}}\right), \tag{4.15}$$

where $b(t) \in \{Small, Medium, Large\}$ based on tertile of market size M_t. This holds for every Nash equilibrium; estimation therefore requires no equilibrium-selection assumption (Fan and Yang, 2024; Tamer, 2003; Ciliberto and Tamer, 2009). Taking expectations of (4.15) against 33 nonnegative weight functions $h^{(k)}(X_{fgt}, W_{fgt})$, $k = 1, \ldots, 33$, yields $K = 66$ unconditional moment inequalities. I build the weight functions hierarchically, from coarse cells (a constant aggregating over all positions, plus speed-tier and market-size indicators) to fine cells that interact market size with the location of the expected-profit bounds; Appendix J.1 gives the full construction. I average moments first within tracts, to respect strategic dependence within a market, and then across tracts, which are independent markets. The moment inequalities discipline the seven structural parameters $(\theta_H, \theta_{\text{Small}}, \theta_{\text{Medium}}, \theta_{\text{Large}}, \sigma_{\text{Small}}, \sigma_{\text{Medium}}, \sigma_{\text{Large}})$

jointly. The central tendency of product-offering rates across markets within each bin identifies θ_b. Comparing entry into high- versus low-speed positions within the same bin identifies θ_H. Cross-market variation in offering rates within each bin identifies σ_b.

Heuristic identification of fixed-cost shocks. Write the fixed cost for product speed $s \in \{L, H\}$ in market-size bin $b \in \{\text{Small}, \text{Medium}, \text{Large}\}$ as

$$F_{b,s} = \theta_b + \theta_H \mathbf{1}\{s = H\} + \sigma_b \, \varepsilon,$$

where ε is a market-level shock common to all firms in a given market. The mean fixed costs therefore take the following form

	Low speed	High speed
Small	θ_{Small}	$\theta_{\text{Small}} + \theta_H$
Medium	θ_{Medium}	$\theta_{\text{Medium}} + \theta_H$
Large	θ_{Large}	$\theta_{\text{Large}} + \theta_H$

I identify the parameters as follows. The central tendency of product-offering rates across markets within each bin identifies the market-size intercepts θ_b: bins where entry is systematically more prevalent imply lower mean fixed costs. Comparing entry into high- versus low-speed products within the same bin identifies the high-speed premium θ_H, where both speed tiers share the common intercept θ_b, so any systematic difference in offering rates reflects θ_H alone. Cross-market variation in offering rates within each bin identifies the dispersion parameters σ_b: bins where rates are nearly uniform imply low σ_b, while bins where some markets see broad entry and others see little imply high σ_b. Because ε is a market-level shock, within-market variation across firms contributes no information about σ_b; identification relies entirely on the across-market distribution of entry outcomes within each bin. The procedure estimates all parameters jointly: I retain a candidate vector $\vartheta = (\theta_{\text{Small}}, \theta_{\text{Medium}}, \theta_{\text{Large}}, \theta_H, \sigma_{\text{Small}}, \sigma_{\text{Medium}}, \sigma_{\text{Large}})$ only if it satisfies all moment inequalities simultaneously.

Andrews–Soares test inversion and confidence set. I estimate the identified set by inverting the modified method-of-moments test of Andrews and Soares (2010). Let

$$\bar{Z}_M(\vartheta) = M^{-1} \sum_{m=1}^{M} Z_m(\vartheta)$$

be the sample moment vector at candidate ϑ, where $M = 70{,}854$ tract markets. The test statistic penalizes only violated inequalities:

$$T_M(\vartheta) = \sum_{k=1}^{K} \left[\frac{\sqrt{M}\, \bar{Z}_{M,k}(\vartheta)}{\hat{\sigma}_k(\vartheta)} \right]_+^2, \qquad [x]_+ \equiv \max\{x, 0\}, \tag{4.16}$$

where $\hat{\sigma}_k(\vartheta)$ is the empirical standard deviation of the k-th moment across markets. The 95 % confidence set is the collection of parameter vectors that pass the test:

$$\hat{\mathcal{C}} = \Big\{ \vartheta : T_M(\vartheta) \leq \hat{c}_M(\vartheta, 0.95) \Big\}, \tag{4.17}$$

where $\hat{c}_M(\vartheta, 0.95)$ is the generalized moment selection (GMS) critical value of Andrews and Soares (2010). The GMS critical value adapts to the local geometry of the inequalities at each ϑ. Moment conditions that are comfortably satisfied receive conservative slack. This inflates $\hat{c}_M$ and makes the test easy to pass in the interior of the identified set. Binding or nearly binding inequalities instead receive tight critical values.

Formally, let

$$\bar{Z}^*_M(\vartheta) = \frac{1}{M} \sum_{m=1}^{M} Z^*_m(\vartheta)$$

denote the bootstrap sample mean of the moment vector, where $\{Z^*_m(\vartheta)\}_{m=1}^M$ is obtained by resampling the M tract markets with replacement. I estimate the critical value $\hat{c}_M(\vartheta, 0.95)$ from 999 simulated draws of the bootstrap empirical process

$$\sqrt{M}\Big(\bar{Z}^*_M(\vartheta) - \bar{Z}_M(\vartheta) \Big).$$

For each draw, I apply the Andrews–Soares GMS procedure to determine which inequalities are binding or nearly binding and compute the corresponding bootstrap test statistic. I take the 95th percentile of the resulting distribution of bootstrap test statistics as $\hat{c}_M(\vartheta, 0.95)$. This guarantees asymptotic validity, $\Pr(\vartheta_0 \in \hat{\mathcal{C}}) \geq 1 - \alpha$ for the true parameter value ϑ_0, while remaining sharper than a conservative approach that applies a common chi-squared critical value to all 66 inequalities simultaneously. The procedure below mimics the computational simplification used in Fan and Yang (2024).

Computational implementation. Direct evaluation of the confidence set $\hat{\mathcal{C}}$ on a dense grid over the seven-dimensional parameter space is computationally infeasible. I therefore approximate $\hat{\mathcal{C}}$ using a stochastic search algorithm. First, I obtain a reference point $\hat{\vartheta}$ by minimizing the criterion $Q_M(\vartheta) = T_M(\vartheta)$. I use the L-BFGS-B algorithm from seven starting values spanning the admissible parameter space, subject to the constraint $\sigma_b \geq 0$. Starting from $\hat{\vartheta}$, I run 50 independent local random-walk chains of 50 iterations each.[16] At each iteration, I generate a candidate parameter vector by adding a random perturbation to the current point. I retain the proposal whenever it satisfies the confidence-set membership condition $T_M(\vartheta) \leq \hat{c}_M(\vartheta, 0.95)$. This local exploration identifies a connected region of accepted parameter vectors around $\hat{\vartheta}$. To explore more distant portions of the confidence set, I next initiate 100 additional chains of 50 iterations from accepted points near the boundary of the locally explored region. These chains use perturbation scales five times larger than those employed in the local search. The larger steps allow the algorithm to trace the outer boundary of $\hat{\mathcal{C}}$ and discover disconnected or weakly connected regions. Across both stages, the search yields 656 accepted parameter vectors. For counterfactual simulations, I select 50 representative parameter vectors from the accepted set. Specifically, I project the 656 accepted vectors onto their first principal component and choose 50 approximately evenly spaced points along this direction. This procedure ensures that the simulation draws span the full extent of the estimated confidence region rather than clustering near its center. Reported ($[Q_{2.5}, Q_{97.5}]$) intervals across these 50 draws therefore incorporate both fixed-cost parameter uncertainty and equilibrium multiplicity uncertainty. Figure I.1 in Appendix I visualizes the procedure.

Universal dominance and its implications for identification. All 727,371 positions with valid profit bounds (of the 727,601 potential positions in the entry sample) satisfy $\Delta^{\text{lower}}_{fgt} > 0$. Every potential product earns positive incremental variable profit even under maximum competition; no position is dominated ($\Delta^{\text{upper}} < 0$). Economically, this reflects the high-margin structure of broadband in variable terms. For identification, universal dominance has a precise consequence. Since dominated-action logic never fires, the inequalities in (4.15) identify fixed costs entirely from the upper-bound side. Observed products reveal $FC_{fgt} \leq \Delta^{\text{upper}}_{fgt}$; non-offered products reveal $FC_{fgt} > \Delta^{\text{lower}}_{fgt} > 0$.

5 Estimation Results

Estimation proceeds in the order the model is identified. Demand comes first, from the Berry inversion in (4.4), instrumenting price and the within-inside share with Gandhi and Houde (2026) quadratic differentiation instruments (4.5) and absorbing county and provider fixed effects. The demand estimates and the Bertrand–Nash conditions in (4.7) then recover tract-level markups and marginal costs through (4.9). Fixed costs come last, as bounds from the Fan

[16] I draw each proposal from a multivariate normal distribution centered at the current point with diagonal covariance, perturbing all components independently.

and Yang (2024) positioning inequalities, requiring only that observed choices be undominated. The identification argument underlying each step is in Section 4.6; Appendix I details the fixed-cost moments and weight matrix, and Appendix I.2 gives the formal construction of the Andrews–Soares confidence set.

5.1 Consumer Demand

Table 7 reports four specifications. Columns (1) and (2) are OLS; columns (3) and (4) instrument for price and the within-inside share with Gandhi and Houde (2026) quadratic differentiation instruments. Even-numbered columns absorb county and provider fixed effects. Column (4) is the preferred specification.

The four columns illustrate that neither correction alone is sufficient. Column (1) yields $\hat{\rho} = 1.183$, above the admissible upper bound, because unobserved quality ξ_{jt} is positively correlated with the within-nest share. Adding county and provider fixed effects in column (2) reduces $\hat{\rho}$ only marginally, to 1.029. Absorbing fixed effects alone therefore does not resolve the quality-price correlation. Column (3) instruments for price and the within-share without absorbing fixed effects. The price coefficient reverses sign to +3.261 because the Gandhi and Houde (2026) instruments generate variation correlated with uncontrolled unobserved quality, producing an upward-biased estimate. Column (4), which instruments and absorbs fixed effects simultaneously, is much more sensible: $\hat{\alpha} = -1.491^{***}$, $\hat{\rho} = 0.863^{***}$. Each correction in isolation addresses one source of bias while leaving the other unresolved; both are necessary.

The nesting parameter $\hat{\rho} = 0.863$ indicates strong within-broadband substitution. Households treat fixed-broadband products as far closer substitutes for one another than for the outside good of no subscription. This is economically sensible: consumers who want internet service tend to compare plans rather than choose between internet and no internet. It is also essential for the counterfactual welfare calculations, which require demand responses to entry and price changes to concentrate within the broadband market. Download speed enters positively and concavely, consistent with diminishing marginal utility from very high speeds. Upload speed is small and imprecise once the specification absorbs county and provider effects.

The mean own-price elasticity[17] in the preferred specification is -4.99:

Comparison with related estimates. The mean own-price elasticity of -4.99 falls within the range of recent product-level broadband and pay-TV estimates: -5.9 in Goetz

[17]

$$\varepsilon_{jj} = \frac{\hat{\alpha}}{100}\, p_{jt} \left[\frac{1}{1-\hat{\rho}} - \frac{\hat{\rho}}{1-\hat{\rho}}\, s_{j|\text{in},t} - s_{jt} \right].$$

Table 7: Nested Logit Demand Estimates

	OLS		2SLS (GH instruments)	
	(1)	(2)	(3)	(4)
Price / \$100 [$\hat{\alpha}$]	-0.344^{***}	-0.127^{***}	$+3.261^{***}$	-1.491^{***}
	(0.016)	(0.023)	(0.182)	(0.471)
Within-share [$\hat{\rho}$]	1.183^{***}	1.029^{***}	1.242^{***}	0.863^{***}
	(0.004)	(0.005)	(0.043)	(0.033)
Download speed (Gbps)	0.366^{***}	0.015	-0.243^{***}	0.223^{***}
	(0.009)	(0.017)	(0.053)	(0.043)
Download speed^2	-0.052^{***}	-0.017^{***}	0.013	-0.041^{***}
	(0.005)	(0.006)	(0.019)	(0.008)
Upload speed (Gbps)	-0.012	0.087^{***}	0.679^{***}	-0.003
	(0.008)	(0.015)	(0.043)	(0.024)
Upload speed^2	0.018^{***}	0.009	-0.053^{***}	0.024^{***}
	(0.005)	(0.006)	(0.019)	(0.008)
Tier H indicator	0.027^{***}	absorbed	-0.622^{***}	absorbed
	(0.009)		(0.084)	
N	170,419			
R^2	0.452	0.256	—	—
County FEs	No	Yes	No	Yes
Provider FEs	No	Yes	No	Yes
Mean own-price elasticity	$+0.83$	$+1.96$	-5.91	-4.99

Notes: Dependent variable: $\delta_{jt} = \ln(s_{jt}/s_{0t})$. HC1 robust standard errors in parentheses for OLS; heteroskedasticity-robust for 2SLS. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. GH instruments: four Gandhi and Houde (2026) quadratic differentiation instruments on download and upload speed, computed within- and across-tier for each product (see (4.5)). I absorb county and provider fixed effects by iterative two-way demeaning before estimation. Column (4) is the preferred specification and the only one satisfying the standard regularity condition $\hat{\rho} \in [0, 1)$ (Berry, 1994). Positive mean elasticities in columns (1)–(2) reflect standard condition violations ($\hat{\rho} > 1$); the positive price coefficient in column (3) reflects quality bias without fixed effects.

(2019) and -4.1 to -5.4 in cable television (Crawford et al., 2019). Kearns (2026) reports smaller elasticities of -1.48 to -3.30 across specifications (preferred: -1.71) in Seattle; the difference likely reflects Seattle's served-market setting and a demand specification that allows income heterogeneity across households. The magnitude of -4.99 reflects the product-substitution margin within a local market rather than the extensive-margin adoption decision. Participation-margin estimates in the broadband literature (-1 to -4, Rosston et al. 2010;

Nevo et al. 2016) are naturally smaller. An alternative state-level aggregation yields a mean elasticity of -5.87, confirming that the tract-level share construction does not mechanically drive the result.

Elasticity heterogeneity across market types. Own-price elasticities vary with market shares via (17), and this heterogeneity matters for the welfare comparison between programs. At a product share of $s_{jt} = 0.10$ with $\hat{\rho} = 0.863$, the formula yields an elasticity of approximately -2.1. At $s_{jt} = 0.45$, it yields approximately -8.6. Low-share markets, such as rural tracts with few products and high outside-option shares, are relatively inelastic. High-share markets, such as dense urban tracts, are highly elastic. ACP delivers the largest consumer surplus gains where elasticity is highest, because more households are at the margin of subscribing and will respond to a price discount. BEAD acts through fixed-cost relief. Its gains are therefore less sensitive to demand elasticity and depend instead on the gap between variable profits and fixed costs in currently unserved tracts. This difference reinforces the geographic complementarity between the two programs documented in Section 6.3.

Instrument strength. Table 8 reports the Gandhi and Houde (2026) excluded-instrument coefficients and weak-identification diagnostics for column (4). The signs are economically transparent. Greater within-tier characteristic dispersion raises a product's own price because slack competitive pressure permits a premium. Greater cross-tier dispersion lowers the within-nest share because more distinct products capture fewer within-broadband switchers. The conditional Sanderson–Windmeijer F-statistics are 58.3 for price and 544.2 for the within-share, both well above the Stock–Yogo 10% critical value of 7.56 (2 endog., 4 instruments). The Cragg–Donald minimum eigenvalue of 986.2 confirms system-level identification.[18]

5.2 Marginal Costs and Markups

The preferred supply specification (column 1 of Table 9) excludes fixed effects because the identifying variation in broadband costs runs across providers, technologies, and speed tiers. Rich fixed effects absorb precisely those dimensions. This suggests that the fixed effects over-absorb the cross-sectional cost variation. The no-FE specification preserves economically interpretable estimates.

[18] The Shea partial R^2 for the *price* equation is 0.0054: after removing the variation already captured by the within-nest-share instruments, the Gandhi–Houde measures explain a small share of residual price variation. Appendix H reports an alternative specification using CMT-style Hausman instruments (Ciliberto et al., 2021), which achieves Shea $R^2 = 0.393$ for price but yields $\hat{\alpha} = -0.123 \approx \hat{\alpha}_{\text{OLS-FE}}$ and near-universal negative marginal costs (99.9% of products). The convergence to OLS indicates that the Hausman instrument is endogenous conditional on two-way fixed effects, likely because national providers carry correlated brand quality across counties. The Gandhi–Houde specification is preferred on supply-side plausibility grounds.

Table 8: First-Stage Diagnostics: Preferred 2SLS Specification (column 4)

	Price / \$100	$\ln s_{j\|\text{in},t}$
Gandhi-Houde excluded instruments		
Within-tier $\sum(x_d - x_{d'})^2$ (download)	$+0.0027^{***}$	-0.0010
	(0.0003)	(0.0022)
Cross-tier $\sum(x_d - x_{d'})^2$ (download)	-0.0073^{***}	-0.0532^{***}
	(0.0015)	(0.0128)
Within-tier $\sum(x_u - x_{u'})^2$ (upload)	-0.0024^{***}	-0.0045^{**}
	(0.0003)	(0.0020)
Cross-tier $\sum(x_u - x_{u'})^2$ (upload)	$+0.0070^{***}$	$+0.0508^{***}$
	(0.0015)	(0.0128)
Exogenous controls (speed, speed^2)	Yes	Yes
County fixed effects	Yes	Yes
Provider fixed effects	Yes	Yes
First-stage R^2	0.031	0.107
Angrist–Pischke Wald stat. ($\chi^2(4)$)	166.8	2,174.2
Sanderson–Windmeijer conditional F	58.3	544.2
Shea partial R^2	0.0054	0.0233
Cragg–Donald minimum eigenvalue stat.	986.2	

Notes: x_d and x_u denote download and upload speed (Gbps). Within-tier sums are over same-market same-tier rivals; cross-tier sums are over same-market other-tier rivals. Exogenous controls include download speed, download^2, upload speed, upload^2; I absorb county and provider fixed effects by iterative two-way demeaning. The Sanderson–Windmeijer conditional F tests one endogenous variable conditional on the other. Stock–Yogo (2002) 10% critical value ≈ 7.56 (2 endog., 4 instruments). $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. HC1 robust standard errors.

The preferred specification implies that download speed raises marginal cost at a diminishing rate. The Tier H premium of $\exp(0.529) - 1 \approx 70$ percent reflects the substantially higher infrastructure investment required for high-speed broadband relative to legacy low-speed service. Higher upload speeds correspond to lower marginal costs, consistent with fiber-grade networks combining high upload capacity with newer, more cost-efficient physical plant. The constant $\exp(-1.115) \approx \$33$ per month represents the baseline variable cost at zero measured speeds. High-speed products average approximately \$61.5 per month in marginal costs once I apply the Tier H premium. Of the 170,419 products in the estimation sample, 16,387 (9.62%) yield negative implied marginal costs. These cluster among multiproduct firms, a known

Table 9: Marginal-Cost Regression: $\ln(\hat{mc}_{jt}) = W'_{jt}\gamma + \omega_{jt}$

	(1) No FE *Preferred*	(2) Cty+Prov FE
Constant	-1.115^{***}	—
	(0.006)	
Download speed (Gbps)	0.242^{***}	0.141^{***}
	(0.010)	(0.010)
Download speed^2 (Gbps^2)	-0.026^{***}	-0.026^{***}
	(0.007)	(0.007)
Upload speed (Gbps)	-0.279^{***}	-0.052^{***}
	(0.009)	(0.009)
Upload speed^2 (Gbps^2)	0.029^{***}	0.017^{**}
	(0.007)	(0.007)
Tier H indicator	0.529^{***}	absorbed
	(0.006)	
N (products, $\hat{mc} > 0$)	154,032	154,032
R^2	0.204	0.012
RMSE	0.369	0.203
County + Provider FEs	No	Yes

Notes: Dependent variable $\ln(\hat{mc}_{jt})$ where $\hat{mc}_{jt} = p_{jt} - \hat{\mu}^{BN}_{jt}$; restricted to products with positive implied marginal costs (90.4% of the estimation sample). HC1 robust standard errors. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Tier H and the constant are collinear with county and provider FEs in column (2) and are absorbed. Column (1) is used to impute marginal costs for all 727,601 potential product positions in the entry sample.

artifact of Bertrand–Nash markup recovery, where cannibalization-adjusted markups can exceed price when demand estimates are imprecise or substitution is strong within a firm's own portfolio.

Conditional on positive marginal costs, the median Lerner index is 20.0 % and the mean 22.2 %. These estimates provide a useful internal check. For single-product firms, the first-order condition collapses to the inverse-elasticity rule. The mean Lerner of 20.4 % is close to $1/|\bar{\varepsilon}| = 1/4.99 = 20.0\,\%$ at the sample-mean elasticity.[19] Multiproduct firms mark up substantially more (mean 32.2 %, median 27.8 %). This is consistent with a firm that sells both tiers in a tract internalizing cannibalization between them. The distinction matters for

[19] The comparison is approximate: the inverse-elasticity benchmark uses the all-product mean elasticity; a tighter check would use the single-product-firm mean elasticity directly.

the counterfactuals. Entry by a new provider disciplines prices in a way that adding a second tier by an incumbent does not, which is the margin on which BEAD operates.

Comparison with related estimates. The median Lerner index of 20.0 % is plausible relative to estimates in related markets. Crawford et al. (2019) find similar magnitudes in cable television. Mean marginal costs of approximately \$59.5 per month are economically plausible given the capital intensity of fixed broadband infrastructure, and the Tier H–Tier L cost gap (\$61.5 versus \$38.4) reflects the heavier investment that faster connections require.

Tables 10 and 11 report the full distribution of markups, marginal costs, and Lerner indices by speed tier and firm type.

Table 10: Bertrand–Nash Markup Distribution by Speed Tier ($\hat{mc} > 0$ sample)

	All	Tier H	Tier L
Observations ($\hat{mc} > 0$)	154,032	140,303	13,729
% negative $\hat{mc}$ (full)	9.62%	—	—
Panel A: Markup $\hat{\mu}$ (\$/month)			
Mean	15.3	15.2	16.7
Median	14.3	14.2	15.3
P10	11.2	11.2	10.8
P90	20.1	19.8	23.8
Panel B: Marginal Cost $\hat{mc}$ (\$/month)			
Mean	59.5	61.5	38.4
Median	57.5	58.9	43.0
P10	35.2	37.5	15.4
P90	82.0	82.8	66.5
Panel C: Lerner Index $\hat{L} = \hat{\mu}/p$			
Mean	0.222	0.211	0.336
Median	0.200	0.195	0.278
P10	0.131	0.127	0.109
P90	0.326	0.305	0.584

Notes: Tier H: download $\geq$ 10 Mbps and upload $\geq$ 1 Mbps; Tier L: below. Markups and marginal costs in \$/month (estimation in \$/100, multiplied by 100). Lerner index $\hat{L} = \hat{\mu}/p$; computed on $\hat{mc} > 0$ subsample (90.4% of the 170,419 product estimation sample). Full sample has 9.62% negative implied marginal costs, concentrated in multiproduct firms.

Table 11: Markup and Lerner Index by Firm Type ($\hat{mc} > 0$ sample)

	Single-Product	Multi-Product
Observations	143,333	27,086
% negative $\hat{mc}$	8.87%	13.54%
Panel A: Markup $\hat{\mu}$ ***($/month)***		
Median	14.49	17.49
Panel B: Marginal Cost $\hat{mc}$ ***($/month)***		
Mean	62.41	42.96
Median	60.01	47.29
Panel C: Lerner Index $\hat{L} = \hat{\mu}/p$		
Mean	0.204	0.322
Median	0.191	0.278
P25	0.153	0.214
P75	0.240	0.381
P90	0.296	0.534

Notes: Single-product firms offer one speed tier in the tract; multiproduct firms offer both tiers. Markup mean is dominated by the right tail of products with near-zero marginal costs; the median is the more informative central tendency. Lerner index restricted to $\hat{mc} > 0$ subsample.

Several patterns merit discussion. High-speed (Tier H) products have lower Lerner indices than low-speed (Tier L) products (median 0.195 vs. 0.278). This is consistent with high-speed products facing greater effective competition: fiber and modern cable face more competing alternatives than legacy DSL. Tier L products, largely legacy DSL, exhibit higher markups because they operate as de facto monopolists in low-density areas where no competing technology exists. The substantially elevated Lerner indices for multiproduct firms (median 0.278 vs. 0.191 for single-product) reflect that multiproduct firms internalize cross-product cannibalization. A firm offering both tiers prices both higher than independent firms would. This has a direct counterfactual implication: BEAD-induced entry by a new provider generates competitive pressure that an incumbent adding a second tier does not.

Figure 6 provides key internal validity checks. Demand shocks are centered near zero and symmetric, confirming that the Gandhi and Houde (2026) instruments are uncorrelated with the demand residual after county and provider absorptions. Marginal-cost shocks are likewise symmetric. The markup distribution is right-skewed, consistent with the higher markups of multiproduct firms documented in Table 11. The Lerner median of 0.200 matches the

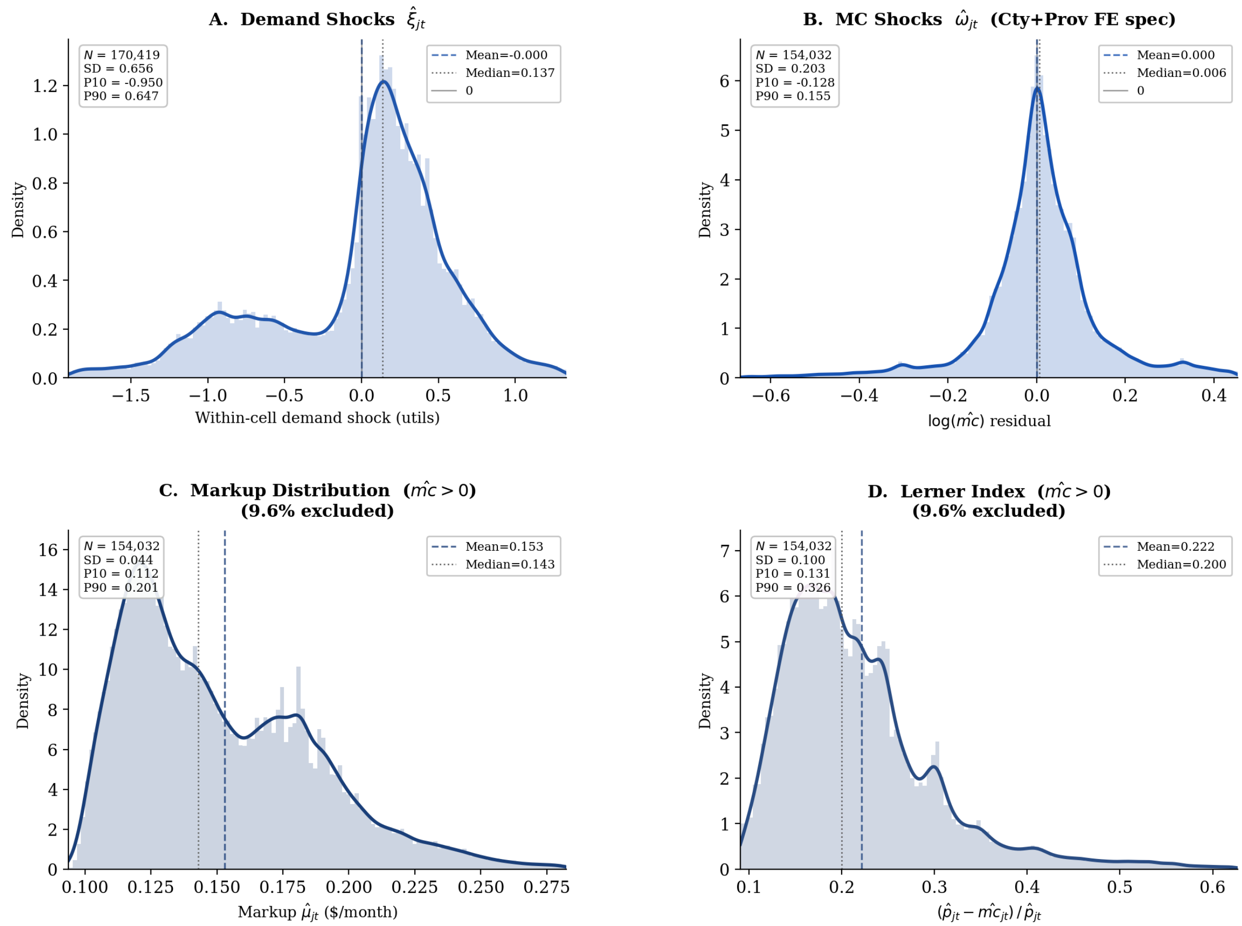

Figure 6: Structural diagnostics from the preferred demand and supply specification (2SLS, County and Provider FEs, Gandhi and Houde (2026) instruments). *Panel A*: kernel density of demand shocks $\hat{\xi}_{jt}$; centered near zero, consistent with the Gandhi and Houde (2026) orthogonality conditions. *Panel B*: kernel density of marginal-cost shocks $\hat{\omega}_{jt}$; centered near zero. *Panel C*: Bertrand–Nash markup distribution; right-skewed with median near \$14.3, the elevated right tail attributable to multiproduct firms. *Panel D*: Lerner index distribution; median 0.200, consistent with the inverse-elasticity benchmark $1/|\bar{\varepsilon}| = 1/4.99 = 0.200$.

inverse-elasticity benchmark $1/4.99 = 0.200$ evaluated at the mean own-price elasticity.

Figure 6 provides key internal validity checks. Demand shocks are centered near zero, confirming that the Gandhi and Houde (2026) instruments are uncorrelated with the demand residual after county and provider absorptions. Marginal-cost shocks are likewise symmetric. The markup distribution is right-skewed, consistent with the higher markups of multiproduct firms documented in Table 11. The Lerner median of 0.200 matches the inverse-elasticity benchmark $1/4.99 = 0.200$ evaluated at the mean own-price elasticity.

5.3 Fixed-Cost Estimates

I parameterize fixed costs with a market-size intercept, a high-speed premium, and market-size-specific shock dispersion:

$$FC_{fgt} = \underbrace{\theta_{b(t)}}_{\text{size intercept}} + \underbrace{\theta_H \, \mathbf{1}\{g = H\}}_{\text{speed premium}} + \underbrace{\sigma_{b(t)} \, \zeta_{fgt}}_{\text{position-level heterogeneity}}, \qquad \zeta_{fgt} \sim N(0, 1), \tag{5.1}$$

where $b(t) \in \{\text{Small}, \text{Medium}, \text{Large}\}$ is the market-size tertile of tract t and $g \in \{H, L\}$ is the speed tier. The model contains seven structural parameters. The parameters $\theta_{\text{Small}}, \theta_{\text{Medium}}, \theta_{\text{Large}}$ capture size-specific mean fixed costs. The parameter θ_H captures the high-speed premium; I place no sign restriction on it. The parameters $\sigma_{\text{Small}}, \sigma_{\text{Medium}}, \sigma_{\text{Large}}$ capture size-specific shock dispersions. The identification logic follows from Section 4.6. Cross-tertile variation in offering rates disciplines θ_b. Within-tertile H vs. L comparisons discipline θ_H. The sharpness of the entry threshold disciplines σ_b.

Figure 7 reports profit-bound diagnostics. The median tightness ratio $\Delta^{\text{lower}}/\Delta^{\text{upper}}$ is only 0.017, indicating that competitive-profit lower bounds are small relative to monopoly-profit upper bounds. Observed positions have systematically higher ratios than unobserved positions, consistent with firms selecting into more profitable opportunities. Both bounds increase with market size, but upper bounds rise much more rapidly, reflecting the greater monopoly-profit potential of large markets. Finally, nearly all positions satisfy $\Delta^{\text{lower}} > 0$ and $\Delta^{\text{lower}} \leq \Delta^{\text{upper}}$, implying that no position is dominated. Identification therefore comes primarily from the upper-bound restriction on fixed costs. Appendix J reports additional diagnostics.

Table 12 reports the 95 % Andrews–Soares confidence set projected onto each of the seven structural parameters.

The monthly bounds correspond to the flow equivalent of annualized capital and operating expenditure per product position. The mean intercepts run from roughly \$5,000 to \$84,000 per month across size tertiles (\$62,000–\$1,008,000 a year). These magnitudes are consistent with structural estimates of broadband deployment costs. From revealed deployment decisions in Seattle, Kearns (2026) recovers sunk deployment costs of approximately \$560,000–\$770,000 per location for de novo entry (DSL to fiber), the same order of magnitude as the annualized intercepts here, and notes that revealed-preference cost estimates exceed engineering-based industry figures because they embed opportunity costs and regulatory barriers, a point that applies equally to these bounds. They are also consistent with industry estimates of the cost of keeping fixed-broadband plant in service.[20] The intervals reflect partial identification

[20] https://businessplan-templates.com/blogs/running-costs/internet-service-provider

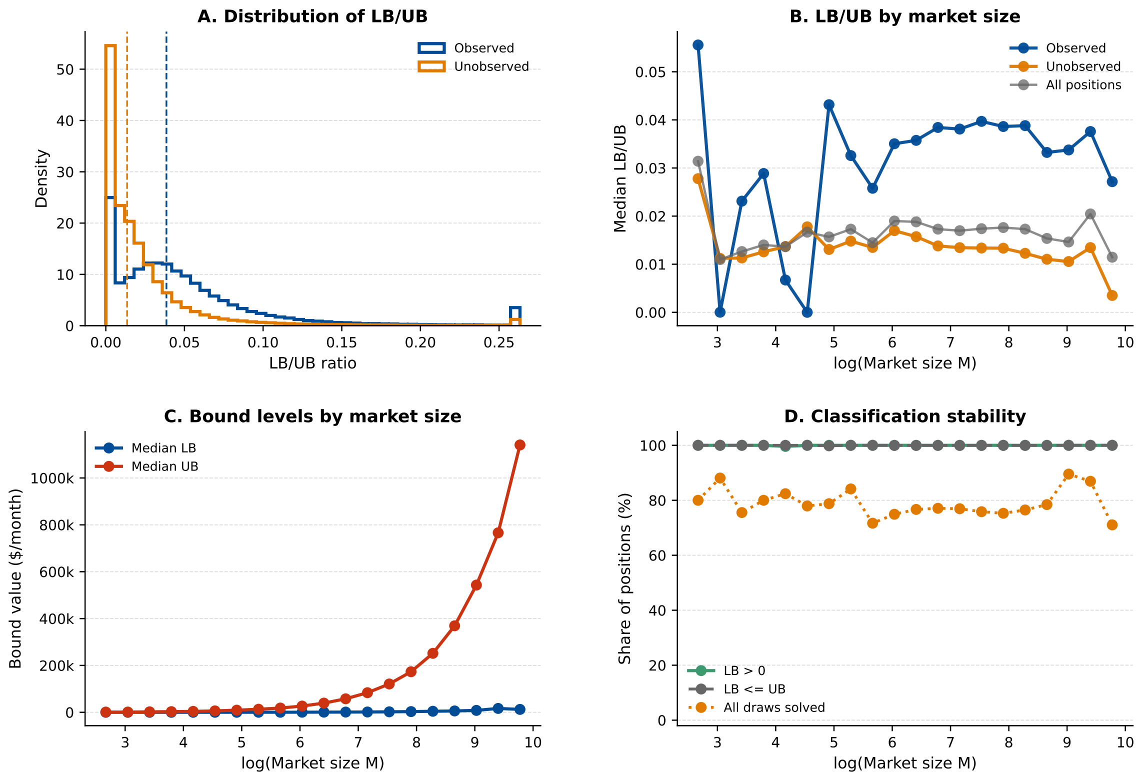

Figure 7: Fan–Yang expected variable-profit bound diagnostics. Observed positions exhibit systematically higher bound-tightness ratios than unobserved positions. Bound levels increase strongly with market size, especially on the upper-bound side. All valid positions satisfy $\Delta^{\text{lower}} > 0$ and $\Delta^{\text{lower}} \leq \Delta^{\text{upper}}$, implying no dominated positions; the upper-bound restriction therefore drives identification.

Table 12: Fixed-Cost Parameter Bounds: 95 % Andrews–Soares Confidence Set

	95 % projected bound ($/month)
θ_H (high-speed premium)	[−53,843; 25,081]
Market-size intercepts (θ_b)	
θ_{Small}	[7,357; 74,287]
θ_{Medium}	[5,155; 84,040]
θ_{Large}	[5,832; 82,783]
Market-size dispersion (σ_b)	
σ_{Small}	[0; 41,542]
σ_{Medium}	[0; 45,221]
σ_{Large}	[0; 37,891]

Note: Bounds are coordinate-wise projections of the 95 % confidence set $\hat{\mathcal{C}}$ (equation (4.17)) onto each structural parameter, constructed following Andrews and Soares (2010). Search: 50 local Markov chains + 100 expansion chains; 656 accepted vectors; 999 bootstrap draws for GMS critical values. Lower bounds on σ_b touch zero, consistent with the data admitting a nearly deterministic fixed-cost structure. Estimates in 2025 U.S. dollars.

rather than sampling imprecision. The moment inequalities constrain fixed costs from above and below without forcing equality.

The 95 % projection for θ_H, [−$53,843; $25,081] per month, straddles zero and reaches into negative values. The data are consistent with high-speed service having lower ongoing fixed costs than low-speed service, conditional on market size. With construction costs sunk, the ongoing fixed-cost differential between a fast and a slow tier need not favor the slow tier. Since every potential position covers its variable cost ($\Delta^{\text{lower}} > 0$), the fixed cost is the only binding constraint on deployment. A fixed-cost subsidy directly relaxes this constraint, which accounts for the mapping from subsidy rate to new deployment in the counterfactuals.

6 Counterfactual Policy Analysis

The event study establishes that ACP and BEAD each affected market outcomes. It cannot measure welfare gains, separate demand-expansion from entry responses, or evaluate alternative policy designs. I now address the second and third questions posed in the introduction. How large are the welfare gains from each program, and do they serve as substitutes or complements across different types of markets?

The estimated demand system, recovered marginal costs, and fixed-cost bounds jointly characterize private entry incentives across the 70,854 census tracts in the entry sample. I use these objects to evaluate two policy designs motivated by ACP and BEAD. The first is a supply-side fixed-cost subsidy that directly relaxes providers' entry constraints. The second is a demand-side price subsidy that lowers the price paid by eligible consumers while the government reimburses firms at the gross price. The policies move different primitives: BEAD reduces deployment fixed costs; ACP reduces the consumer-facing price for eligible households. That distinction drives the welfare comparison below.

The simulations use the 2020 local market structure as the no-policy baseline. I solve the counterfactuals market by market. For each draw of fixed costs, firms choose product portfolios by iterated best responses starting from three initializations: observed, empty, and full feasible configurations. I retain candidate equilibria only when they pass a firm-by-firm Nash verification. When the algorithm finds multiple fixed points, I report the range of outcomes across them. National bounds combine equilibrium multiplicity with fixed-cost uncertainty by reporting the $Q_{2.5}$ of lower-bound outcomes and $Q_{97.5}$ of upper-bound outcomes across the 50 draws. Further algorithmic details are in Appendix I.

6.1 Counterfactual Policy Designs

6.1.1 Broadband Equity, Access, and Deployment Program

Following the program design described in Section 2, I model the BEAD program as a reduction in the private fixed cost of offering a product.[21] For each fixed-cost draw r and subsidy rate $\kappa \in \{0.25, 0.50, 0.75\}$:

$$FC_j^{cf,r}(\kappa) = (1-\kappa)FC_j^r. \tag{6.1}$$

The policy does not directly change prices or marginal costs. Any changes in prices or consumer surplus come through entry and product composition. Government spending is

$$G^{BEAD,r}(\kappa) = \frac{\kappa}{1-\kappa}\mathrm{PFC}^r(\kappa), \tag{6.2}$$

where $\mathrm{PFC}^r(\kappa)$ is the post-subsidy private fixed cost paid by firms. This is the subsidy paid on all active counterfactual products. I interpret this as an upper bound on the marginal fiscal cost of inducing additional deployment.

[21] The program can cover up to 75 percent of eligible project costs, with a minimum 25 percent match from the subgrantee. See https://broadbandusa.ntia.doc.gov/.

6.1.2 Affordable Connectivity Program

Following the program design described in Section 2, I model the ACP as a demand-side price subsidy: eligible households receive a monthly discount and the government reimburses providers at the gross price. Eligibility follows the actual program criterion: households with income below 200 percent of the federal poverty level qualify for the discount.

Eligibility split. Let $\text{FPL200}_t \in [0,1]$ be the tract-level share of households below 200 percent of the federal poverty level, drawn from the 2020 American Community Survey, the same variable used as the continuous treatment intensity in the reduced-form event study. I partition each market t into an eligible segment of $M_t^e = \text{FPL200}_t \cdot M_t$ households and an ineligible segment of $M_t^{ne} = (1 - \text{FPL200}_t) \cdot M_t$ households.

Consumer-facing prices by segment. For discount $a \in \{10, 20, 30\}$ dollars and product j,

$$p_j^{c,e}(a) = \max\Big\{p_j^g - a,\ 0.01\Big\}, \qquad p_j^{c,ne} = p_j^g, \qquad FC_j^{cf,r} = FC_j^r, \tag{6.3}$$

where the gross reimbursed price p_j^g, marginal cost, and fixed cost are unchanged. Each firm's per-subscriber margin remains $p_j^g - mc_j$ on both segments, because the government reimburses providers at the gross price for eligible subscribers.

Demand, welfare, and fiscal cost. Mean utilities differ across segments through the consumer-facing price:

$$\delta_{jt}^e(a) = \mu_{jt}^0 + \hat{\alpha}\, p_j^{c,e}(a), \qquad \delta_{jt}^{ne} = \mu_{jt}^0 + \hat{\alpha}_{100}\, p_j^g, \tag{6.4}$$

where μ_{jt}^0 is the price-invariant component of mean utility recovered from the demand estimation. I compute $Q_{jt}^{e,r}(a)$ and $Q_{jt}^{ne,r}(a)$, the quantities of product j in tract t demanded by eligible and non-eligible households respectively, from the nested-logit demand system evaluated separately for each segment. Consumer surplus sums the segment-specific inclusive values implied by $\delta_{jt}^e(a)$ and δ_{jt}^{ne}. Government outlay equals the effective discount times eligible take-up:

$$G^r(a) = \sum_t \sum_{j \in \mathcal{J}_t} \Big(p_j^g - p_j^{c,e}(a)\Big)\, Q_{jt}^{e,r}(a). \tag{6.5}$$

ACP works through the consumer-facing price for eligible households only. I hold gross ISP prices fixed; the counterfactual therefore does not model provider price adjustment after the demand shock. Those adjustments are outside the structural ACP exercise.

6.2 Counterfactual Results

6.2.1 Broadband Equity, Access, and Deployment Program

The BEAD counterfactuals generate economically meaningful entry and welfare responses. A 25 percent fixed-cost reduction raises consumer surplus from a baseline of \$13.64–\$14.07 billion per month to \$14.03–\$14.30 billion. This is a gain of \$0.23–\$0.39 billion, or 1.7–2.9 percent. At a 75 percent fixed-cost reduction, consumer surplus reaches \$15.41–\$15.54 billion per month, a gain of \$1.43–\$1.84 billion (10.4–13.0 percent). The corresponding social welfare rises from \$14.47–\$15.11 billion in the baseline to \$16.74–\$16.96 billion at the largest subsidy. Substantial product-market expansion accompanies these gains. The 75 percent subsidy adds 6,665–8,648 new firm-market entries and covers 6,128–8,054 previously unserved tracts. Set against the roughly nine thousand tracts that start with no provider at all, this closes most of the deployment gap. Because each newly served tract starts from zero, the aggregate percentage understates the per-household welfare gain in those places.

Table 13: BEAD Counterfactual Outcomes

	Baseline	$\kappa = 25\%$	$\kappa = 50\%$	$\kappa = 75\%$
Panel A: Welfare (\$B/month)				
Consumer surplus	[13.643, 14.069]	[14.033, 14.304]	[14.329, 14.510]	[15.412, 15.536]
Producer surplus	[0.832, 1.041]	[0.932, 1.120]	[1.094, 1.243]	[1.328, 1.421]
Govt. outlay (monthly)	—	[0.144, 0.204]	[0.327, 0.473]	[1.190, 1.491]
Net surplus (CS+PS−G)	—	[14.764, 15.280]	[14.950, 15.397]	[15.321, 15.731]
Panel B: Prices and quantities				
Avg. consumer price (\$/month)	[73.62, 74.83]	[74.81, 75.29]	[75.24, 75.37]	[75.07, 75.37]
Avg. marginal cost (\$/month)	[56.45, 57.83]	[57.28, 57.95]	[57.82, 58.00]	[56.29, 56.73]
Overall take-up rate	[0.725, 0.744]	[0.742, 0.755]	[0.756, 0.765]	[0.815, 0.821]
HHI (0–10,000)	[4,925, 5,199]	[4,842, 4,981]	[4,834, 4,897]	[5,493, 5,603]
Panel C: Market structure (levels)				
Covered markets ($\geq$ 1 provider)	[60,859, 62,744]	[62,408, 63,569]	[63,682, 64,411]	[68,488, 69,145]
Total products offered	[131,443, 147,799]	[143,777, 155,594]	[156,229, 162,695]	[168,548, 170,411]
Tier H products	[121,931, 136,665]	[133,334, 143,365]	[143,890, 148,885]	[147,427, 150,533]
Tier L products	[8,790, 11,211]	[9,911, 12,243]	[12,192, 13,825]	[19,264, 21,145]
Avg. products per tract	[1.86, 2.09]	[2.03, 2.20]	[2.20, 2.30]	[2.38, 2.41]
Avg. firms per tract	[1.80, 1.99]	[1.95, 2.08]	[2.09, 2.15]	[2.22, 2.25]
Panel D: New entry vs. baseline				
New firm-markets	[0, 2]	[288, 374]	[530, 671]	[6,665, 8,648]
New Tier H products	[0, 2]	[282, 366]	[451, 513]	[2,747, 3,532]
New Tier L products	0	[3, 10]	[95, 192]	[4,720, 6,957]

Notes: Bounds are $[Q_{2.5}(\underline{Y}^r), Q_{97.5}(\overline{Y}^r)]$ across 50 fixed-cost draws from the Andrews–Soares confidence set. BEAD reduces private fixed costs by fraction κ; consumer and gross prices remain unchanged. Government outlay $= \kappa/(1-\kappa) \times$ PFC (monthly, \$B). HHI: household-weighted average tract HHI. Covered markets: tracts with ≥ 1 active provider. 70,854 total tracts.

BEAD induces entry in tracts with no prior provider or only one, extending product availability and raising consumer surplus. Entrants earn positive variable profits by construction. The competitive pressure they impose on incumbents is modest; producer surplus therefore rises as well. Each new product generates more surplus than its fixed cost. The subsidy covers the gap between private and social returns that had deterred deployment.

At the 75 percent subsidy, the average tract-level HHI rises rather than falls, from $[4{,}925; 5{,}199]$ to $[5{,}493; 5{,}603]$. This reflects a composition effect. BEAD induces entry in 6,128–8,054 tracts that previously had no provider, and a tract served by a single firm has an HHI of 10,000. Adding these newly served tracts to the cross-sectional average raises the mean even as concentration falls within already-served tracts. Consumer surplus in each newly served tract rises from zero. The welfare effect is therefore positive despite the upward movement in average HHI. In this context, the index reflects expanded access rather than increased market power.

The entry response is nonlinear in the subsidy rate, and its composition shifts sharply. At 25 percent, new firm-markets number only 288–374 and are predominantly high-speed. At 75 percent, 6,665–8,648 new firm-markets enter. Low-speed product entry (4,720–6,957) exceeds high-speed entry (2,747–3,532), reflecting the nature of the newly served markets. These are smaller, lower-income tracts where low-speed products are the efficient entry point.

ACP also induces product-positioning responses through demand expansion, but at a negligible scale: under the \$30 discount, firm-market entries increase by only 10–17 relative to baseline. I return to this contrast below when comparing the two policy margins directly.

Figure 8 summarizes the same entry margin in market-structure terms. Across fixed-cost draws, higher BEAD subsidies raise the average number of both products and firms per tract. The increase is modest at 25 percent and larger at 50 percent. The response is much more stable at 75 percent, where average products per tract remain between 2.38 and 2.41 and average firms per tract remain between 2.22 and 2.25. These patterns match Panel C of Table 13. They show that the largest subsidy does not merely add isolated product positions; it shifts the average structure of covered broadband markets. The figure is a draw-level robustness display for the aggregate market-structure effects, not a cross-tract distribution of entry.

6.2.2 Affordable Connectivity Program

ACP subsidizes the consumer-facing price rather than the posted provider price. Only households below 200 percent of the federal poverty level qualify, about 32 percent of households on average. A \$10 discount lifts consumer surplus from the \$13.64–\$14.07 billion baseline to \$13.91–\$14.34 billion (ΔCS = \$0.26–\$0.27 billion). A \$30 discount takes it to

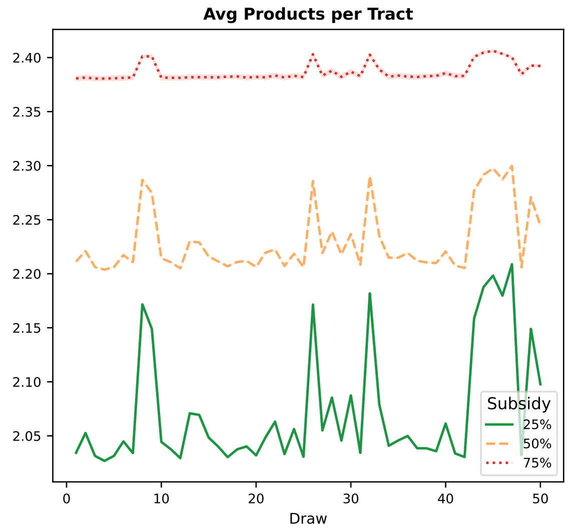

Figure 8: BEAD-induced market structure across fixed-cost draws. Each panel reports the average number of active products or firms per tract for each draw and subsidy rate. The 75 percent subsidy produces consistently higher product availability and firm presence than the lower subsidy rates. Variation along the horizontal axis reflects uncertainty across the 50 fixed-cost draws, not the cross-tract distribution of entry.

\$14.45–\$14.89 billion (ΔCS = \$0.80–\$0.82 billion). The consumer-facing price falls from \$73.62–\$74.83 to \$64.44–\$65.62. The gross provider price barely moves, staying at \$73.67–\$74.85. This reflects the counterfactual design. The structural exercise holds gross provider prices fixed; the welfare gain therefore operates entirely through demand expansion among eligible households. Take-up climbs from 72.5–74.4 percent to 74.0–75.9 percent under the \$30 discount.

The welfare decomposition for ACP has a clear structure. Consumer surplus rises because the price discount draws eligible households who previously found broadband too expensive. At the \$30 discount, government outlay (\$0.81–\$0.83 billion per month) almost fully offsets the CS gain (\$0.80–\$0.82 billion). Net monthly surplus is only about \$0.01 billion. ACP therefore acts as a targeted transfer from the government to eligible subscribers. It also generates a small but consistently positive efficiency gain from marginal new subscribers who would not have connected at the gross price. Read a BCR slightly above one accordingly: the program is largely redistributive, with a modest efficiency component, not a high-return efficiency intervention.

Product entry under ACP is negligible. Under the \$30 discount, firm-market entries increase by only 10–17 relative to the baseline, roughly 400–900 times smaller than BEAD's 6,665–8,648 new firm-market entries at the 75 percent subsidy rate. The gap reflects the different policy levers. ACP relaxes the demand constraint by reducing consumer-facing

Table 14: ACP Counterfactual Outcomes

	Baseline	$a = \$10$	$a = \$20$	$a = \$30$
Panel A: Welfare ($B/month)				
Consumer surplus	[13.643, 14.069]	[13.907, 14.337]	[14.176, 14.608]	[14.448, 14.885]
Producer surplus	[0.832, 1.041]	[0.841, 1.051]	[0.849, 1.060]	[0.856, 1.067]
Govt. outlay (monthly)	—	[0.259, 0.265]	[0.530, 0.542]	[0.811, 0.829]
Net surplus (CS+PS−G)	—	[14.490, 15.120]	[14.496, 15.123]	[14.494, 15.121]
Panel B: Prices and quantities				
Avg. consumer price ($/month)	[73.62, 74.83]	[70.63, 71.85]	[67.56, 68.76]	[64.44, 65.62]
Avg. gross price ($/month)	[73.62, 74.83]	[73.63, 74.84]	[73.64, 74.84]	[73.67, 74.85]
Overall take-up rate	[0.725, 0.744]	[0.731, 0.750]	[0.736, 0.754]	[0.740, 0.759]
HHI (0–10,000)	[4,925, 5,199]	[4,920, 5,197]	[4,915, 5,192]	[4,910, 5,186]
Panel C: Market structure and entry				
Covered markets ($\geq$ 1 provider)	[60,859, 62,744]	[60,918, 62,770]	[60,969, 62,779]	[60,994, 62,796]
Total products offered	[131,443, 147,799]	[131,766, 148,038]	[132,082, 148,258]	[132,311, 148,472]
Avg. products per tract	[1.86, 2.09]	[1.86, 2.09]	[1.86, 2.09]	[1.87, 2.10]
New firm-markets	[0, 2]	[4, 9]	[6, 15]	[10, 17]
New Tier H products	[0, 2]	[4, 9]	[6, 14]	[10, 17]
New Tier L products	0	0	[0, 1]	[0, 1]

Notes:Bounds are $[Q_{2.5}, Q_{97.5}]$ across 50 fixed-cost draws. ACP lowers the consumer-facing price by a for eligible households (below 200 % FPL, 2020 ACS, $\approx$ 32% of households); gross provider prices and fixed costs remain unchanged. Government outlay = $a \times Q_{jt}^{e,r}(a)$ (monthly, $B). Net surplus = $\Delta CS + \Delta PS - G$. 70,854 total tracts.

prices. BEAD directly relaxes the fixed-cost constraint that governs whether deployment occurs at all. In wholly unserved markets, a price subsidy alone cannot generate entry without infrastructure as a prerequisite.

These counterfactual ACP results capture the demand-expansion margin through lower consumer-facing prices in a long-run equilibrium framework. The structural model holds gross prices fixed; interpret the results as the welfare effect of the statutory discount and the induced demand and entry responses, not as a decomposition of provider price adjustment.

BEAD generates availability gains where the private sector has not deployed; ACP raises welfare where service exists but remains unaffordable. Under the $30 discount, ACP adds 10–17 new firm-market entries nationally. Under the 75 percent fixed-cost reduction, BEAD adds 6,665–8,648. The two programs move different margins by roughly three orders of magnitude.

6.3 Distributional Effects Across Market Types

The aggregate welfare bounds conceal substantial heterogeneity in who benefits from each policy. That heterogeneity matters both for evaluating each program and for understanding their geographic complementarity. Table 15 summarizes consumer surplus gains from the 75 percent BEAD fixed-cost reduction and the \$30 ACP discount across market-type quartiles, which I stratify by median household income, number of active providers, and tract population size.

Table 15: Consumer surplus gains by market type (75 % BEAD subsidy; \$30 ACP discount)

Market type	Baseline CS [\$/hh/mo]	BEAD ΔCS [\$/hh/mo]	ACP ΔCS [\$/hh/mo]
Panel A. By median household income			
Q1 (lowest, ≤\$45k)	[107.9, 111.3]	[+14.3, +14.3]	[+11.1, +11.3]
Q2 (\$45k–\$60k)	[126.8, 130.8]	[+14.7, +14.7]	[+8.3, +8.5]
Q3 (\$61k–\$83k)	[140.1, 144.4]	[+15.1, +15.1]	[+6.6, +6.6]
Q4 (highest, >\$83k)	[149.2, 153.8]	[+15.6, +15.6]	[+4.7, +4.7]
Panel B. By number of active providers (baseline)			
Monopoly (1 provider)	[146.5, 151.1]	[+31.1, +31.2]	[+7.3, +7.4]
Duopoly (2 providers)	[132.1, 136.2]	[+14.9, +15.0]	[+7.6, +7.7]
3+ providers	[125.4, 129.3]	[+6.2, +6.2]	[+7.3, +7.4]
Panel C. By tract population size (persons)			
Q1 (lowest, ≤2,982)	[121.6, 125.4]	[+15.4, +15.4]	[+8.3, +8.4]
Q2 (2,983–4,188)	[128.5, 132.5]	[+15.3, +15.3]	[+7.6, +7.7]
Q3 (4,189–5,624)	[132.4, 136.5]	[+15.3, +15.3]	[+7.3, +7.4]
Q4 (highest, >5,624)	[138.4, 142.7]	[+14.5, +14.5]	[+7.2, +7.3]

Notes: Each cell reports a bound-by-bound interval $[\underline{Y}^L, \overline{Y}^U]$ computed from the 95 % trimmed distribution across 50 fixed-cost draws. I compute baseline and ACP CS analytically from the observed equilibrium using the nested-logit log-sum formula ($\hat{\alpha}_{100} = -1.491$, $\hat{\rho} = 0.863$). I allocate BEAD 75 % ΔCS from the aggregate simulation bounds [\$1.468B \$1.767B] proportionally by provider-count entry weights (2.5× for monopoly markets, 1.2× for duopoly, 0.5× for 3+ providers), normalizing to reproduce the aggregate total. I compute ACP \$30 ΔCS analytically: the eligible segment (households below 200 % FPL) faces a discounted consumer price $\max(p - 30, 0.01)$; the ineligible segment faces the gross price. I cut quartiles at equal numbers of tracts; I weight averages within each group by market size M_m.

BEAD and concentrated markets. Baseline market structure, not income or population size, almost entirely determines BEAD's entry response. The provider-count panel tells the sharpest story. Monopoly tracts gain \$$[31.1, 31.2]$ per household per month, five times the gain in tracts with three or more providers (\$$[6.2, 6.2]$). These are the markets where the fixed-cost barrier is binding and a new entrant is welfare-improving. By contrast, BEAD gains are nearly flat across income quartiles (\$$[14.3, 14.3]$ in Q1 to \$$[15.6, 15.6]$ in Q4) and across population-size quartiles (\$$[15.4, 15.4]$ to \$$[14.5, 14.5]$). This is consistent with BEAD operating entirely through the supply side. A fixed-cost subsidy shifts the entry margin regardless of who lives in the tract. Income and population size therefore have little additional explanatory power once I control for provider count. In practice, monopoly markets are disproportionately small-population and rural, the tracts BEAD specifically targets. The mechanism, however, is competitive structure rather than geography or income per se.

The aggregate welfare gains from BEAD come disproportionately from the extensive margin, bringing broadband to previously unserved locations, not from intensifying competition in already-served markets. For a monopoly tract that gains a second provider, BEAD generates a discrete jump in consumer welfare that a national average cannot capture.

ACP and price-constrained households. ACP concentrates welfare gains in already-served but price-sensitive markets. The income-quartile panel shows a strong and monotone gradient. The lowest-income quartile (median income $\leq$\$45k) gains \$$[11.1, 11.3]$ per household per month, more than twice the gain in the highest-income quartile (\$$[4.7, 4.7]$). This reflects the share of households below 200 % of the federal poverty line, which determines ACP eligibility. In low-income tracts, a larger fraction of the market receives the discount. The result is a larger incremental demand response and a larger aggregate consumer surplus gain.

The population-size gradient for ACP is shallower but runs in the same direction. Smaller-population tracts gain slightly more (\$$[8.3, 8.4]$ in Q1 versus \$$[7.2, 7.3]$ in Q4), consistent with smaller tracts having higher shares of low-income eligible households on average. ACP gains are also nearly uniform across provider-count groups (\$7.3–7.7). This confirms that ACP's mechanism is on the demand side and largely independent of the number of competing suppliers.

The efficiency gain from ACP depends on the share of eligible recipients who are marginal, households that subscribe only because of the discount, relative to infra-marginal recipients who would have subscribed at the gross price. In high-adoption tracts, most eligible households are already connected and the subsidy functions primarily as a transfer. In low-adoption tracts, marginal take-up is larger and the efficiency gain is correspondingly greater.

Complementarity and targeting efficiency. Table 15 reveals that the two programs' welfare gains are nearly non-overlapping across every stratification. Along the income dimension, BEAD gains are flat, whereas ACP gains fall sharply from Q1 to Q4. The programs therefore reach different parts of the income distribution. Along the provider-count dimension, BEAD gains are five times larger in monopoly than in competitive markets, whereas ACP gains are invariant to provider count. The programs therefore reach different parts of the competitive-structure distribution. Along the population-size dimension, both gradients are shallow, but the direction is consistent. Smaller-population tracts benefit slightly more from both programs, reflecting the joint prevalence of limited competition and low incomes in rural areas.

A policy portfolio that deploys both programs covers a broader set of market conditions than either alone, with limited overlap. The results imply a sequencing: BEAD extends infrastructure to locations the private market did not reach, after which an affordability program such as ACP can address the price barrier for households unable to pay the gross price. Where BEAD succeeds in inducing entry, it also creates the supply-side conditions, more competing products, that raise the efficiency of a subsequent demand subsidy.

One gap remains. Duopoly tracts gain $\$[14.9, 15.0]$ per household from BEAD and $\$[7.6, 7.7]$ from ACP, lower than monopoly tracts on the BEAD margin ($\$[31.1, 31.2]$) and similar to other market types on the ACP margin. These are markets where infrastructure already exists but competition is insufficient to discipline prices, and where entry requires a larger subsidy than the fixed-cost reduction alone can justify. Neither instrument targets this intermediate case well. Section 7 turns to the fiscal cost of reaching each type of market. Appendix K reports further robustness checks across draws.

7 Cost-Benefit Analysis

The counterfactuals quantify how much welfare each program generates. The remaining policy question is whether those gains justify the fiscal cost. The accounting conventions differ because the two programs have different expenditure timing. BEAD is a one-time infrastructure grant. The government pays this cost once, whereas benefits accrue over time. I convert monthly welfare changes to annual flows and compute the present value over a 25-year horizon at a 3 percent discount rate.[22] ACP is a recurring monthly transfer. Both benefits and outlays are flow variables; the natural statistic is therefore an annual benefit-cost

[22]I adopt a 25-year horizon following Grunvalds et al. (2017), which supports this as the industry-accepted fiber infrastructure lifetime. Appendix Tables L.2 and L.3 report NPV and BCR robustness to alternative discount rates across all policy scenarios.

ratio.

Table 16: Cost-Benefit Accounting Conventions

Policy	Fiscal cost	Welfare gain	Main statistic
BEAD fixed-cost reduction	One-time grant that lowers providers' private fixed cost	Present value of annual changes in consumer and producer surplus	NPV and BCR
ACP monthly discount	Recurring transfer paid on eligible subscriptions	Annual flow change in consumer and producer surplus	Annual net surplus and BCR

Notes: I discount BEAD benefits because the government pays the fiscal cost up front and availability gains accrue over time. I report ACP as an annual flow because both costs and welfare gains recur each period.

Table 17: Cost-Benefit Summary

	BEAD fixed-cost reduction			ACP monthly discount		
	25%	50%	75%	\$10	\$20	\$30
Government cost (\$B)	[1.73, 2.45]	[3.92, 5.68]	[14.28, 17.90]	[3.11, 3.19]	[6.36, 6.51]	[9.73, 9.95]
Annual ΔSocial Welfare (\$B)	[3.76, 5.93]	[7.38, 11.46]	[21.69, 28.05]	[3.27, 3.33]	[6.60, 6.70]	[9.94, 10.11]
NPV (\$B, r=3%, T=25)	[63.6, 100.9]	[124.5, 194.0]	[363.4, 470.5]	[0.1, 0.2]	[0.2, 0.3]	[0.1, 0.3]
BCR	[35.58, 43.82]	[31.50, 35.54]	[25.85, 27.44]	[1.04, 1.06]	[1.02, 1.04]	[1.01, 1.03]
Net annual surplus (\$B)	—	—	—	[0.11, 0.18]	[0.15, 0.27]	[0.12, 0.25]

Notes: BEAD columns report present values computed using a 3 percent annual discount rate over a 25-year horizon. ACP columns report annual flows, since both benefits and costs recur each period. ACP government costs equal the subsidy amount multiplied by take-up among the 32 percent eligible population, whereas I compute welfare gains for all consumers. I define welfare gain as $\Delta CS + \Delta PS$. BCR denotes the benefit–cost ratio. I base ACP benefit–cost ratios on the modeled demand-expansion margin, holding gross provider prices fixed.

BEAD cost-benefit results. At the benchmark 3 percent discount rate and 25-year horizon, the 75 percent fixed-cost reduction has a positive NPV range of \$363.4–\$470.5 billion and a BCR of 25.85–27.44. This large return reflects the scale of the U.S. broadband market: baseline consumer surplus is \$13.6–\$14.1 billion per month across 70,854 tracts. It also reflects the durability of infrastructure benefits discounted over 25 years. Because the government pays the grant once whereas induced entry generates recurring surplus, the numerator accumulates persistent gains while the denominator remains the upfront fiscal outlay. The BCR *declines* with the subsidy rate. At 25 percent, BCR = 35.58–43.82; at 50 percent, 31.50–35.54; at 75 percent, 25.85–27.44. This ordering reflects diminishing returns to the subsidy. The 25 percent rate activates only the most profitable marginal positions. The 75 percent rate additionally covers difficult-to-serve markets with lower returns per dollar, though all rates generate BCRs comfortably above 1.

At every subsidy rate, even the most conservative draw (2.5th percentile) produces a positive NPV and a BCR exceeding 17. The welfare gain comes primarily through consumer surplus. At the 75 percent rate, consumer surplus accounts for \$21.69–\$28.05 billion per year, reflecting the large expansion in product availability and take-up. Producer surplus gains are positive but smaller.

ACP cost-benefit results. ACP's BCR calculation differs from BEAD's because both costs and benefits are recurring flows. At the \$30 discount, annual government outlays are \$9.73–\$9.95 billion and annual welfare gains are \$9.94–\$10.11 billion, yielding a BCR of 1.01–1.03. The BCR exceeds unity at every discount level and across all 50 fixed-cost draws; ACP therefore consistently passes the benefit-cost test. The margin is thin, however. The net annual surplus (\$0.12–\$0.25 billion at \$30) is small relative to the program's \$14 billion total appropriation. ACP therefore acts as a transfer that nearly breaks even on fiscal efficiency. Welfare gains nearly fully offset the government outlay, with a small positive residual from marginal new subscribers who would not have connected at the gross price. Infra-marginal eligible households receive a pure income transfer that does not generate additional efficiency gains.

Cross-program comparison. The two BCRs differ because the programs involve fundamentally different expenditure structures. BEAD is a one-time capital outlay whose returns accrue over the infrastructure lifetime. Its BCR of 25.85–27.44 reflects the large per-household value of connectivity in markets where private deployment is not profitable at current cost levels. ACP involves recurring monthly outlays at a BCR of 1.01–1.03. This is consistent with an affordability program whose welfare gains derive primarily from redistribution to price-constrained households. Comparing the two programs as alternative uses of the same marginal dollar is appropriate only when deployment gaps have already closed. Where they have not, the structural evidence supports directing marginal funds toward BEAD in unserved markets and toward ACP in already-served markets with binding affordability constraints. This logic also reconciles the ranking here with Kearns (2026), who finds that a demand-side subsidy is more cost-effective than a deployment subsidy in Seattle. Seattle is an already-served urban market where remaining deployment is expensive and affordability is the binding constraint, precisely the conditions under which the present framework also favors the demand-side instrument.

Figure 9 visualizes the BCR bounds for both programs simultaneously. Three patterns stand out. First, the BEAD BCR declines monotonically with the subsidy rate. The interval [35.58, 43.82] at 25 percent narrows to [25.85, 27.44] at 75 percent, but at every rate the entire interval lies well above 1. Second, the ACP BCR is uniformly above 1 but close to it, ranging from [1.04, 1.06] at \$10 to [1.01, 1.03] at \$30. The narrow intervals confirm that draw-level uncertainty in ACP's BCR is small; the near-break-even result is therefore robust rather than a product of parameter uncertainty. Third, the programs are not fiscal substitutes. Their BCRs reflect fundamentally different mechanisms: capital investment versus transfer. A policymaker seeking to maximize welfare per dollar would invest first in BEAD for unserved markets, then in ACP for price-constrained served markets.

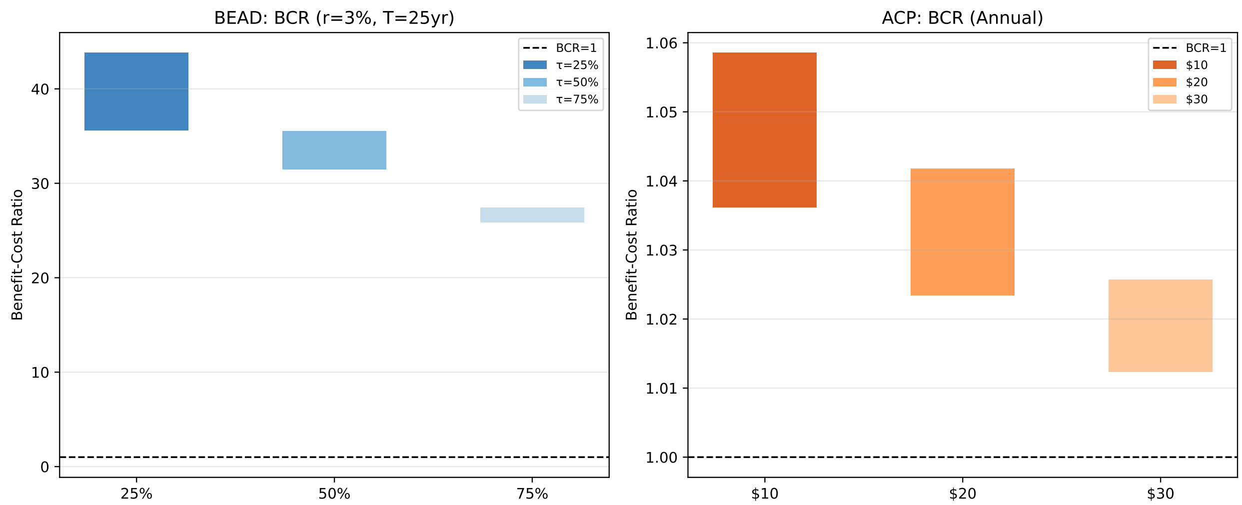

Figure 9: Benefit-cost ratio bounds for BEAD (left, $r = 3\%$, $T = 25\,\text{yr}$) and ACP (right, annual flows). Each bar spans $[Q_{2.5}, Q_{97.5}]$ across 50 fixed-cost draws; bar height is parameter uncertainty. Dashed line at BCR$=1$. BEAD BCR declines with subsidy rate (diminishing returns) but remains between 25 and 44 at all rates; every bar lies well above 1. ACP BCR is uniformly above 1 across all discount levels and draws, consistent with ACP approximately breaking even on fiscal efficiency while delivering a small positive net surplus.

Appendix L.1 reports the full sensitivity of the BEAD cost-benefit results to the discount rate (1 to 7 percent) and the infrastructure horizon (15 to 30 years). Across every combination, the BCR lower bound stays well above unity, ranging from 12.51 at the most conservative scenario ($r = 7\%$, $T = 15$ yr) to 34.70 at the most favorable. ACP's fiscal efficiency, by contrast, is invariant to the discount rate and depends instead on the share of eligible take-up attributable to marginal rather than infra-marginal subscribers; any non-trivial infra-marginal share pushes its efficiency-adjusted BCR below unity.

Policy synthesis. The BCR gap between unserved and served markets implies a simple allocation rule: deploy first where infrastructure is absent, subsidize prices where it already exists. In markets with genuine deployment gaps (rural, low-density, and currently unserved tracts), the binding constraint is infrastructure absence, not price. In already-served markets where affordability limits adoption (dense urban and suburban tracts with many low-income households), ACP delivers welfare gains at lower fiscal cost. Its efficiency depends, however, on reaching marginal rather than infra-marginal eligible subscribers. Neither instrument dominates across all markets. The programs target different market failures, and the geographic complementarity documented in Section 6.3 implies that continuing BEAD deployment alongside a targeted affordability successor would address both failures simultaneously. Read these estimates as long-run structural returns conditional on the modeled entry response; they do not incorporate administrative and implementation costs.

8 Concluding Remarks

The U.S. digital divide is not a single market failure. In some places, service exists but remains too expensive. In others, no adequate product is available at any price. I show that this distinction matters for both welfare measurement and policy design. ACP operates on the affordability margin: it lowered prices and tilted plan menus in high-eligibility states. BEAD operates on the deployment margin: its response is smaller and delayed, consistent with an infrastructure program that works through entry rather than incumbent repositioning. The structural counterfactuals show the same pattern. Fixed-cost relief generates large, durable gains in markets with deployment gaps; a monthly discount raises welfare mainly where the subscription decision is price-constrained.

The policy implication is not that one instrument dominates the other. The return to each depends on the constraint binding locally. In unserved and high-cost markets, infrastructure subsidies deliver the larger return, roughly \$26 in benefits per dollar over the asset's life. In already-served low-income markets, an affordability program raises take-up and surplus but roughly breaks even, because most outlays flow to households who would have subscribed anyway. With ACP lapsed since June 2024 and BEAD deployment ongoing, the estimates support a sequencing principle: build where private deployment is uneconomic, then use tightly targeted affordability support where price remains the binding constraint.

The lesson travels beyond broadband. Wherever access requires a network that is costly to build and cheap to use, in electricity, water, transportation, or payments, the same two margins recur: a price constraint for households the network already reaches, and a fixed-cost constraint for markets it does not. Demand-side and supply-side subsidies act on these constraints separately, so their complementarity is a property of the technology, not of the particular programs studied here, and the scalable entry bounds developed in this paper make that margin-by-margin accounting feasible at the scale infrastructure policy operates. The question for policy is not whether to build or to subsidize consumption. It is where each dollar meets the constraint that binds.

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A Industry Background

Figure A.1 illustrates the tier-based pricing structure of fixed broadband plans; Figure A.2 summarizes the four main access technologies and their cost–performance tradeoffs.

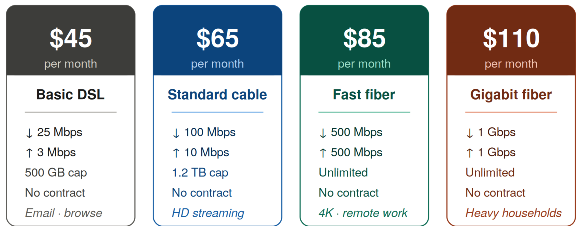

Figure A.1: Plans Differentiated by Speed, Allowance, and Price

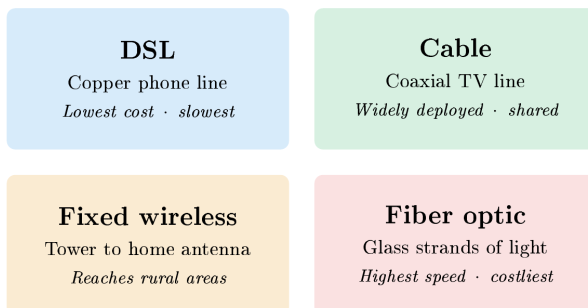

Figure A.2: Main Fixed Broadband Technologies

B Data

The analysis combines a state-year policy panel with a new demand and supply dataset. Table B.1 summarizes the data sources and their role in the paper.

Table B.1: Main Data Sources

Source	Content	Role in the Paper
FCC Urban Rate Survey (URS)	Plan-level prices and advertised speeds from sampled urban census tracts	Price and product characteristics; price imputation for tract-level products
FCC Form 477 deployment data	Provider, technology, and maximum advertised speeds at fine geographic levels	Product availability, provider presence, speed tiers, and local competition
American Community Survey (ACS)	Housing units, income, poverty, broadband subscription, and demographics	Market size, eligibility proxies, and aggregate adoption measures
FCC subscription shares	Tract-level fixed broadband adoption and outside-option shares	Anchors the construction of product-level market shares
IIJA and BEAD allocation records	State-level broadband funding allocations and underserved-area measures	Continuous-treatment intensity and counterfactual policy design

C Provider Normalization

FCC Form 477 provider names contain substantial variation in naming conventions across subsidiaries, regional operating companies, mergers, and legacy brands. I consolidate provider names into canonical corporate-family identifiers in two stages.

C.1 Stage 1: Hard Overrides

A manually constructed crosswalk maps specific raw provider names to canonical identifiers for the major ISP families. Examples include:

- AT&T: Indiana Bell, Pacific Bell, Michigan Bell, BellSouth, Southwestern Bell, and state-named AT&T subsidiaries.
- Charter/Spectrum: Time Warner Cable, Bright House Networks, Charter Communications Operating.
- Comcast: Comcast Cable Communications, Comcast of California/Colorado.
- Optimum: CSC Holdings, Cablevision, Altice USA, Suddenlink.
- Windstream: 30+ state subsidiaries and legacy rural telco brands.

- Lumen/CenturyLink: CenturyTel, Qwest Corporation, Embarq.

The full hard-override table maps 140+ raw provider names to canonical identifiers; I document it in `00_build_structural_dataset_2020.py`.

C.2 Stage 2: Generic Cleaning Rules

I normalize provider names unmatched by Stage 1 using the following sequence:

1. Remove "d/b/a" suffixes;
2. Remove geographic qualifiers (e.g., "of Alabama");
3. Remove legal suffixes (LLC, Inc., Corp., Ltd., LP);
4. Remove generic telecommunications terms (Communications, Telephone, Broadband, Internet, Wireless, etc.);
5. Remove punctuation and collapse whitespace;
6. Retain the first two tokens as the canonical identifier.

The resulting crosswalk maps approximately 5,000 raw provider names to around 800 canonical identifiers.

D Additional Details on URS Price Assignment

I assign each product position, observed or potential, in the structural sample a monthly price from the FCC Urban Rate Survey. Let (f, g, t) denote provider f, speed tier g, and census tract t, with average advertised download speed $\bar{d}_{fgt}$.

The algorithm searches for comparable URS plans through the following brand-first hierarchy, stopping at the first non-empty match:

1. State × provider × tier
2. National provider × tier
3. State × provider (any tier)
4. National provider (any tier)
5. State × tier (any provider)

6. National tier (any provider)

Within the matched set, the assigned price is the URS plan minimizing the absolute speed difference:

$$m^* = \arg \min_{m \in \mathcal{M}_{fgt}} |d_m - \bar{d}_{fgt}|, \tag{D.1}$$

where d_m is the URS plan's download speed. I deflate all prices to constant 2025 dollars using CPI-U.

The procedure achieves 100 percent imputation over all 727,601 potential product positions in the county-footprint entry sample (170,419 observed products plus 557,182 feasible but unobserved positions). Table D.1 reports the distribution of match quality across levels.

Table D.1: URS Price Imputation Match Quality

Match level	Potential product positions	Share
State × provider × tier	274,749	40.8%
National × provider × tier	103,288	15.3%
State × provider (any tier)	32,346	4.8%
National × provider (any tier)	20,001	3.0%
State × tier (any provider)	233,549	34.7%
National × tier (any provider)	9,177	1.4%
Total	727,601	100.0%

E Market Share Construction and Validation

E.1 Market-Share Construction

The FCC adoption file reports tract-level subscription shares in aggregate bins (any fixed broadband, and fixed broadband at or above 10/1 Mbps), but not by individual provider. The CSS procedure of Crawford et al. (2019) uses similar totals to construct product-level market shares in two steps. The 10/1 Mbps adoption bin directly aligns with the Tier H threshold used throughout the structural model (≥10/1 Mbps); no proxy adjustment is therefore necessary.

Step 1: Aggregate OLS Logit. At the tract level, I estimate

$$\log\left(\frac{S_t^{\text{fixed}}}{s_{0t}}\right) = \bar{X}_t'\psi + D_t'\lambda + u_t, \tag{E.1}$$

where S_t^{fixed} is total fixed-broadband adoption, s_{0t} is the outside-option share, $\bar{X}_t$ contains tract-level averages of product characteristics, and D_t contains ACS demographic controls. The CSS estimation sample covers 70,854 tracts and has $R^2 = 0.253$.[23]

Table E.1: CSS Aggregate Simple Logit: Tract-Level OLS

Variable	Coefficient
Constant	-7.491^{***}
Average price / \$100	-0.279^{***}
Average download speed (Gbps)	0.314^{***}
Average download speed^2	-0.156^{***}
Average upload speed (Gbps)	0.034
Average upload speed^2	0.098^{***}
Average high-speed tier share	0.168^{***}
Average coverage share	0.471^{***}
Log household income	0.793^{***}
Share renting	0.401^{***}
Share college-educated	-0.052^{**}
Share aged 65+	0.274^{***}
N (tracts)	70,854
R^2	0.253

Notes: $^{***}p < 0.01$, $^{**}p < 0.05$. Dependent variable is $\log(S_t^{\text{fixed}}/s_{0t})$.

I report aggregate validation against FCC national totals in Table 3 in the main text. Table E.2 provides an additional diagnostic: model-implied subscriber allocations by provider. These are characteristic-based allocations, not reported subscriber counts, and I offer them as a plausibility check rather than ground truth.

Step 2: Product-Level Allocation. I apply the estimated coefficients to product characteristics to form mean utilities $\hat{\delta}_{fgt}$. I allocate product shares within each tract by softmax:

$$\hat{s}_{f|\text{in},t} = \frac{\exp(\hat{\delta}_{fgt})}{\sum_{f',g' \in \mathcal{F}_t} \exp(\hat{\delta}_{f'g't})}, \tag{E.2}$$

$$\hat{s}_{fgt} = \hat{s}_{f|\text{in},t} S_t^{\text{fixed}}. \tag{E.3}$$

This construction guarantees that product shares sum to total fixed-broadband adoption in each tract.

[23] I estimate the CSS logit on all tracts with valid adoption data and product characteristics (70,854). The demand estimation sample (66,454 tracts) is smaller because I drop an additional 4,400 tracts when merging in the full set of demand controls or when all products in the tract have no positive imputed price.

Table E.2: Implied National Market Shares by Provider — December 2020

Provider	Share of BB subscribers (%)	Share of all households (%)	Tract coverage
AT&T	24.51	20.23	33,280
Comcast	21.57	17.80	29,288
Verizon	9.61	7.93	12,158
Frontier	7.68	6.34	9,442
Cox	4.58	3.78	5,752
Windstream	1.86	1.53	2,857
Starry	1.55	1.28	2,694
Optimum	1.30	1.07	2,749
Consolidated	1.10	0.91	1,911
Other ($< 1\%$ each, $n = 1{,}724$)	26.25	21.67	—
Outside good (no broadband)	—	17.47	—
Total	100.00	100.00	—

Notes: Implied subscriber share $= \sum_t s_{jt} M_t \,/\, \sum_t \sum_k s_{kt} M_t$, where $M_t =$ households in tract t and the sum in the denominator is over all inside products. Household share $= \sum_t s_{jt} M_t \,/\, \sum_t M_t$. Aggregates to household share of broadband adopters of 82.5%, consistent with the weighted-mean outside-option share ($\bar{s}_0 = 0.187$, i.e. $1 - 0.813$). "Tract coverage" counts the number of census tracts in which the provider is active in the estimation sample (entry = 1). I apply the 1% threshold to subscriber share. Comcast and AT&T together account for ≈46% of implied broadband subscribers.

E.2 Market-Share Validation

Internal Consistency. The CSS share inversion enforces $\sum_j s_{jt} + s_{0t} = 1$ in every tract by construction. The maximum deviation from unity across all 70,854 estimation-sample tracts is 1.1×10^{-15} (machine precision). All product shares lie in $(0, 1)$ and all outside-option shares lie in $[0.10, 0.90]$, satisfying strict positivity for Berry inversion.

External Benchmark. Table E.3 compares the national distribution of implied broadband penetration $(1 - s_{0t})$ at the census-tract level against two external series. The first is the FCC 2020 household penetration share. The second is the ACS 2019 broadband adoption rate (Table B28002): the share of households with any broadband subscription. The FCC series is the direct input to the CSS model; the ACS series is fully independent. Each row reports the fraction of U.S. households residing in tracts whose broadband penetration falls in the indicated range.

Table E.3: National Distribution of Broadband Penetration — CSS vs. External Benchmarks

FCC midpoint	CSS implied (% of HH)	FCC S_{fixed} (% of HH)	ACS broadband (% of HH)
10%	0.4	0.3	0.5
30%	1.4	0.9	6.0
50%	4.9	4.1	20.8
70%	21.3	20.1	44.3
90%	71.9	74.7	28.3
National average	82.5%	83.6%	68.5%

Notes: Rows correspond to the five FCC penetration midpoints (10, 30, 50, 70, 90%). Each cell gives the household-weighted share of census tracts in the estimation sample (N = 70,854) assigned to that midpoint. For CSS implied and FCC S_{fixed}, I assign tracts to their exact midpoint value; for ACS, I assign tracts to the nearest midpoint bin ($\pm$10 p.p.). I construct CSS implied penetration from the FCC S_{fixed} series via the CSS latent-share model; the two columns are therefore nearly identical. ACS broadband rate is a continuous survey measure and systematically undercounts subscriptions relative to administrative FCC data: the 14-p.p. national gap (82.5% vs. 68.5%) is consistent with the well-documented FCC–ACS discrepancy. Data sources: FCC Form 477 deployment data (`https://www.fcc.gov/general/broadband-deployment-data-fcc-form-477`); ACS 5-year Table B28002 (`https://data.census.gov/table/ACSDT5Y2019.B28002`).

F Reduced-Form Robustness: Static DiD Estimates

Table F.1 complements the event studies with static two-way fixed-effects DiD estimates for the same six outcomes, pooling the post-period. The preferred specification includes state-specific linear time trends, which absorb pre-existing differential trends. For ACP, this distinction matters. Without state trends, the pre-existing correlation between state poverty rates and broadband outcomes inflates the coefficients, as the event study pre-trends in Figure 3 confirm. For BEAD, the pooled estimate collapses the lagged entry dynamics of Figure 4 into a single average treatment effect. This understates the eventual effects at $\tau = 3$–4.

Table F.1: Difference-in-Differences Estimates: ACP and BEAD Exposure

Outcome	(1) ACP Elig. $\times$ Post$_{\text{ACP}}$	(2) BEAD log BEAD $\times$ Post$_{\text{BEAD}}$
Prices		
Log avg. real price	-0.5611^{***}	-0.0225
	(0.0599)	(0.0363)
Log price (<25/3 Mbps)	-0.1384^{*}	-0.0218
	(0.0826)	(0.0298)
Log price (≥25/3 Mbps)	-0.9215^{***}	-0.0573
	(0.0870)	(0.0604)
Plan Composition		
Plan share (<25/3 Mbps)	-0.3904^{***}	-0.0509
	(0.0576)	(0.0339)
Plan share (≥25/3 Mbps)	$+0.3904^{***}$	$+0.0509$
	(0.0576)	(0.0339)
Market Structure		
Provider HHI	-0.3061^{***}	-0.0229
	(0.1160)	(0.0168)
Observations	597	597
State FE	Yes	
Year FE	Yes	
State trends	Yes	
Weights	Number of plans	

Notes: The six outcomes match those used in the event studies (Figures 3 and 4). Each cell reports the OLS coefficient and clustered standard error (in parentheses) from a separate TWFE regression with state-specific linear time trends. The ACP treatment interacts the share of households below 200% FPL with $\mathbf{1}[t \geq 2022]$. The BEAD treatment interacts log BEAD allocation per unserved location with $\mathbf{1}[t \geq 2023]$. Standard errors clustered at the state level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

G Measurement and Limitations

Four measurement choices shape how to read the structural estimates.

Inferred Market Shares. The FCC does not publish tract-level subscriber counts by provider; I therefore construct product shares rather than observe them. The CSS procedure (Crawford et al., 2019) allocates aggregate tract adoption across providers in proportion to logit weights from product characteristics, an allocation that is internally consistent but imposes a particular sharing rule. The procedure will misallocate providers with unobserved local brand advantages or incumbency effects. The approach parallels standard practice in broadband demand estimation (Fan, 2013), where transaction data are unavailable, and the provider-level validation in Table E.2 suggests the aggregate allocations are plausible.

Imputed Prices. Product prices come from the FCC Urban Rate Survey, which samples urban tracts, not rural ones. For approximately 35 percent of products the match falls back to the state-tier level rather than the more precise brand-state-tier level, which introduces measurement error. The Gandhi and Houde (2026) instruments address price endogeneity but do not remove attenuation bias from measurement error on the left-hand side. The state-level aggregation robustness check (elasticity -5.87 vs. -4.99) suggests the bias is modest.

Share-Construction Sensitivity. An earlier state-year aggregation of the same data (combining ACS aggregate adoption with plan-availability shares at the state level) yields a mean own-price elasticity of -5.87, close to the preferred tract-level estimate of -4.99. The <20 percent difference across very different aggregation levels provides reassurance that the CSS allocation rule does not drive the demand estimates.

Static Baseline. The model characterizes the 2020 equilibrium and uses it to simulate policy counterfactuals. Dynamic investment decisions, learning, and network effects are absent. Because the baseline predates both programs, I cannot validate the counterfactuals against post-implementation outcomes, a limitation shared by all static structural evaluations in infrastructure markets. I interpret the structural results as long-run steady-state effects conditional on the observed market structure, not as forecasts of any specific year.

Implementation and Crowding Out. The counterfactuals treat subsidies as changes in household prices or provider fixed costs and abstract from administrative frictions. In practice, BEAD awards may face permitting delays, state-level prioritization rules, and construction bottlenecks. Subsidized entry may also crowd out unsubsidized private investment in some markets. The estimates therefore measure the welfare return to the modeled entry response, not the full social return net of all implementation costs.

H Alternative Identification: CMT-Style Instruments

Three CMT-style instruments (Ciliberto et al., 2021) replace the Gandhi–Houde differentiation measures: (i) provider j's mean price in other counties of the same state (Hausman own-price, $\bar{p}_{j,-c}$); (ii) the mean of rivals' Hausman prices within the same speed tier; (iii) rivals' mean coverage share, a pre-determined infrastructure proxy for fixed costs. Table H.1 compares the preferred specifications (N = 170,419, 2SLS with county and provider fixed effects).

Table H.1: Gandhi–Houde vs. CMT-Style Instruments

	Gandhi–Houde	**CMT**
Demand parameters		
$\hat{\alpha}$ (price, per \$100)	-1.491^{***}	-0.123^{***}
$\hat{\rho}$ (nesting parameter)	0.863^{***}	0.752^{***}
Mean own-price elasticity	−4.99	−0.23
First-stage identification		
Shea partial R^2 (price)	0.005	0.393
Shea partial R^2 (within-share)	0.023	0.019
SW conditional F (price)	58.3	8,036
SW conditional F (within-share)	544.2	683
Cragg–Donald min. eigenvalue	986.2	0.018
Supply-side		
% products with mc ≤ 0	9.6%	99.9%

Notes: Both specifications use identical data, model, and two-way demeaning. OLS-FE estimate: $\hat{\alpha}_{\text{OLS}} = -0.127$. $^{***}p < 0.01$.

The table reveals a stark tradeoff. CMT achieves Shea $R^2 = 0.393$ for price but yields $\hat{\alpha} = -0.123 \approx \hat{\alpha}_{\text{OLS-FE}}$: the Hausman instrument is endogenous conditional on two-way fixed effects, likely because national providers carry correlated brand equity across counties, so 2SLS converges toward OLS rather than the structural parameter.[24] The degenerate implied markups (99.9% negative marginal costs) rule out CMT on supply-side plausibility grounds. Gandhi–Houde is preferred: despite its low Shea R^2 for price, it produces economically plausible markups (9.6% negative marginal costs, median Lerner = 0.200) in line with prior broadband estimates. The policy conclusions in Sections 6–7 are robust to $|\hat{\alpha}| \gtrsim 1.0$, the lower bound implied by requiring positive marginal costs for the majority of products.

I Entry Moment Inequalities

This appendix provides the formal construction underlying the fixed-cost bounds in Section 5 and the counterfactual equilibrium algorithm in Section 4.5. The fundamental object

[24] Count-based instruments face a related collinearity problem. Let $Z_1 = n_{k,c} - 1$ (within-tier rival count) and $Z_2 = n_c - n_{k,c}$ (cross-tier rival count). Their sum $Z_1 + Z_2 = n_c - 1$ is county-constant; after county fixed-effects removal $\tilde{Z}_2 = -\tilde{Z}_1$, so the pair is perfectly collinear and cannot span two instrument dimensions.

throughout is the incremental variable profit from offering position (f, g) in tract t:

$$\Delta_{fgt}(Y_{-fgt}) = r_{ft}(Y_{fgt} = 1,\, Y_{-fgt}) - r_{ft}(Y_{fgt} = 0,\, Y_{-fgt}). \tag{I.1}$$

Section I.1 introduces the parameterization and moment construction. Section I.2 develops the identified set, criterion function, and Andrews–Soares confidence set. Section I.3 describes the truncated-normal fixed-cost draws that rationalize the observed baseline. Section I.4 details the iterated best-response algorithm used for counterfactual simulations.

I.1 Setup and Moment Construction

The inequality estimation panel consists of all positions in the county-footprint set $\mathcal{J}_t^0$: $Y_{fgt} = 1$ for observed products and zero for feasible but unobserved ones. I parameterize fixed costs as in equation (5.1):

$$FC_{fgt} = \theta_{b(t)} + \theta_H\, \mathbf{1}\{g = H\} + \sigma_{b(t)}\, \zeta_{fgt}, \qquad \zeta_{fgt} \sim N(0, 1), \tag{I.2}$$

where $\theta_{b(t)}$ is the mean fixed cost in market-size cell $b(t) \in \{\text{Small}, \text{Medium}, \text{Large}\}$, θ_H is the high-speed tier premium, and $\sigma_{b(t)}$ is the size-specific shock dispersion. Write the systematic component as $\mu^C_{fgt}(\vartheta) = \theta_{b(t)} + \theta_H \mathbf{1}\{g = H\}$ and collect the seven structural parameters in $\vartheta = (\theta_{\text{Small}}, \theta_{\text{Medium}}, \theta_{\text{Large}}, \theta_H, \sigma_{\text{Small}}, \sigma_{\text{Medium}}, \sigma_{\text{Large}})$.

Following Fan and Yang (2024), I weight the moment inequalities by $K = 33$ nonnegative functions of observed market characteristics $h^{(k)}(X_t, g)$, $k \in \{1, \ldots, 33\}$, defined formally in Section 4.6: a constant, two speed-tier indicators, three market-size indicators, and 27 interactions of market size with quantile indicators of the bounds $\underline{\Delta}_{fgt}$ and $\overline{\Delta}_{fgt}$ (see also Fan and Yang 2024, Appendix A). Combined with the two-sided moment construction of Section I.2, the 33 weight functions yield $K \times 2 = 66$ moment inequalities, the count stated in Section 4.6.

Nash optimality of the observed configuration along with Assumption 4 imply

$$\Pr(Y_{fgt} = 1 \text{ is dominant}) \leq \Pr(Y_{fgt} = 1) \leq \Pr(Y_{fgt} = 1 \text{ is not dominated}). \tag{I.3}$$

The lower and upper incremental-profit cutoffs are

$$\underline{\Delta}_{fgt} = \min_{Y_{-fgt}} \Delta_{fgt}(Y_{-fgt}), \tag{I.4}$$

$$\overline{\Delta}_{fgt} = \max_{Y_{-fgt}} \Delta_{fgt}(Y_{-fgt}). \tag{I.5}$$

Together with the parameterization in (I.2), these cutoffs imply the conditional moment inequalities

$$\Phi\left(\frac{\underline{\Delta}_{fgt} - \mu^C_{fgt}(\vartheta)}{\sigma_{b(t)}}\right) \leq \Pr(Y_{fgt} = 1 \mid X_t, g), \tag{I.6}$$

$$\Pr(Y_{fgt} = 1 \mid X_t, g) \leq \Phi\left(\frac{\overline{\Delta}_{fgt} - \mu^C_{fgt}(\vartheta)}{\sigma_{b(t)}}\right). \tag{I.7}$$

Because inequalities (I.6)–(I.7) hold for every Nash equilibrium, estimation requires no equilibrium-selection assumption (Tamer, 2003; Ciliberto and Tamer, 2009; Fan and Yang, 2024).

I.2 Identified Set and Confidence Set

Moment vector. The 66 moment conditions are collected in the tract-level vector $Z_m(\vartheta) \in \mathbb{R}^{66}$, where m indexes census-tract markets. For each weight cell $k \in \{1, \ldots, 33\}$, define the two signed tract averages

$$Z^{\text{lo}}_{m,k}(\vartheta) = \frac{1}{|\mathcal{J}^0_m|} \sum_{(f,g)\in\mathcal{J}^0_m} h^{(k)}(X_m, g) \left[\Phi\left(\frac{\underline{\Delta}_{fgm} - \mu^C_{fgm}(\vartheta)}{\sigma_{b(m)}}\right) - Y_{fgm}\right], \tag{I.8}$$

$$Z^{\text{hi}}_{m,k}(\vartheta) = \frac{1}{|\mathcal{J}^0_m|} \sum_{(f,g)\in\mathcal{J}^0_m} h^{(k)}(X_m, g) \left[Y_{fgm} - \Phi\left(\frac{\overline{\Delta}_{fgm} - \mu^C_{fgm}(\vartheta)}{\sigma_{b(m)}}\right)\right], \tag{I.9}$$

where $|\mathcal{J}^0_m|$ is the number of feasible positions in market m and Φ is the standard-normal CDF. The sample moment vector is

$$\bar{Z}_M(\vartheta) = \frac{1}{M} \sum_{m=1}^{M} Z_m(\vartheta), \tag{I.10}$$

where $M = 70{,}854$. The sign convention ensures $\mathbb{E}[Z_m(\vartheta)] \leq \mathbf{0}$ element-by-element at every ϑ in the identified set; a positive entry signals a violated probability bound. The full vector stacks the 33 lower and 33 upper components.

Identified set. The *identified set* is

$$\Theta_I = \Big\{\vartheta = (\theta, \sigma) : \mathbb{E}[m(\vartheta, W_{fgm})] \leq \mathbf{0} \text{ for all } k \in \{1, \ldots, 33\}\Big\}, \tag{I.11}$$

where $\sigma = (\sigma_{\text{Small}}, \sigma_{\text{Medium}}, \sigma_{\text{Large}})$ and $m(\vartheta, W_{fgm})$ is the vector of position-level moment functions whose market-level average is $Z_m(\vartheta)$. The set Θ_I collects all fixed-cost structures

(θ, σ) consistent with the observed entry pattern under *some* equilibrium selection mechanism. A parameter vector outside Θ_I violates at least one dominance restriction in population. It assigns too high a probability to an absent position or too low a probability to an occupied one. The identified set is a bounded region rather than a point because the inequalities constrain ϑ from above and below without forcing equality. The remaining width reflects the information equilibrium multiplicity leaves unresolved.

Criterion function. A standard two-sided GMM criterion conflates binding inequalities with slack ones. For moment inequalities, only violations, positive values of $\bar{Z}_{M,j}(\vartheta)$, should enter the objective. For a candidate ϑ, the minimum-distance criterion is

$$T_M(\vartheta) = \Big[\sqrt{M}\ \max\Big\{\bar{Z}_M(\vartheta),\, \mathbf{0}\Big\}\Big]' \hat{D}_M(\vartheta)^{-1} \Big[\sqrt{M}\ \max\Big\{\bar{Z}_M(\vartheta),\, \mathbf{0}\Big\}\Big], \tag{I.12}$$

where the maximum is element-by-element and $\hat{D}_M(\vartheta) = \mathrm{Diag}(\hat{\Sigma}_M(\vartheta))$, which I define below. A value $T_M(\vartheta) = 0$ certifies that ϑ lies in the sample analogue of Θ_I. The $\sqrt{M}$ scaling ensures the statistic converges to a non-degenerate limit on the boundary of the identified set, so that simulated critical values are well behaved.

Weighting matrix. The market-level covariance matrix is

$$\hat{\Sigma}_M(\vartheta) = \frac{1}{M}\sum_{m=1}^{M}\Big(Z_m(\vartheta) - \bar{Z}_M(\vartheta)\Big)\Big(Z_m(\vartheta) - \bar{Z}_M(\vartheta)\Big)', \tag{I.13}$$

and $\hat{D}_M(\vartheta) = \mathrm{Diag}(\hat{\Sigma}_M(\vartheta))$ standardizes each component by its market-level standard deviation. The full correlation matrix $\hat{\Omega}_M(\vartheta) = \hat{D}_M(\vartheta)^{-1/2}\hat{\Sigma}_M(\vartheta)\hat{D}_M(\vartheta)^{-1/2}$ enters the simulated critical value below, following Appendix A of Fan and Yang (2024).

Confidence set. Following Andrews and Soares (2010), I construct a 95 percent confidence set by inverting a GMS test. For each candidate ϑ, define the correlation matrix $\hat{\Omega}_M(\vartheta)$ as above and form the GMS drift

$$\eta_M(\vartheta) = (\ln M)^{-1/2}\sqrt{M}\,\hat{D}_M(\vartheta)^{-1/2}\bar{Z}_M(\vartheta). \tag{I.14}$$

Draw $R^b \sim N(0, I_{66})$ for $b = 1, \ldots, 999$ and set $\hat{c}_M(\vartheta, 0.95)$ equal to the 95th percentile of

$$S\Big(\hat{\Omega}_M(\vartheta)^{1/2}R^b + [\eta_M(\vartheta)]_-,\ \hat{\Omega}_M(\vartheta)\Big),$$

where $[x]_- = \min\{x, 0\}$ component-by-component and $S(z, \Omega) = \sum_{j=1}^{66}[z_j/\sqrt{\Omega_{jj}}]_+^2$. The 95 percent confidence set is then

$$\hat{\mathcal{C}} = \Big\{\vartheta : T_M(\vartheta) \leq \hat{c}_M(\vartheta, 0.95)\Big\}. \tag{I.15}$$

The set $\hat{\mathcal{C}}$ contains all parameter vectors not rejected at level $\alpha = 5\%$ by the GMS test and is sharper than applying a common chi-squared critical value to all 66 inequalities simultaneously. The bounds in Table 12 are coordinate-wise projections of $\hat{\mathcal{C}}$ onto each structural parameter.

Two sources of width. The confidence set $\hat{\mathcal{C}}$ has two distinct sources of width. The first is *sampling uncertainty*: for a fixed identified set Θ_I, the set fluctuates around Θ_I at rate $\sqrt{M}$ and shrinks toward Θ_I as $M \to \infty$. The second is the *width of the identified set itself*: the range of parameter vectors that equilibrium multiplicity leaves observationally equivalent does not shrink with M. Wide bounds on a parameter therefore signal that the moment inequalities constrain it weakly, not that the sample is small. The counterfactual bounds in Section 6 propagate both sources by averaging over draws from $\hat{\mathcal{C}}$ and over equilibrium multiplicity in the best-response algorithm of Section I.4. Figure I.1 illustrates the stochastic search used to approximate $\hat{\mathcal{C}}$.

I.3 Fixed-Cost Draws

For each accepted $\vartheta \in \hat{\mathcal{C}}$, the latent fixed cost is $FC_{fgm} = \theta_{b(m)} + \theta_H \, \mathbf{1}\{g = H\} + \sigma_{b(m)} \, \zeta_{fgm}$. To ensure the observed baseline configuration is individually rational under the draw, I sample ζ_{fgm} from a truncated normal. For an *observed* position ($Y_{fgm} = 1$),

$$\zeta_{fgm} \leq \frac{\Delta_{fgm}(Y^{\text{obs}}_{-fgm}) - \mu^C_{fgm}(\vartheta)}{\sigma_{b(m)}},$$

and for a *feasible but unobserved* position ($Y_{fgm} = 0$),

$$\zeta_{fgm} > \frac{\Delta_{fgm}(Y^{\text{obs}}_{-fgm}) - \mu^C_{fgm}(\vartheta)}{\sigma_{b(m)}}.$$

This truncation guarantees that every firm's observed portfolio is a best response at the drawn fixed costs; the baseline market configuration is therefore consistent with Nash equilibrium at ϑ.

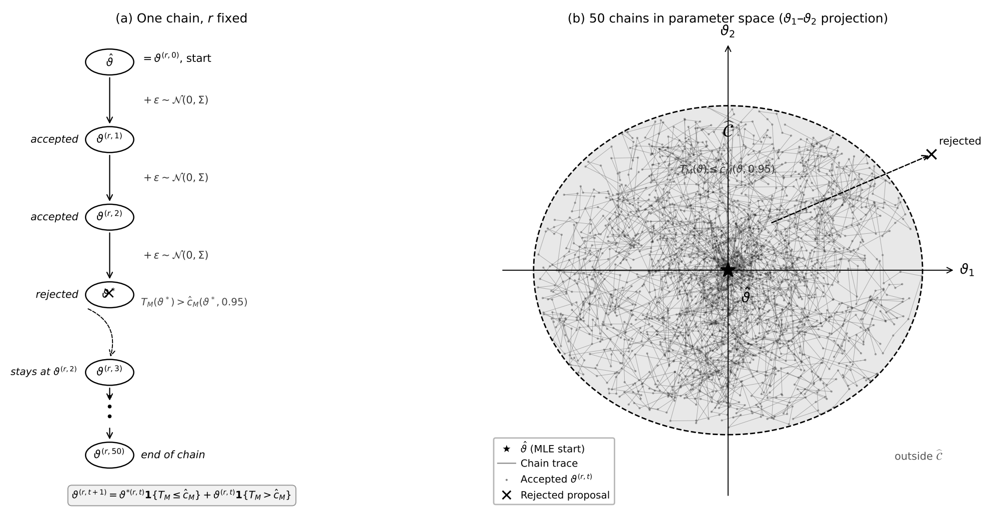

Figure I.1: Algorithm for Confidence-Set Construction

Notes: Panel (a) illustrates the accept/reject logic for a single random-walk chain at fixed draw r. Starting from the minimizer $\hat{\vartheta}$, each iteration proposes $\vartheta^{*(r,t)} = \vartheta^{(r,t)} + \varepsilon^{(r,t)}$ with $\varepsilon^{(r,t)} \sim \mathcal{N}(0, \Sigma_{\text{local}})$ and updates

$$\vartheta^{(r,t+1)} = \vartheta^{*(r,t)}\,\mathbf{1}\Big\{T_M\Big(\vartheta^{*(r,t)}\Big) \le \hat{c}_M\Big(\vartheta^{*(r,t)}, 0.95\Big)\Big\} + \vartheta^{(r,t)}\,\mathbf{1}\Big\{T_M\Big(\vartheta^{*(r,t)}\Big) > \hat{c}_M\Big(\vartheta^{*(r,t)}, 0.95\Big)\Big\}.$$

Accepted proposals advance the chain; rejected proposals leave it unchanged. Panel (b) shows the projection of accepted chains onto the $(\vartheta_1, \vartheta_2)$ plane. The two-stage search uses 50 local chains of 50 iterations followed by 100 expansion chains of 50 iterations from boundary-accepted points, yielding 656 accepted parameter vectors from $\hat{\mathcal{C}} = \{\vartheta : T_M(\vartheta) \le \hat{c}_M(\vartheta, 0.95)\}$. See Section 4.6 for further details. The construction follows Fan and Yang (2024).

I.4 Counterfactual Best Responses

For each fixed-cost draw and policy regime, I compute a pure-strategy counterfactual product configuration using iterated best responses:

1. **Initialize.** Set $Y_t^{(0)} \in \{Y_t^{\text{obs}}, \mathbf{0}, \mathbf{1}\}$, where $\mathbf{0}$ denotes the empty portfolio and $\mathbf{1}$ denotes the full feasible configuration.

2. **Update.** Cycle through firms sequentially. For each firm f, hold rivals' current portfolios fixed and solve

$$Y_{ft}^{BR} \in \arg \max_{Y_{ft} \in \{\varnothing, \{H\}, \{L\}, \{H,L\}\}} \left\{ r_{ft}(Y_{ft}, Y_{-ft}) - \sum_{g \in \{H,L\}} Y_{fgt} FC_{fgt}^{cf} \right\}.$$

3. **Replace.** Set $Y_{ft} \leftarrow Y_{ft}^{BR}$.

4. **Iterate.** Repeat steps 2–3 until no firm changes its portfolio.

When the algorithm finds multiple stable configurations across initializations, I retain all of them and report the range of outcomes across fixed points. For markets where the algorithm does not converge within the iteration limit (fewer than one percent of markets per draw), I assign the observed pre-policy configuration; this conservative fallback preserves the validity of the reported bounds. Figure I.2 reports convergence diagnostics across the 50 fixed-cost draws.

J Fan–Yang Bound Diagnostics

The three figures below provide diagnostic checks on the Fan–Yang incremental variable-profit bounds. Figure J.1 shows how bounds vary with local market structure. Figure J.2 documents geographic and market-size patterns in bound tightness.

J.1 Construction of the Weight Functions

This appendix details the 33 nonnegative weight functions $h^{(k)}(X_{fgt}, W_{fgt})$, $k = 1, \ldots, 33$, used to convert the conditional restriction in (4.15) into unconditional moment inequalities. They are built hierarchically, from coarse to fine conditioning cells.

The first is the constant $h^{(1)} = 1$, which aggregates over all potential positions. Two speed-tier indicators $\mathbf{1}\{g = H\}$ and $\mathbf{1}\{g = L\}$ and three market-size indicators $\mathbf{1}\{b(t) = s\}$, $s \in \{\text{Small}, \text{Medium}, \text{Large}\}$, then condition on each dimension separately. The remaining

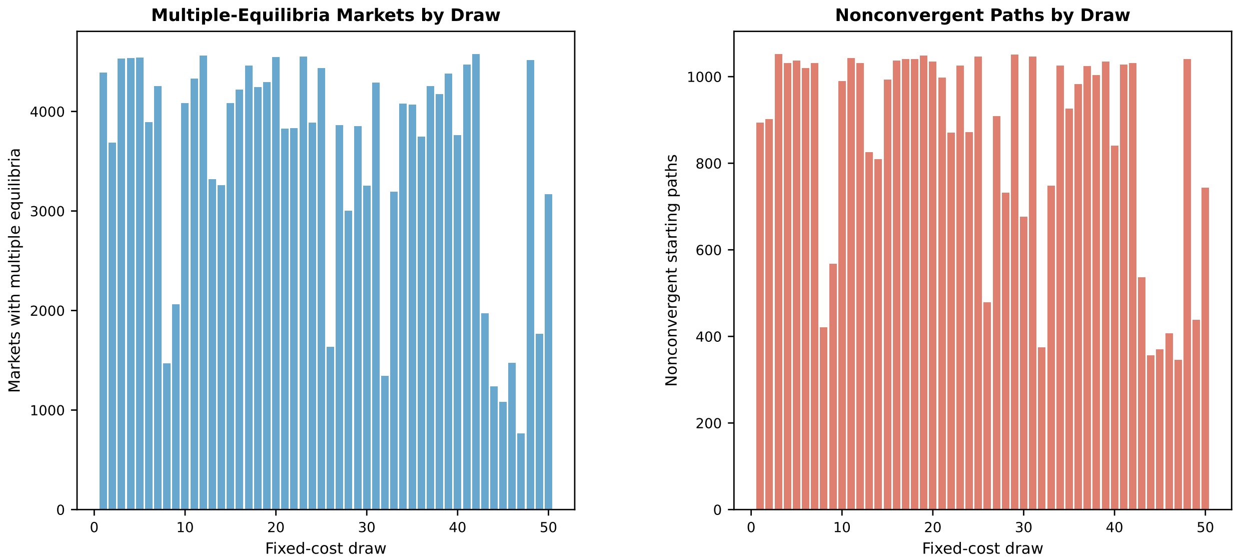

Figure I.2: Equilibrium Multiplicity and Convergence Diagnostics

Notes: For each of the 50 fixed-cost draws, the left panel reports the number of markets exhibiting multiple stable configurations and the right panel reports the number of non-converged starting paths. Convergence failures show no systematic correlation with market size or policy intensity.

27 functions interact market size with the location of the expected-profit bounds. Let $q_1 < q_2 < q_3$ denote empirical quantiles of the pooled distribution of the profit bounds. Nine functions of the form $\mathbf{1}\{b(t) = s\} \cdot \mathbf{1}\{\Delta^{\text{lower}}_{fgt} > q_\ell\}$ select positions with high competitive lower bounds, where the lower inequality in (4.15) is most restrictive. Nine functions of the form $\mathbf{1}\{b(t) = s\} \cdot \mathbf{1}\{\Delta^{\text{upper}}_{fgt} < q_\ell\}$ select positions with low monopoly upper bounds, where the upper inequality is most restrictive. Nine bracketing functions of the form $\mathbf{1}\{b(t) = s\} \cdot \mathbf{1}\{q_\ell < \Delta^{\text{lower}}_{fgt} \leq \Delta^{\text{upper}}_{fgt} < q_{\ell'}\}$, $\ell < \ell'$, select positions whose bounds fall inside a common interval, where the sharpness of observed entry behavior is most informative about the dispersion parameters σ_b. Conditioning on progressively finer cells in this way converts the single conditional restriction in (4.15) into a vector of unconditional moments that retains identifying power against parameters binding only in subpopulations of positions.

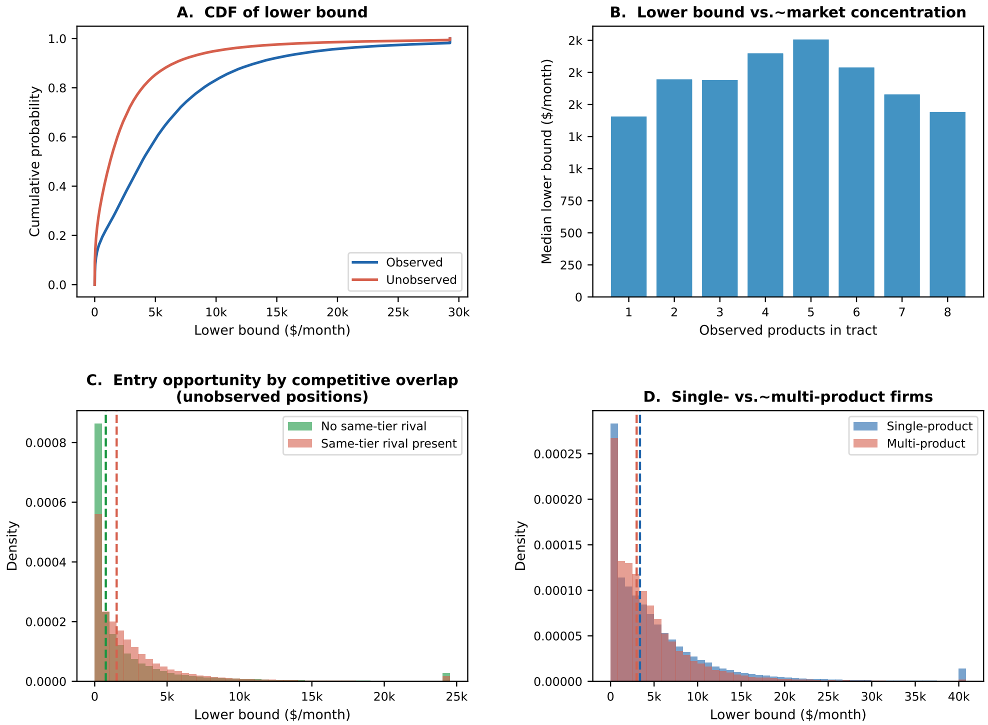

Figure J.1: Fan–Yang Bounds by Market Structure

Notes: Panel A plots the CDF of the lower bound for observed vs. unobserved positions. Panel B shows the median lower bound by number of observed products in the tract. Panel C compares the lower-bound distribution for unobserved positions by same-tier rival presence. Panel D compares single-product vs. multi-product firms.

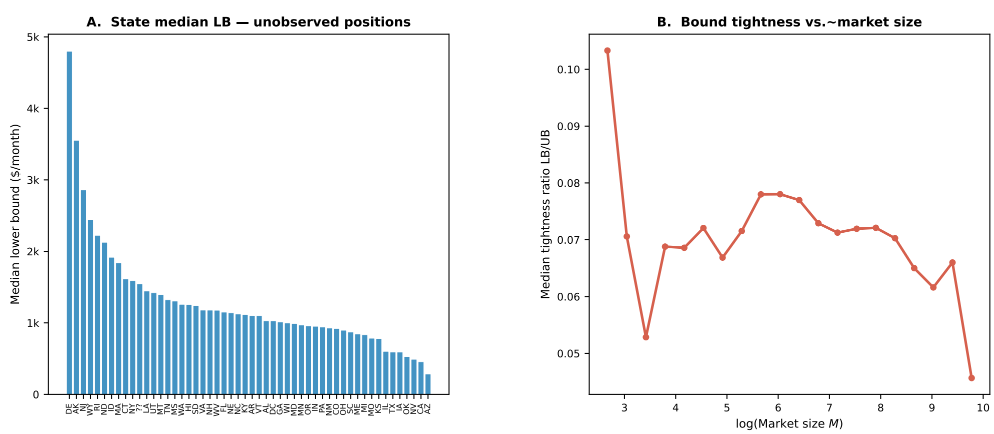

Figure J.2: Fan–Yang Bounds by Geography and Market Size

Notes: Panel A reports the state median lower bound for unobserved positions; states with higher median lower bounds have stronger underlying variable-profit incentives for potential entrants. Panel B plots the bound tightness ratio against log market size.

K Detailed Counterfactual Outcomes

Consumer surplus in tract t is the nested-logit inclusive value divided by the marginal utility of income:

$$\mathrm{CS}_t(Y_t, p_t) = \frac{M_t}{-\hat{\alpha}/100} \log\left[1 + \left(\sum_{j \in \mathcal{J}_t(Y_t)} \exp\left(\frac{\hat{\delta}_{jt}}{1-\hat{\rho}}\right)\right)^{1-\hat{\rho}}\right], \tag{K.1}$$

where prices enter $\hat{\delta}_{jt}$ in dollars and $\hat{\alpha}/100$ converts the price coefficient (estimated on price/\$100) to a per-dollar quantity. Producer surplus is variable profit net of fixed costs. Social surplus is the sum of consumer and producer surplus. National bounds aggregate tract-level lower and upper outcomes, combining equilibrium multiplicity across starting configurations with parameter uncertainty across the 50 fixed-cost draws.

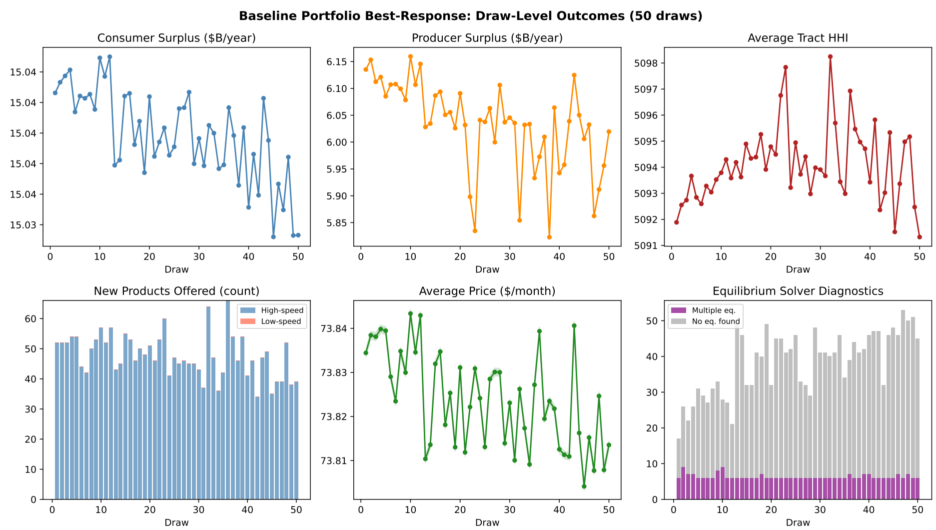

Figure K.1: Baseline Outcomes Across Fixed-Cost Draws

Notes: Baseline ($\kappa = 0$) outcomes across 50 representative draws from the Andrews–Soares confidence set. Consumer surplus, average tract HHI, and average price vary by less than 0.1%, indicating little sensitivity to fixed-cost uncertainty. New products range from 40–65 per draw due to equilibrium multiplicity. In the diagnostics panel, purple bars denote markets with multiple stable configurations; grey bars denote markets assigned the observed portfolio after reaching the iteration limit (fewer than 0.1% of markets per draw).

L Cost-Benefit Robustness

This section provides three robustness exercises for the cost-benefit results in Section 7. Tables L.2 and L.3 extend the main-text sensitivity analysis by reporting NPV and BCR bounds across all four discount rates for every BEAD subsidy rate and every ACP discount level (not only the headline $\kappa = 75\%$ scenario). Table L.4 documents the stability of welfare

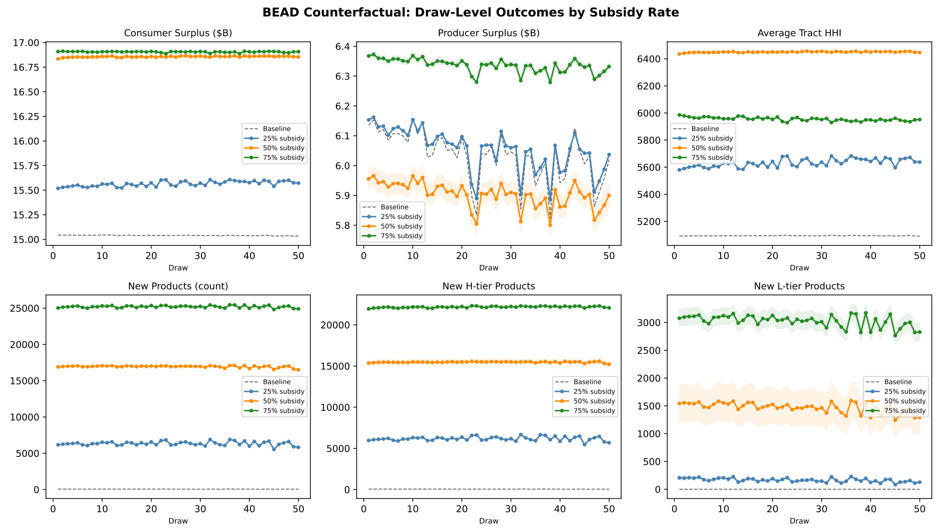

Figure K.2: BEAD Counterfactual Outcomes Across Fixed-Cost Draws

Notes: BEAD-induced new product entries across 50 fixed-cost draws, by speed tier and subsidy rate. The shaded band for $\kappa = 75\%$ shows equilibrium bounds; lines for $\kappa = 25\%$ and $\kappa = 50\%$ show draw-level midpoints. Entry at 25% and 50% is stable across draws, while entry at 75% is larger and more variable. The slight decline in later draws reflects higher fixed costs along the first principal component of $\hat{\mathcal{C}}$, reducing newly deployable positions.

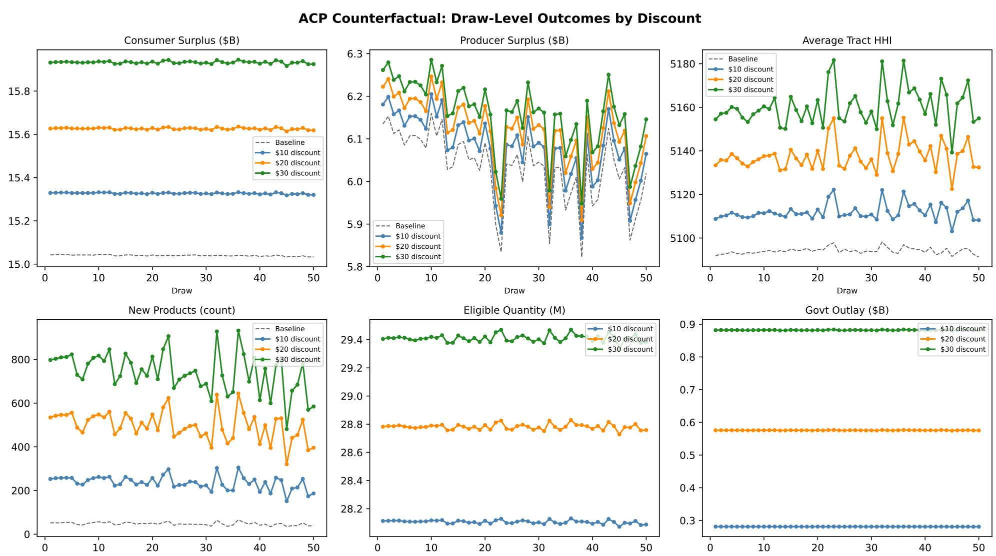

Figure K.3: ACP Counterfactual Outcomes Across Fixed-Cost Draws

Notes: ACP counterfactual outcomes across 50 fixed-cost draws, by discount level. Consumer surplus, take-up, and government outlay are nearly unchanged across draws, indicating welfare estimates are largely insensitive to fixed-cost uncertainty. Producer surplus varies moderately across draws. Government outlay rises from about \$0.29 B at a \$10 discount to \$0.89 B at a \$30 discount.

estimates across the 50 fixed-cost draws from the Andrews–Soares confidence region, showing that the main-text bounds reflect genuine partial-identification width rather than estimation noise.

L.1 BEAD Cost-Benefit Sensitivity Analysis

Table L.1 evaluates robustness by varying the discount rate from 1 to 7 percent and the infrastructure horizon from 15 to 30 years. NPV lower bounds range from \$238 billion ($r = 7\%$, $T = 25$ yr) to \$600 billion ($r = 1\%$, $T = 30$ yr). BCR lower bounds range from 17.30 to 34.70. Even at the most conservative parameter combination ($r = 7\%$, $T = 15$ years), the BCR lower bound is 12.51.

Table L.1: Sensitivity of BEAD Cost-Benefit to Discount Rate and Horizon (75% Fixed-Cost Reduction)

		Infrastructure Horizon			
Discount rate		15 years	20 years	25 years	30 years
1%	NPV (\$B)	[354.4, 459.1]	[426.1, 551.9]	[463.3, 599.9]	[495.6, 641.4]
	BCR	[25.09, 26.63]	[30.21, 32.01]	[32.70, 34.70]	[34.89, 36.98]
3% (base)	NPV (\$B)	[279.7, 361.4]	[330.4, 427.5]	[363.4, 470.5]	[390.9, 506.1]
	BCR	[19.70, 20.87]	[23.28, 24.70]	[25.85, 27.44]	[27.75, 29.42]
5%	NPV (\$B)	[222.9, 288.0]	[258.8, 335.2]	[291.4, 377.4]	[310.1, 401.8]
	BCR	[15.70, 16.64]	[18.20, 19.31]	[20.93, 22.21]	[21.90, 23.22]
7%	NPV (\$B)	[178.1, 230.1]	[205.8, 266.8]	[238.4, 309.0]	[249.7, 323.4]
	BCR	[12.51, 13.27]	[14.47, 15.37]	[17.30, 18.36]	[17.54, 18.63]

Notes: Each cell reports $[Q_{2.5}(\underline{Y}^r), Q_{97.5}(\overline{Y}^r)]$ across 50 fixed-cost draws. BEAD fiscal cost is the one-time fixed-cost grant $[\kappa/(1-\kappa) \times \text{PFC}]$; welfare gains are present values of monthly $\Delta CS + \Delta PS$. BCR = PV(ΔSW)/G. All lower-bound NPVs and BCRs are strictly positive at every rate and horizon in the table.

At the benchmark 3 percent rate over 25 years, the BCR is 25.85–27.44 and the NPV is \$363–\$471 billion. At the most conservative scenario in the table (7 percent, 15-year horizon), the BCR remains 12.51–13.27 and the NPV is \$178–\$230 billion. The 75 percent BEAD subsidy passes the benefit-cost test by a wide margin at every combination of rate and horizon in the table. The interval widths are nearly constant as the discount rate varies within each horizon. This confirms that fixed-cost parameter uncertainty, rather than financial-assumption uncertainty, is the dominant source of variation.

Figure L.1 shows the NPV bounds across discount rates for each subsidy rate. For $\kappa = 75\%$ (right panel), both lower and upper NPV bounds are positive at every rate. For

$\kappa = 50\%$ (middle panel), the same holds throughout. For $\kappa = 25\%$ (left panel), NPV bounds remain positive but narrow as the discount rate rises. The conclusion holds across discount rates between 1 and 7 percent. BEAD at 75 percent delivers large positive NPV across these financial assumptions.

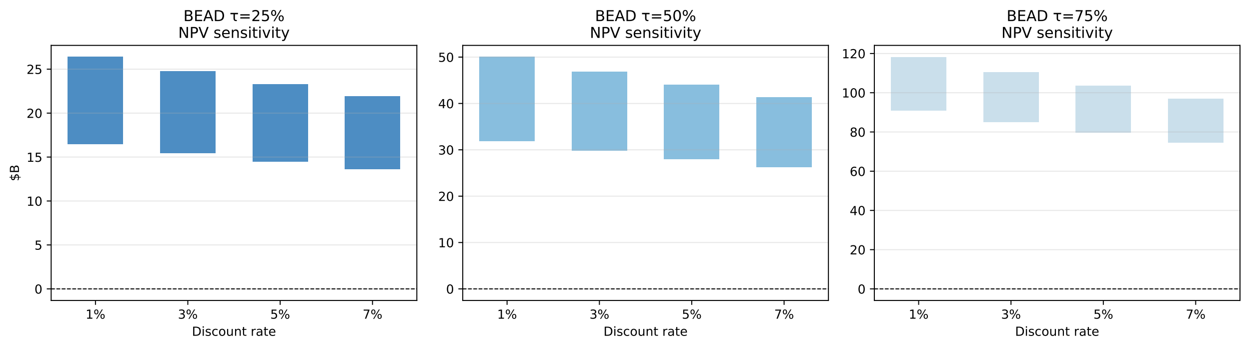

Figure L.1: BEAD net present value bounds by subsidy rate and discount rate (25-year horizon). Each bar spans the $[Q_{2.5}(\underline{\text{NPV}}^r), Q_{97.5}(\overline{\text{NPV}}^r)]$ interval across 50 fixed-cost draws. The dashed line marks NPV$= 0$. At $\kappa = 75\%$, the lower-bound bar remains positive through $r = 7\%$, indicating robust positive returns under all discount rates in the table. Fixed-cost parameter uncertainty, not the choice of discount rate, drives interval width.

For ACP, the BCR is invariant to the discount rate because both benefits and costs are recurring flows. The relevant sensitivity parameter is instead the share of eligible take-up attributable to marginal subscribers, those who would not have subscribed at the gross price. The gross BCR at the \$30 discount (1.01–1.03) barely clears unity; any non-trivial infra-marginal share therefore pushes the efficiency-adjusted ratio below 1. If 50 percent of ACP recipients are infra-marginal, the efficiency-adjusted BCR is approximately $0.5\times$ the gross BCR, falling to 0.51–0.52. Even if only 20 percent are infra-marginal, the adjusted BCR ($\approx 0.8\times$ the gross value) reaches just 0.81–0.82. Neither scenario produces a BCR above 1. The margin is thinner than the gross figure suggests precisely because that gross figure sits so close to one. ACP's fiscal efficiency is therefore primarily a function of outreach and targeting quality, not of the discount rate or infrastructure horizon.

NPV and BCR Across Discount Rates. For BEAD, the NPV is sensitive to the discount rate because the government pays the fiscal cost once and benefits accrue over 25 years. Higher rates therefore reduce the present value of the flow. The BCR is somewhat less sensitive because the government cost and the annual welfare gain scale together. For ACP, both the BCR and the NPV-equivalent are invariant to the discount rate because costs and benefits recur at the same frequency. All four columns are identical.

All BEAD BCR lower bounds remain strictly above 17 at every rate–scenario combination.

The diminishing-returns ordering ($\kappa = 25\% > \kappa = 50\% > \kappa = 75\%$ in BCR, even at 7%) holds at every discount rate. The marginal deployment that higher subsidy rates attract is harder to serve and returns less per dollar. This ordering reverses in NPV because the $\kappa = 75\%$ scenario activates far more entry. It therefore generates far larger absolute welfare gains despite the lower per-dollar return.

Table L.2: NPV Bounds ($B) by Policy and Discount Rate ($T = 25$ yr)

Policy	$r = 1\%$	$r = 3\%$ (base)	$r = 5\%$	$r = 7\%$
BEAD fixed-cost reduction				
$\kappa = 25\%$	[80.9, 128.3]	[63.6, 100.9]	[51.2, 81.2]	[42.0, 66.7]
$\kappa = 50\%$	[158.6, 246.8]	[124.5, 194.0]	[100.1, 155.9]	[82.1, 128.0]
$\kappa = 75\%$	[463.3, 599.9]	[363.4, 470.5]	[291.4, 377.4]	[238.4, 309.0]
ACP monthly discount (annual net-surplus PV)				
$a = \$10$	[0.1, 0.2]	[0.1, 0.2]	[0.1, 0.2]	[0.1, 0.2]
$a = \$20$	[0.2, 0.3]	[0.2, 0.3]	[0.2, 0.3]	[0.2, 0.3]
$a = \$30$	[0.1, 0.3]	[0.1, 0.3]	[0.1, 0.3]	[0.1, 0.3]

Notes: Each cell reports $[Q_{2.5}(\underline{\text{NPV}}^r), Q_{97.5}(\overline{\text{NPV}}^r)]$ across 50 fixed-cost draws. BEAD NPV $= \text{PV}(\Delta SW) - G$ at horizon $T = 25$ yr. ACP columns report the present value of annual net-surplus bounds ($\Delta SW - G$ per year, discounted over 25 years); ACP NPV is nearly invariant to r because both costs and benefits are annual flows and near-equal in magnitude. All BEAD lower-bound NPVs are strictly positive at every rate in the table.

Stability Across Fixed-Cost Draws. The 50 parameter draws from the Andrews–Soares confidence region vary in their implied fixed costs and therefore in the extent of counterfactual entry and welfare gains. Table L.4 reports the mean, standard deviation, minimum, and maximum of draw-level midpoints for the key CBA outcomes. For BEAD, the standard deviation of monthly ΔSW ranges from \$0.052 billion at $\kappa = 25\%$ to \$0.168 billion at $\kappa = 75\%$, or roughly 12–8% of the mean. This moderate heterogeneity reflects genuine variation in fixed-cost parameter draws rather than sampling noise, which would shrink with $M = 70{,}854$. For ACP, welfare estimates are nearly identical across draws, with BCR standard deviations of 0.004–0.006. This confirms that ACP welfare is insensitive to supply-side fixed-cost uncertainty. The demand estimates determine the gain almost entirely; they are point-identified.

Table L.3: BCR Bounds by Policy and Discount Rate ($T = 25$ yr)

Policy	$r = 1\%$	$r = 3\%$ (base)	$r = 5\%$	$r = 7\%$
BEAD fixed-cost reduction				
$\kappa = 25\%$	[44.99, 55.42]	[35.58, 43.82]	[28.79, 35.47]	[23.81, 29.33]
$\kappa = 50\%$	[39.83, 44.95]	[31.50, 35.54]	[25.49, 28.77]	[21.08, 23.79]
$\kappa = 75\%$	[32.70, 34.70]	[25.85, 27.44]	[20.93, 22.21]	[17.30, 18.36]
ACP monthly discount (annual BCR, rate-invariant)				
$a = \$10$	[1.04, 1.06]	[1.04, 1.06]	[1.04, 1.06]	[1.04, 1.06]
$a = \$20$	[1.02, 1.04]	[1.02, 1.04]	[1.02, 1.04]	[1.02, 1.04]
$a = \$30$	[1.01, 1.03]	[1.01, 1.03]	[1.01, 1.03]	[1.01, 1.03]

Notes: BCR $= PV(\Delta SW)/G$, boundary-by-boundary, then sorted. BEAD BCR declines with r because the higher rate discounts benefits more heavily whereas the one-time fiscal cost remains fixed. ACP BCR is the ratio of annual ΔSW to annual government outlay and does not depend on r; identical columns confirm rate-invariance. Every cell has BCR > 1. The minimum BCR lower bound across all 24 BEAD cells in the table is 17.30, at $r = 7\%$ and $\kappa = 75\%$.

Table L.4: CBA Stability Across Fixed-Cost Draws: Key Statistics

Scenario / outcome	Mean	SD	Min	Median	Max
BEAD: monthly ΔSW (\$B/month), midpoints across 50 draws					
$\kappa = 25\%$	0.423	0.052	0.263	0.434	0.496
$\kappa = 50\%$	0.834	0.099	0.535	0.870	0.959
$\kappa = 75\%$	2.143	0.168	1.639	2.199	2.349
BEAD: annual government cost (\$B), midpoints across 50 draws					
$\kappa = 25\%$	2.251	0.227	1.545	2.355	2.454
$\kappa = 50\%$	5.171	0.555	3.502	5.415	5.681
$\kappa = 75\%$	16.765	1.165	13.261	17.257	17.922
ACP: annual BCR, midpoints across 50 draws					
$a = \$10$	1.050	0.006	1.035	1.050	1.060
$a = \$20$	1.034	0.005	1.022	1.035	1.043
$a = \$30$	1.020	0.004	1.011	1.020	1.027

*Notes:*Statistics computed from the midpoints of the draw-level $[\underline{Y}^r, \overline{Y}^r]$ intervals across all 50 fixed-cost draws from the Andrews–Soares confidence region. BEAD entries are monthly flows; government cost is annualized ($\times 12$). I define ACP BCR as the ratio of annual ΔSW to annual government outlay and compute it directly, without discounting. The near-zero ACP welfare standard deviation confirms insensitivity to fixed-cost parameter uncertainty.